

DEVELOPMENT OF LOW-COST CUFFLESS CONTINUOUS BLOOD PRESSURE MONITORING SYSTEM USING BLOOD PRESSURE TRENDING (BPT) AND PULSE TRANSIT TIME (PTT) TECHNIQUES

An Undergraduate CAPSTONE Project
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Faculty of Engineering American International University - Bangladesh

A CAPSTONE Project submitted to the Faculty of Engineering, American International University - Bangladesh (AIUB) in partial fulfillment of the requirements for the degree of Bachelor of Science in their mentioned respective programs.

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DECLARATION

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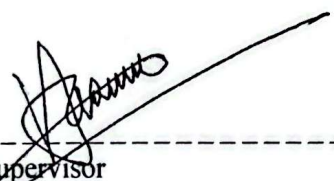
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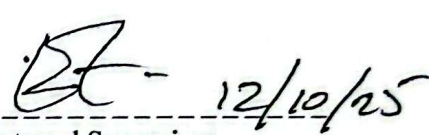
The CAPSTONE Project titled **DEVELOPMENT OF LOW-COST CONTINUOUS CUFFLESS BLOOD PRESSURE MONITORING SYSTEM USING BLOOD PRESSURE TRENDING (BPT) AND PULSE TRANSIT TIME (PTT) TECHNIQUES** has been submitted to the following respected members of the Board of Examiners of the Faculty of Engineering in partial fulfillment of the requirements for the degree of Bachelor of Science in the respective programs mentioned below on **October 12, 2025** by the following students and has been accepted as satisfactory.

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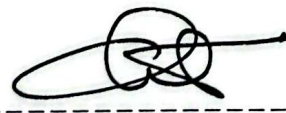
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ABSTRACT

This project aimed to create a practical and low-cost cuffless continuous blood pressure monitoring system using non-invasive sensing methods. We developed two prototypes to test different approaches for continuous monitoring. The first prototype used Blood Pressure Trending (BPT) by combining a MAX32664 biometric hub, a MAX30102 pulse oximeter sensor, and a MAX30205 temperature sensor. The second prototype applied the Pulse Transit Time (PTT) method by merging ECG and PPG signals obtained through the MAX86150 module. This data was processed using Raspberry Pi Zero and ESP8266 microcontrollers. Both prototypes provided real-time continuous readings and additional features like heart rate, oxygen saturation, and temperature monitoring. Results were displayed on LCD modules and supported wireless transmission. We validated the performance of these systems against traditional cuff-based sphygmomanometers. The BPT prototype showed consistent and reliable measurements that closely matched reference devices, achieving accuracy within medically acceptable limits. The PTT-based prototype effectively extracted ECG and PPG features to estimate systolic and diastolic blood pressure trends. However, issues like motion artifacts and the need for individual calibration affected its precision. Despite these challenges, both systems met their initial goals of providing continuous, user-friendly, and affordable monitoring solutions. The innovation of this work lies in dual-prototype design and the integration of several physiological parameters into a compact, wearable format. Unlike traditional cuff-based devices, the proposed systems remove discomfort and allow for ongoing monitoring. This feature makes them particularly beneficial for long-term patient care, remote health applications, and telemedicine after the pandemic. The results of this project show that affordable cuffless monitoring is feasible and contributes to the progress of wearable healthcare technologies while laying the groundwork for further clinical validation and future improvements.

Chapter 1

INTRODUCTION

1.1. Overture

The development of non-invasive sensing technologies has led to new ways of continuously monitoring health, especially for measuring blood pressure (BP). Traditional cuff-based sphygmomanometers are the clinical standard, but they can be uncomfortable and only provide discontinuous readings. This makes them unsuitable for tracking the changing nature of BP throughout the day. Cuffless blood pressure monitoring marks an important shift, aiming to offer continuous, discreet, and comfortable data trends. Among the methods explored, Pulse Transit Time (PTT) stands out as a very promising approach for estimating blood pressure without using a cuff. This concept is based on the well-known link between blood pressure and the speed of the pulse wave moving through the arteries.

The reason for choosing this type of project is the necessity of continuous monitoring of blood pressure in ambulatory setting or in home monitoring.. Hypertension (HTN) is an increasingly important medical and public health problem. In Bangladesh, approximately 20% of adult and 40–65% of elderly people suffer from HTN. High incidence of metabolic syndrome, and lifestyle-related factors like obesity, high salt intake, and less physical activity may play important role in the pathophysiology of HTN [1]. The association of angiotensin-converting enzyme (ACE) gene polymorphism and low birth weight with blood pressure has been studied inadequately [1]. Studies have found relationship between mass arsenic poisoning and HTN [1]. Hypovitaminosis D presumably plays role in the etiopathogenesis of HTN in Bangladeshi population. South Asians appear to respond to antihypertensive therapy in a similar manner to the Whites [1]. Large-scale, preferably, nation-wide survey and clinical research are needed to explore the different aspects of HTN in Bangladesh [1]. Therefore, continuous monitoring approach of Blood pressure is highly preferable which gives a clear idea about the overall condition.

Our project's main goal is to create a practical and affordable cuffless continuous blood pressure monitoring system. To fulfil this goal, we used various non-invasive sensing technologies to track blood pressure trends accurately. The motivation for this project grew significantly due to the challenges revealed during the COVID-19 pandemic. The demand for regular monitoring of patient vitals, along with strict infection control measures, made contact-based methods a possible risk for virus

transmission. It also increased the workload for healthcare staff. A cuffless, contact-minimizing system directly addresses these issues by allowing remote and continuous monitoring. This reduces the need for direct physical contact between patients and healthcare workers. It not only improves safety for both patients and staff but also enables the monitoring of isolated or quarantined people without affecting their care. The main objective is to create a reliable system that combines clinical accuracy with low cost and user-friendly design. This will make health monitoring more accessible to a larger population. The expected results of this research will greatly help in creating new solutions for personal healthcare and preventative medicine. Developing this system will allow individuals to access timely information to manage their cardiovascular health actively. It will also give doctors valuable long-term data to enhance diagnosis and treatment. This technology has the potential to be used in wearable devices like smartwatches and chest straps, making it easy to fit into daily routines.

There are many models that use PTT (Pulse Transit time) for measure blood pressure. In linear models there are many sub linear models like Simple linear model, Inverse linear model, Log-linear model and multiple linear regression model [2]. In non-linear models there are also various models that use non-linear methods such as Polynomial/Spline model, Exponential model, Physiological model and Machine Learning model [3]. There are also some models that use MBP (mean blood pressure) to measure Blood pressure [4]. These models can be linear and non-linear. Certainly, using a PTT-based method for cuffless BP estimation faces challenges. The link between PTT and BP varies from person to person. It can be affected by factors like arterial stiffness, vascular tone, and where the body measurement is taken. This requires personalized calibration. In addition, motion artifacts significantly threaten the signal quality of both ECG and PPG sensors. Addressing these challenges we tried to develop two models for cuffless blood pressure measurement that give us continuous monitoring and user friendly. The first model is BPT (blood pressure trending). The main theme of this model is first take calibration data of the individual then used the MAX32664 sensor to measure blood pressure that use calibration data as a baseline to give the value for blood pressure measurement. The second model use PTT (Pulse Transit Time). The main theme of this model is using the MAX86150 to extract ECG and PPG signal then measure PTT and using a linear regression to measure Blood pressure.

1.2. Engineering Problem Statement

The section explains the engineering problem of the project and the reason for making this project.

The problem statements are: -

Problem 1: The problem that we are addressing in this project is noninvasive continuous blood pressure monitoring. Continuous blood pressure can be monitored through invasive techniques. It involves placing a catheter directly into an artery [5]. But invasive blood Pressure Measurement is technically complicated and hard to set up. It can cause damage to blood vessels. There is a risk of infection since it involves inserting a tube into an artery. The process is painful for the patient. This method is usually used only for very sick patients. It is inconvenient for regular monitoring. Our project eliminates this painful process to make it user friendly.

Problem 2: Another problem of measuring blood pressure requires cuff-based method. Cuff blood pressure monitoring involves wrapping an inflatable cuff around the arm or wrist, which then inflates to temporarily stop blood flow in the brachial artery. As air slowly releases from the cuff, blood flow and pressure are measured. This process can be manual or automatic with a digital device [6]. Using cuff device to measure blood pressure can be uncomfortable for some people. As our project is cuffless so this cause of uncomfortable can be totally avoidable.

1.3. Related Research Works

1.3.1. Earlier Research

Kumar and Ayub et al. [7] has establish a non-invasive technique for estimating blood pressure (BP) using electrocardiogram (ECG) and Photoplethysmogram (PPG) signals. Before presenting their proposed approach, they discussed several conventional methods of blood pressure measurement to highlight the limitations that motivated their research. The Korotkoff method, which has been the traditional auscultatory technique since 1905, relies on the detection of arterial sounds produced during the deflation of a cuff. Although reliable, this method is cumbersome and not suitable for continuous monitoring. Similarly, the oscillometric method, which measures small fluctuations in the cuff rather than Korotkoff sounds, still depends on an inflatable cuff, limiting its practicality for real-time applications. Automated indirect methods follow the same principle, inflating and deflating the cuff automatically, and although convenient, they share the same limitation of being cuff-based. Beyond cuff-based approaches, the authors also reviewed the pulse wave velocity (PWV) method, which uses the speed of pulse

propagation through the arteries to estimate BP. This technique relies on recording ECG and PPG signals to determine PWV, and although it eliminates the cuff, it still requires careful calibration and monitoring. Based on this background, the authors proposed a methodology centered on pulse transit time (PTT) as a direct and non-invasive indicator of blood pressure variation. The proposed method defined PTT as the time interval between two characteristic points: the R-peak of the ECG, which represents the time when blood leaves the aortic valve, and the peak of the PPG waveform measured at peripheral sites such as the finger, ear lobe, or toe. By capturing both ECG and PPG signals simultaneously, the researchers calculated PTT and examined its relationship with blood pressure. Importantly, they distinguished between two variations of PTT: PTT_b, measured from the ECG R-peak to the bottom of the PPG waveform, and PTT_t, measured from the ECG R-peak to the top of the PPG waveform. These were respectively correlated with systolic blood pressure (SBP) and diastolic blood pressure (DBP), recognizing that different parts of the PPG waveform reflect different aspects of arterial pressure. To establish a mathematical model, the authors applied linear regression analysis using MATLAB 7.12.0 (R2011a). The regression model was designed to capture the inverse relationship between PTT and BP, where a shorter PTT corresponds to higher BP due to increased arterial stiffness, and a longer PTT indicates lower BP. The fitting tools in MATLAB were used to derive the relationship between PTT_b and SBP, and between PTT_t and DBP. Data for this study were collected both from clinical recordings at PGI Hospital, Lucknow, [4] and from the MIMIC II waveform database, ensuring a combination of real patient data and validated reference signals [4]. This method gives them very low mean errors 0.29 mmHg for SBP and 0.18 mmHg for DBP which comply the standard of ANSI/AAMI sphygmomanometer certification standard of ± 5 mmHg. This study shows us a way pf using PTT for cuff less, no-invasive BP estimation gives us a very convenient real-time monitoring alternative to cuff-based and invasive techniques. But the work relies on a small set and linear regression modelling that is bound to limit on diverse population [7]. Also, this study does not identify whether the method has effect on physiological factors such as age, height, or arterial stiffness. While this study gives us a statical relationship between PTT and BP, but it does not have robustness and clinical reliability in practical healthcare applications.

Shriram et al. [8] investigated to develop a way to measure continuous systolic blood pressure by using ECG and PPG signal. This study involved 23 healthy volunteers (aged 18–60 years) and focused on systolic blood pressure estimation. They computed PTT as the interval between the ECG R-wave and the PPG signal. Blood pressure is calculated by using the following formula.

$$BP = \frac{1}{\alpha} \left[\ln \left(\frac{L^2 dP}{h E_0} \right) - 2 \ln(PTT) \right] \quad (1.1)$$

Where, E_0 is Young's Modulus of arterial wall at zero pressure, h is the brachial artery wall thickness, d is inner radius of brachial artery, ρ is the blood density, L is distance of heart from fingertip and PTT is the pulse transit time [5]. Following this method, systolic blood pressure measured. Then the result was compared against standard Sphygmomanometer and Oscillometric devices. This method achieved a very high coefficient (0.983) with reference standards. The mean difference was -0.34 ± 3.1 mmHg when compared with Sphygmomanometer, demonstrating accuracy within the medically acceptable limits. This approach offers a non-invasive, cuffless, and portable alternative suitable for long-term monitoring applications. But there are some limitations of this study. This study only measured Systolic blood pressure but leaving diastolic blood pressure which reduce any possibility for analysis of hypertension management. Another limitation of this study is that the technique that has been used required a stable sensor placement and controlled environment. The volunteers of this study comparatively small to give us a valid medical reason to employ it globally. Most of the volunteers is female with few diseased cases, limiting generalizability. Furthermore, while the method is suitable for continuous monitoring, it does not evaluate measurement uncertainty or explore real-time integration into medical devices, which are crucial for clinical adoption [8].

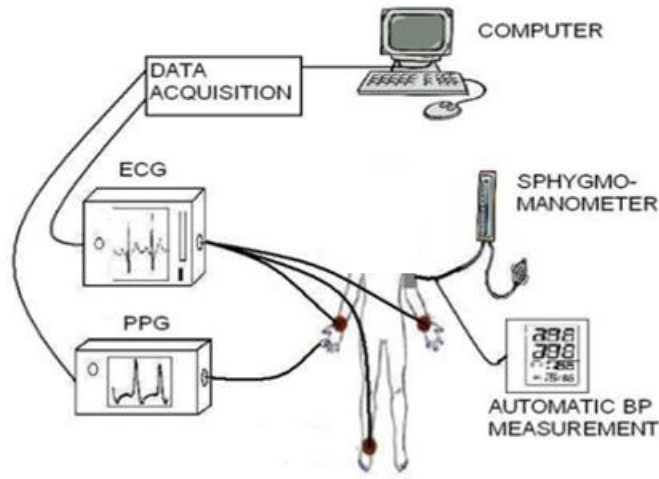


Fig 1.1: Basic Block Diagram of the System [8]

They collected data for 23 subjects. The parameters measured are SBP, HR by ECG and PPG, PPG amplitude for various age group and sex. SBP was measured on 23 subjects aged 18-60 years, without known cardiovascular disease except hypertension [8].

Bose et al. [9] has focused on continuous non-invasive estimation of blood pressure by incorporating pulse transit time (PTT), hemodynamic factors (HDF), and regulatory factors (RF) derived from ECG and PPG signals. The traditional research only relies on PTT but in this study, they also approach additional physiological parameters such as systolic amplitude, pulse width, and heart rate for aiming the dynamic regulation of blood pressure. They use two models for blood pressure measurement.

$$BP = (a_0 * PTT) + (a_1 * AMP) + (a_2 * PW) + (a_3 * HR) + a_4 \quad (1.2)$$

Here AMP is the systolic amplitude of PPG, PW is pulse width of PPG and HR is heart rate.

$$BP = (a_0 * PTT) + (a_1 * AMP) + (a_2 * PW) + (a_3 * HR) + (a_4 * \Delta PTT) + (a_5 * \Delta AMP) + (a_6 * \Delta PW) + a_7 \quad (1.3)$$

where ΔPTT , ΔAMP , ΔPW are beat to beat changes in PTT, systolic amplitude and pulse width respectively. Eq. (1.2) which uses PTT and HDF as input features is considered as model-I whereas (1.3) which uses PTT, HDF and RF as input features is considered as model-II. Constants a_0 , a_1 , and a_2 were subject dependent and required calibration procedure to stabilize the system. The evaluation was conducted using ECG, PPG, and arterial blood pressure signals from the MIMIC-II database. Outliers are defined as those observations that deviate too much from the other observations. These outliers can be due to patient movements. The identification and removal of outliers is important to avoid the wrong computation of the model parameters. This is done by taking the z-score of the input features. The z-score is given by

$$x_{score} = \frac{x - \text{mean}(x)}{\text{std}(x)} \quad (1.4)$$

where $\text{mean}(x)$ and $\text{std}(x)$ are the mean and the standard deviation of the input feature x . The data points for which the value of x_{score} is greater than 2.5 and less than -2.5 were considered as outliers and they were removed for further processes.

To obtain the best parameter set for an individual, the 75% of the data is used as training dataset (TrainDB) and 25% of the data as testing dataset (TestDB). The linear regression analysis of the form $y = ax + b$ was performed on TrainvalDB to obtain the unknown coefficients (a) from model I and II. From the model each person blood pressure measured separately. The model demonstrated improved accuracy that achieved a reduction of 6.36% in mean absolute error (MAE) and 4.98% in root mean square error (RMSE) for systolic BP, and 12.28% in MAE and 28.23% in RMSE for diastolic BP, compared with PTT-only models. But this study also has limitation, the method in this study only validated for elderly patients (aged 62–84) from the MIMIC-II dataset, where arterial stiffness is more prominent, raising concerns about generalizability across younger and healthier populations. Their approach relies on subject specific calibration as regression coefficient vary between individuals. The addition of

hemodynamic and regulatory factors improved estimation performance but then the system still requires multiple signal features and computational processing that can limit its application in portable or real-time monitoring devices. Furthermore, the study lacks an uncertainty analysis of the measurement process, which is essential for comparing results with other research groups.

Yoon et al. [10] has focused on developing a compact device for non-constrained blood pressure monitoring using ECG and PPG signal. The device was built around an 8-bit ADuC812 microcontroller and was designed to use low power, running continuously for up to 72 hours on a 3V battery. ECG signals were obtained using electrodes and processed with a high CMRR amplifier, notch filters, and band-pass filtering. PPG signals were captured from backscattered light on the finger using an LED and photodetector. All measured signals were stored on an SD card and could be sent in real time to a PC for monitoring. The experimental trials involved five healthy male volunteers aged 25 to 30, with BMI ranging from 22 to 27. Measurements were conducted in a quiet environment at a controlled room temperature of 23°C plus or minus 2°C. Each subject refrained from smoking and caffeine for at least six hours before the test. ECG and PPG were recorded simultaneously for 60 seconds at a sampling rate of 227 Hz, while blood pressure was measured using a commercial oscillometric BP device for reference. Tests were performed in both resting and post-exercise conditions, with the latter induced by 100 step climbs, and repeated three times a day over five consecutive days. The recorded signals were analyzed to extract key parameters: pulse transit time from ECG R-peak to PPG onset (PTT_{foot}), pulse transit time to the maximum derivative of PPG (PTT_{dp}), systolic upstroke time (Sys_{t1}), diastolic time (Dia_{t2}), and the width of two-thirds pulse amplitude (2/3 wt). Linear regression analysis in the form of $y = ax + b$ was set up for systolic blood pressure (SBP) and diastolic blood pressure (DBP) respectively. The study compared total calibration, using data from all subjects, against individual calibration, using subject-specific data with a leave-one-out cross-validation method. The findings showed that PTT_{dp} had the strongest correlation with systolic blood pressure, while Dia_{t2} best predicted diastolic pressure. Importantly, individual calibration provided more accurate estimations of systolic blood pressure, with errors ranging from 4 to 11%, compared to total calibration, which produced errors of 9 to 14%. Diastolic predictions showed little difference between the two approaches. This methodology was important because it showed that cuffless, continuous blood pressure monitoring is possible with a portable system, and that accuracy improves significantly when the model is calibrated for each individual rather than relying on a generalized regression model. However, there are some limitations as the subject was a young healthy male group. If they expand their project volunteers to female or different age group, then lesser regression and higher percentage of error are expected. It was reported that PTT

decreases with aging and pulse waveform is different with aging, sex or illness [8]. While the study was optimized for a device for compactness but the lack of methodology here to quantifying measurement gives uncertainty

1.3.2. Recent Research

Paul C.-P. Chao et al. [3] has focused on BP estimation using cuffless sensors, demonstrated a wide range of techniques and algorithms for accurate estimation of SBP (Systolic blood pressure) and DBP (Diastolic blood pressure). The traditional method consists of ECG and PPG signal analysis, but it faces certain limitations in accuracy and computational complexity because of the dynamic nature of the human cardiovascular system. The method used pulse wave velocity (PWV), such as the Bramwell-Hill equation (1.5), have shown potential in theoretical BP calculation models.

$$PWV = \sqrt{\frac{V \Delta P}{\rho \Delta V}} \quad (1.5)$$

where ρ is the blood density, V is the volume of blood in artery, ΔP is the variance between systolic (SBP) and diastolic blood pressures (DBP), and ΔV is the corresponding blood volume change. Due to the fact that the ρ , V , and ΔV are nearly constant for each subject, Eq. (1.5) can be re-expressed in a different form of

$$SBP - DBP = \frac{\rho \Delta V}{V} - (PWV)^2 = \frac{\rho \Delta V}{V} \left(\frac{L}{PTT} \right)^2 = K_a \frac{1}{PTT^2} \quad (1.6)$$

where PTT is the time period taken by a pressure propagation wave to travel between two arbitrary sites in an artery, called the pulse transit time. On the other hand, L is the associated propagation distance (unit meter) of the pulse pressure wave during a given PTT period (unit second).

$$PWV = L/PTT \quad (1.7)$$

K_a is a parameter to be calibrated by experiments for a subject. So SBP can be expressed by

$$SBP = DBP + K_a \frac{1}{PTT^2} \quad (1.8)$$

They also reported that many studies applied various learning regression techniques to predict blood pressures based on features beyond PPTs and PATs taken from the measured PPG waveforms. The regression algorithms used include multiple linear regression (MLR), support vector machine (SVM), regression tree (RT), random forest (RF), adaptive boosting (AdaBoost), and artificial neural network (ANN). Two main databases are used for training and validation:

MIMIC-II and Queensland [3]. The MIMIC-II database is collected from intensive care unit (ICU) patients and contains 21,442 records [3]. In contrast, Queensland has more than 7,000 records [3]. These application of methods such as pulse transit time (PTT) or pulse arrival time (PAT) has improved the accuracy of SBP and DBP prediction. But the dynamic nature of the human body in real conditions complicates accurate forecasting. Despite the successes, many of the algorithms proposed by the authors exhibit large errors, particularly in predicting DBP. They suggested using algorithms such as adaptive boosting (AdaBoost) and deep recurrent neural networks (BiLSTM), which allowed for more accurate long-term results; however, the high computational complexity of these models limits their practical application. It is important to note that most studies used pre-defined features of the signals, which also impacts the accuracy of the final models. Machine learning systems, such as deep neural networks and supervised learning methods, have shown significant improvements in BP prediction accuracy, but they remain sensitive to feature selection, training data, and the number of processed samples. The main challenges are the need for calibration, physiological dependencies' complexity, and models' high computational complexity. Despite these advances, further improvements in BP prediction require the development of new algorithms that will account for greater variability in physiological factors and optimize computational resources to ensure real-time accuracy [3].

Yan J. et al. [11] has focused on effectively using the fusion of ECG, PPG, and PPW signals along with the Random Forest Regression (RFR) algorithm for predicting BP. This result shows us that demographic factors and physiological signal features such as PWV-a, ECG, and PPW significantly contribute to predicting both SBP and DBP. But this study faces limitations including a lack of sample diversity and challenges in applying the approach for long-term monitoring. In this study, the RFR model demonstrated a very good performance for predicting SBP and DBP. The Mean Absolute Error (MAE) for SBP prediction was 0.90 mmHg, and for DBP, it was 0.64 mmHg. The standard deviation (STD) was 2.47 mmHg for SBP and 1.53 mmHg for DBP. The feature fusion method notably improved prediction performance, outperforming other algorithms like Gaussian Process Regression and Support Vector Machine. But the preferred method has the following significant drawbacks: the selection of optimal features using RFR and Sequential Forward Selection may require large computational resources and large datasets to avoid overfitting; the model may be less reliable for populations not included in the study group (e.g., individuals without cardiovascular diseases); implementing this model in real-time systems may be challenging due to the computational requirements [11].

Ghosh et al. [2] has focused on continuous Blood pressure monitoring based on pulse transit time (PTT). PTT is calculated by the time delay between ECG peak (R-peak) and ppg peak. synchronize ECG and PPG data was collected by three channels of ECG and one channel of PPG. Once data was collected the data was analysis by MATLAB Cross correlation between the adjacent peak points of ECG and PPG signals. BP was measured using a validated Oscillo metric device as the reference standard enabling continuous estimation of both systolic (SBP) and diastolic blood pressure (DBP). For calibration Cuff-based BP measurements was employed in rest condition. linear regression model was used to predict BP continuously

$$BP = \frac{a}{PTT} + b \quad (1.9)$$

The Parameter a and b are subject-dependent parameter. In this study linear regression model was used to determine the coefficients. The results demonstrated that PTT-based BP prediction was most accurate under stationary conditions. Specifically, the root-mean-squared error (RMSE) for SBP was lowest in the seated posture (5.06 mmHg with sparsification vs. 6.51 mmHg without), and DBP prediction also improved significantly when preprocessing was applied (6.36 mmHg seated vs. 7.41 mmHg without). Bland-Altman analysis revealed mean differences of 0.07 ± 5.8 mmHg for SBP and -2.1 ± 7.3 mmHg for DBP in the seated condition, indicating close agreement with the reference device. However, performance degraded substantially during physical activity, with mean errors of 4.4 ± 20.9 mmHg while walking and -10.2 ± 16.0 mmHg while cycling. These findings underscore that the method is feasible for continuous BP monitoring in stationary states but remains unsuitable for ambulatory measurement due to motion-induced artifacts. The study highlights the promise of PTT-based BP estimation, particularly when combined with signal sparsification techniques for preprocessing. Nevertheless, its reliance on a small sample of normotensive subjects, the use of linear regression alone, and limited robustness under dynamic conditions restrict its applicability for real-world ambulatory monitoring.

Ping-Kwan Man. et al [12] highlights the challenges of using PWV for BP estimation. The approach used in this study was designed to systematically review, analyze, and present the development and current status of blood pressure (BP) measurement techniques. The focus was on both contact-based and contactless methods. The paper started by laying a biophysical foundation for BP measurement, concentrating on the cardiovascular system and the mathematical models that explain pulse wave velocity (PWV), arterial compliance, and pulse transit time (PTT). It revisited classical formulas like the Moens-Korteweg equation and Windkessel models to clarify the relationship between arterial elasticity, vessel size, and blood

density. This context helped frame the later data-driven methods. The authors grouped earlier works into three methodological categories: non-machine learning (non-ML), traditional machine learning (TML), and deep learning (DL) approaches. Each category included a detailed description of experimental processes. In non-ML methods, BP prediction mainly depended on waveform analysis, PTT extraction, and basic regression models. TML methods emphasized extracting features from photoplethysmography (PPG) signals, including systolic upstroke time, diastolic time, pulse amplitude ratios, and spectral features. These features were then applied to regression models, support vector machines (SVMs), and k-nearest neighbor (KNN) algorithms. DL methods removed the need for manual feature extraction by inputting waveform data or region-of-interest (ROI) signals into convolutional neural networks (CNNs), recurrent neural networks (RNNs), and hybrid systems. A key step across studies was signal preprocessing. They extensively removed noise and motion artifacts to improve signal quality. Techniques included adaptive filters, Kalman smoothers with accelerometry, wavelet de-noising, independent component analysis (ICA), double-tree complex wavelet transforms, and morphological filtering. In remote photoplethysmography (rPPG), preprocessing was done to extract facial or hand regions from video recordings. Chrominance-based algorithms and blind signal separation techniques were used to isolate hemodynamic signals. The experiments reviewed involved both small-scale and large-scale trials with human subjects. Some studies included fewer than 20 participants, while others analyzed data from more than 1300 individuals. For example, one deep learning method using transdermal optical imaging (TOI) involved extracting 126 features from 17 facial areas and 29 meta-features that considered environmental and demographic factors. Principal component analysis (PCA) was used to reduce dimensionality before training a multi-layer perceptron. In another study, model personalization was tested by adding biometric features like body mass index (BMI) and age into the predictive models. Evaluation methods were consistent and included mean absolute error (MAE), root mean square error (RMSE), and mean error (ME) to compare predicted BP values and reference BP measurements. Some methods also used cross-validation techniques such as leave-two-out or k-fold validation. Experimental conditions varied across different times of day (afternoon versus evening), activity states (rest, exercise, recovery), and lighting conditions to account for physiological and situational differences. In summary, this paper combined theoretical cardiovascular modeling with a thorough review of algorithmic strategies and experimental validations. It included mathematical modeling, contact-based PPG and ECG analysis, contactless rPPG imaging, feature extraction, signal preprocessing, and deep learning frameworks. Throughout all approaches, the study prioritized robustness against noise, subject variability, and environmental factors. This focus ensured that the comparative review delivered a detailed understanding of BP measurement

techniques. But they require highly accurate timing of the events like aortic valve opening. Alternative methods like seism cardiography (SCG) and gyro cardiography (GCG) are focusing on bypassing the complexities of the pulse ejection period (PEP). The solution of this problem could be Remote. It is contactless solution that offers portability and ease of use for continuous BP monitoring.

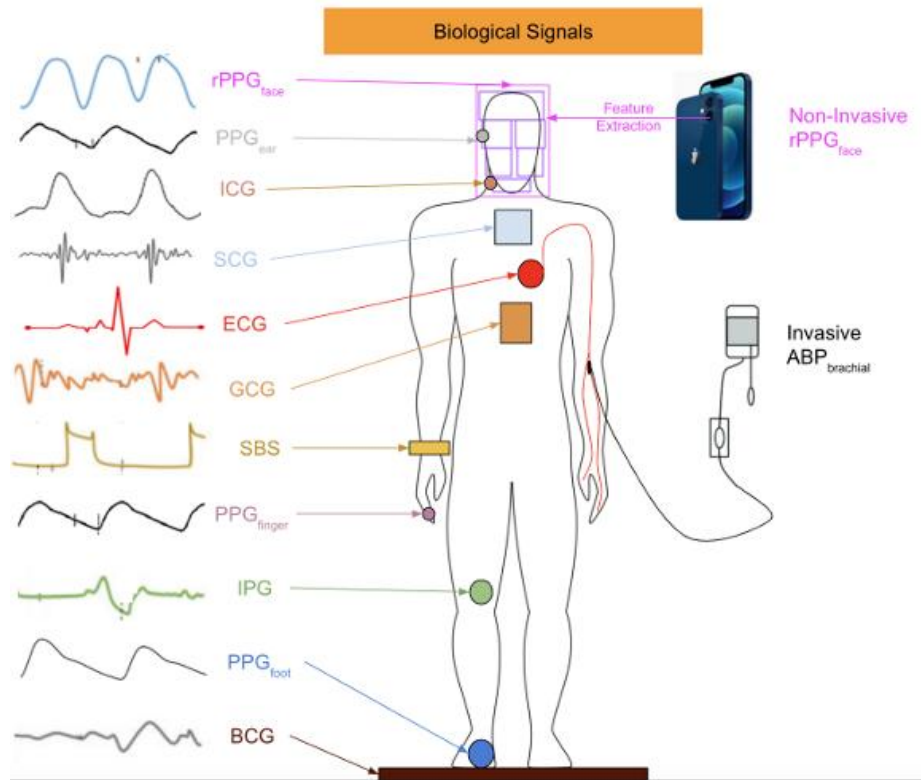


Figure 1.2. Biological signals from various parts of the body that can be used for blood pressure measurement [12].

Vasilevskyi et al. [4] has focused mathematical models that can measure systolic and diastolic blood pressure using ECG and PPG signals acquired from the fingers through the MAX86150 evaluation system. This study has two approaches in MATLAB. One based on linear equations with fixed coefficients from prior studies. Another way is using experimentally derived parameters linked to pulse transit time (PTT). The results were validated on two patients (42- and 66-year-old) with systolic and diastolic values that were determined both linear and non-linear methods. This methodology gives us a way to enhance the functionality of portable devices such as electronic stethoscopes thus enabling cuffless BP monitoring. But the study has a limit in that it has a very small datasheet considering only two patients. Unlike machine learning-based methods this methodology does not dynamically across varying physiological conditions

or larger populations. Despite these limitations this study gives us a structured framework for integrating ECG/PPG-based indirect BP estimation into compact medical devices [4].

1.4. Critical Engineering Specialist Knowledge

The topic of a feasible cost-effective cuffless continuous blood pressure monitoring system requires special knowledge. The specialized critical engineering knowledge about this module is necessary to understand the working procedure and the operation of this module for desired task. There are several issues for this module which need special knowledge of critical engineering. Such as

Electronics and Embedded System Design

The system used embedded systems like ESP8266 and Raspberry Pi zero 2W board. So, knowledge about communication protocol between sensor and board is highly necessary to understand for building the project. For making the prototype portable much knowledge of electronics is also necessary like charging system, buck converter, booster etc.

Biomedical Signal Processing

The knowledge of processing physiological signals such as ECG and PPG is essential for extracting features like R-peaks and PPG foot points. Knowledge of filtering, noise reduction peak detection, and feature extraction techniques is required to achieve reliable Pulse Transit Time (PTT).

Physiological Modeling and Blood Pressure

A very good understanding of cardiovascular physiology and the relationship between arterial characteristics, pulse transit time (PTT), and blood pressure is essential. This knowledge enables the development of precise mathematical models to measure blood pressure.

Algorithm Development

Deep understanding of developing algorithms for data fusion, real time signal analysis and decision making is very important. For improving BP monitoring knowledge on adaptive filtering, machine learning and regression methods are very important.

Signal Processing and Programming:

The cuffless blood pressure monitoring system will acquire ECG and PPG signals using biomedical sensors such as MAX32664 in the first prototype and MAX86150 in the second prototype. These sensors will capture physiological signals and transmit the data to a microprocessor for processing. Since the acquired signals may contain noise and artifacts, proper filtering and peak detection techniques are needed to extract accurate features such as R-peaks from ECG and the foot/peak points from PPG. These features will then be used to calculate Pulse Transit Time (PTT) and estimate blood pressure trends. Signal processing is therefore a crucial area of knowledge for this project. Additionally, microprocessor programming knowledge is required to implement algorithms for data acquisition, real-time analysis,

and communication with output devices. Programming will enable continuous monitoring, validation of normality, and detection of abnormalities in blood pressure levels, making it an essential component of the system design.

1.5. Stakeholders

There are various stakeholders and their needs for the project by examining research and publications. Developing a cost-effective cuffless blood pressure monitoring system with non-invasive sensing technologies like Pulse Transit Time (PTT) and blood pressure trending will significantly impact both clinical and personal healthcare. By collaborating, these stakeholders can help ensure that this technology enhances the lives of millions.

- **Health Service Providers:** Hospitals, clinics, and healthcare professionals will benefit most from this project. The cuffless BP monitoring system will allow for continuous and real-time blood pressure tracking for patients. This will cut down on the need for manual cuff-based measurements all the time. It will increase efficiency in hospital wards, outpatient clinics, and intensive care units. Healthcare providers can monitor more patients remotely and gather reliable data for diagnosis and treatment planning.
- **Patients with Hypertension and Cardiovascular Risk:** Patients dealing with hypertension, hypotension, or other cardiovascular issues will gain directly from this project. Continuous and non-invasive monitoring will be comfortable, remove the hassle of cuff inflation, and allow for early detection of abnormal blood pressure patterns. Patients can use this device at home, making it valuable for preventive care and long-term management.
- **Biomedical Instrument Manufacturers:** Companies that design and produce biomedical devices can take advantage of this technology to broaden their product line. The need for affordable, wearable, and continuous blood pressure monitoring solutions is rising. Integrating non-invasive sensing technologies like ECG-PPG will create opportunities for innovation and market expansion.
- **Biomedical Researchers and Engineers:** Researchers and engineers will benefit from this project as it offers a chance to study signal processing, sensor fusion, and health data analysis. The system can serve as a foundation for further advancements in cuffless monitoring, predictive healthcare, and connections with IoT-based medical devices.

- **Regulatory Bodies:** Regulatory organizations, such as the Directorate General of Drug Administration (DGDA) in Bangladesh and international medical device approval authorities like the FDA and CE, will play a crucial role in assessing the safety, accuracy, and reliability of this cuffless monitoring system. They will ensure the device meets medical standards before it is used in clinical and commercial settings.
- **General Public and Caregivers:** Beyond hospitals, individuals, caregivers, and families will also see benefits from this project. With easy-to-use, non-invasive monitoring, people can track their health at home. This fosters awareness and encourages lifestyle changes to lower long-term health risks.

1.6. Objectives

This project aims to create a practical, cost-effective continuous cuffless blood pressure monitoring system that uses non-invasive sensing technologies. The system will combine methods like blood pressure trending, pulse transit time (PTT), and related signal processing techniques to offer continuous, accurate, and easy-to-use monitoring. The design will focus on affordability, accessibility, and reliability to ensure the technology is suitable for both clinical and personal healthcare applications. By using ECG, PPG, and other physiological signals, the system will avoid the discomfort and limitations of traditional cuff-based devices. This development seeks to support preventive healthcare, early detection of hypertension, and better patient management.

1.6.1. Primary Objectives

- Designing and developing a cuffless blood pressure monitoring system using non-invasive sensing technologies.
- Implementing pulse transit time (PTT) and blood pressure trending methods for continuous measurement.
- Integrating ECG and PPG signal acquisition to extract features and estimate blood pressure.
- Developing strong algorithms for signal processing, noise reduction, and peak detection.
- Validating the system's performance against traditional cuff-based sphygmomanometers.
- Ensuring the system is accurate, reliable, and safe for users during testing and validation.

1.6.2. Secondary Objectives

- Optimizing the design for cost-effectiveness and energy efficiency to ensure extended operation.
- Developing a modular hardware and software structure to support scalability and future upgrades.
- A simple user interface for real-time monitoring, visualization, and data logging.
- Conducting risk assessments and reliability analyses to ensure safe operation in various environments.
- Documenting the entire design, development, and testing process for academic book presentation.

1.7.Organization of Book Chapters

Chapter 2: Project Management The basic principles of project management are the main subject of this chapter. It covers topics like S.W.O.T. analysis, schedule management, cost analysis, P.E.S.T. analysis, professional responsibilities and management principles and economic models. The ideas presented in this chapter provide a foundation for the later chapters and a framework for effective project management.

Chapter 3: Methodology and Modeling - This chapter builds upon the project management principles discussed in Chapter 2 by delving into block diagram, hardware model and modeling techniques used in the projects. It covers topics such as system modeling and simulation.

Chapter 4: Implementation of Project- In this chapter the focus shifts to the practical aspects of implementing a project. It covers topics such required tools and components, implemented models like simulation and hardware models and engineering Solution in accordance with professional practices.

Chapter 5: Results Analysis & Critical Design Review- - The analysis of project results and the critical design review process are discussed in this chapter. It covers topics like simulated results analysis, hardware results, and project results comparison.

Chapter 6: Conclusion- The final chapter summarizes findings, novelty of the work, cultural and societal factors and impacts, limitations of the work, future scopes, economic and cultural factors, and sustainability.

Chapter 2

PROJECT MANAGEMENT

2.1. Introduction

Blood pressure is one of the most important health parameters that is directly related to cardiovascular health and overall well-being. There are many abnormal fluctuations that occurs in blood pressure such as hypertension (a condition where the blood force against the artery walls is consistently too high) and hypotension (is a cardiovascular condition characterized by abnormally reduced blood pressure) etc. These conditions are life threatening, and they can cause heart attack, stroke, and organ damage. Therefore, it is very necessary to have accurate and timely monitoring of Blood pressure [14]. Taking these factors into consideration, the project is based on a continuous blood pressure monitoring system that is also affordable, user-friendly, and accurate.

The capstone project “Development of a Low-Cost, Cuffless, Continuous Blood Pressure Monitoring System Using Blood Pressure Trending (BPT) and Pulse Transit Time (PTT) Techniques” was executed by applying fundamental engineering management principles, including strategic planning, resource allocation, time scheduling, and cost management. At the initial stage we clearly identified the scope, deliverables, potential risk and constraints. Using tools like Gantt charts, work breakdown structure (WBS) and routine time meetings held to aim to complete the project. The progress of the project was systemically tracked and maintained timeliness. The structured management framework ensured consistent quality, accountability, and effective collaboration across all phases of the project for all members.

Our team applied various engineering management principles such as planning, organizing, leading, and controlling with effective economic decisions to navigate technical and nontechnical challenges as a collaborative unit.

2.2. S.W.O.T. Analysis

The S.W.O.T analysis gives us a strategic overview of the internal and external factors that influence the project. This gives us a clear view of Strengths and Weaknesses within the team and system design (internal), and the Opportunities and Threats present in the external environment. This helps us with early-stage planning and decision making and updating the project strategy.

2.2.1. Strengths

Non-invasive, Cuffless Design and Continuous monitoring

The device eliminates the need for a cuff machine and make it sure more comfortable and user friendly compared to traditional sphygmomanometers. Patients can measure blood pressure without any hassle or interruption in daily activities. This encourages regular monitoring and better compliance. This device also gives us continuous blood pressure measurement that is noninvasive. Traditional continuous blood pressure monitoring often requires invasive methods, which can cause discomfort. The proposed device offers a non-invasive alternative, making the process more comfortable for patients.

Multi-parameter Sensing for Accuracy

By incorporating Pulse Transit Time (PTT), Blood Pressure Trending the system reduces the risk of errors that come from relying on a single parameter [15], [16]. The use of multiple sensing modules using better and higher accuracy, robustness and adaptability.

Cost-effectiveness

The initial cost of this prototype is low compared to BP monitors and other advanced medical grade equipment that is using now a days [17]. If there is a chance to mass produce this prototype the cost will be significantly less than what it is now. Particularly in developing countries where accessing technology is limited, this affectability makes it suitable for mass development.

Portability and Continuous Monitoring

The system is lightweight and small thus making it easy portability for patients. As it has the capability for continuous and real time monitoring it gives us long-term data rather than isolated measurement. This makes it highly beneficial for hypertension management and predictive healthcare analysis.

2.2.2. Weaknesses

Calibration Dependency

For the device that uses BPT (blood pressure trending) it requires individual calibration to establish baseline readings [18]. If we don't have proper calibration, the accuracy of the device drops significantly. This requirement makes it less desirable as it limits the ease of use for the first-time user. The device that used the pulse transit time does not need calibration.

Signal Reliability Issues

The signal reliability issues come from the sensor. As the sensor is very sensitive there is a lot of noise that comes from the environment. This makes it hard and must be very complex signal processing. For example, during sweating the sensors may lose proper contact with the skin and lead to unreliable data. Another exam, if there is noise in the environment then the sensor also gives unreliable data.

System Complexity

The integration of multiple sensing sensors requires very precise and advanced signal processing. Thus, make it more technical complexity, increase power consumption, and may require advanced microcontrollers which can be a limitation in low-cost design.

Lack of Standardization

The cuff-based BP monitors are widely used. So, the data is standardized and clinically validated. The cuffless technology is in a research state. Clinical acceptance remains limited due to insufficient large-scale validation studies, making doctors hesitant to rely solely on such devices.

2.2.3. Opportunities

Telemedicine Integration.

After COVID-19 pandemic the remote health care is expanding rapidly [19]. This device plays a very important role in telemedicine by transmitting real-time BP data to healthcare providers. This gives us the opportunity for proactive diagnosis, timely interventions and massively improves patient doctor connectivity.

Low-resource Healthcare Solutions

In developing countries advanced healthcare is limited. An affordable and portable BP monitor can serve as a vital tool for the community. This greatly helps health workers and small clinics significantly improve public health.

AI and IoT Enhancement

This device can integrate with artificial intelligence (AI) and Internet of Things (IoT) systems [20]. This makes it possible for us to evolve beyond monitoring to predictive analytics. Cloud based platforms can analyze long term BP trends, detect anomalies, and send alerts, offering a complete digital health ecosystem.

Fitness and Wellness Industry Expansion

Apart from clinical use the device can be marketed for fitness enthusiasts who want to track cardiovascular health during workouts. Partnering with gyms, wellness apps, and sports technology companies could open up a non-clinical but high-demand market.

2.2.4. Threats

Regulatory Barriers

All medical devices must undergo rigorous approval processes from authorities such as the FDA (U.S.) and CE (Europe) [21]. In Bangladesh the project has to go through Directorate General of Drug Administration (DGDA) for approval. Delays in certification or failure to meet regulatory requirements can stall commercialization.

Strong Market Competition

Popular large tech companies like Apple and Samsung are highly researching and integrating cuffless BP monitoring in their device [22], [23]. Competing against these established companies with very high R&D budgets is a significant challenge.

Calibration and Trust Issues

As the calibration problem remains unsolved, doctors and patients may hesitate to trust the device. This would hinder widespread adoption despite low affordability.

2.3. Schedule Management

Project activities and tasks were structured using a schedule management framework with temporal factors. The necessary actions were detailed for the project's timely and budget-compliant conclusion. Establishing a schedule management system is essential for initiating a project, monitoring its advancement, and guaranteeing completion. The Gantt chart illustrates the project's timetable and phases, including all critical dates.

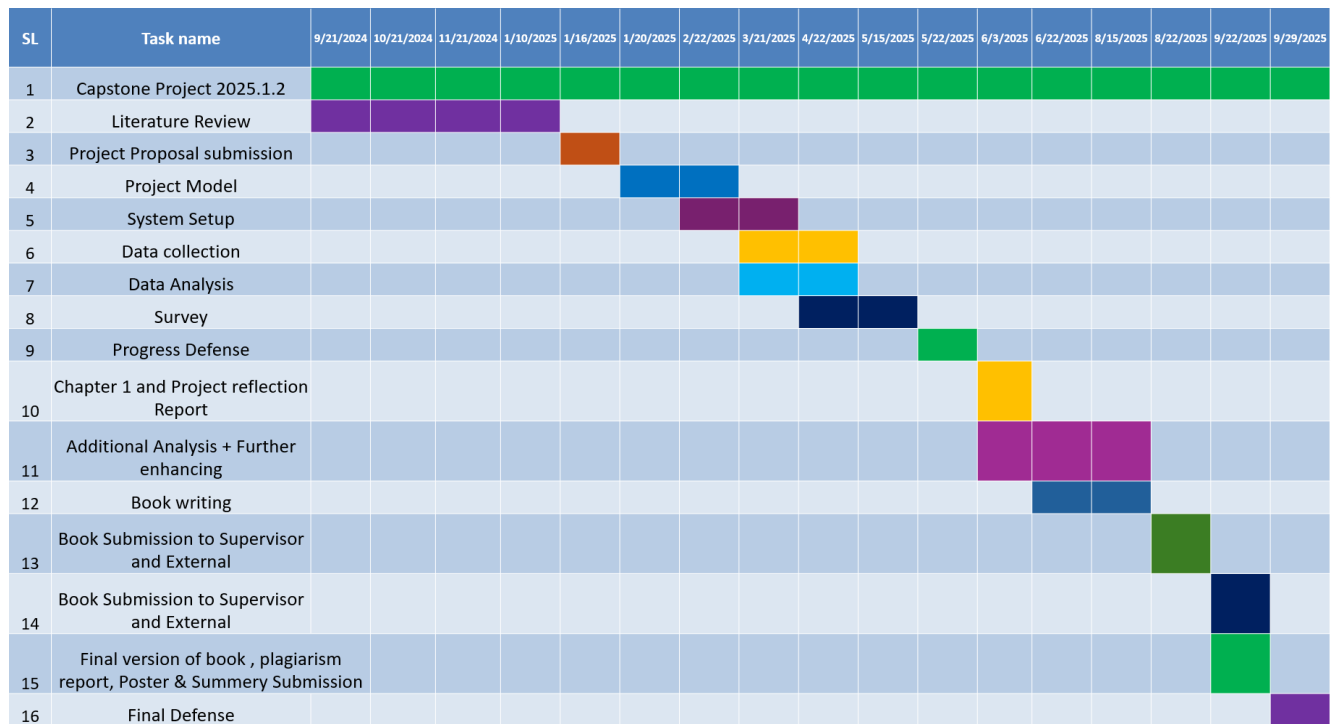


Fig 2.1: Gantt chart

2.4. Cost Analysis

In this project there are two devices that has been developed. The first device uses the principle of blood pressure trending (BPT) and the second device uses the principle of pulse transit time (PTT). The device that used BPT is referred Prototype 1 and the device that used PTT s referred Prototype 2.

Table 2.1: Cost Analysis of the BPT based Prototype 1.

Components	Quantity	Estimated Price (TK)	Final Price (TK)
MAX30205 Wearable Body Thermometer	1	700	650
ESP 8266 Node MCU	1	350	320
Pulse Express Max32664	1	3000	2850
Battery	1	100	80
TP4056 Battery Charging module	1	50	35
LED	2	10	10
1.8"-inch TFT LCD Display	1	600	550
Casing	1	300	230
Miscellaneous		500	500
	Total	5610	5225

Total Estimated Project Cost = BDT. 5610/-

Total Actual Cost of Project = BDT. 5225/-

% of Error in Cost Estimation = $(5610-5225)/5225 \times 100 \%$
= 7.3684%

Table 2.2: Cost Analysis of the PTT based Prototype 2

Components	Quantity	Estimated Price (TK)	Final Price (TK)
Raspberry Pi Zero 2W	1	2500	2200
ESP 8266 Node MCU	1	350	320
MAX86150 PPG and ECG breakout Board	1	4000	3750
3.5 inch LCD TFT Touch Screen Display for Raspberry Pi	1	2400	2150
Battery Adapter	1	350	320
BMS 15W 8.4V 1.5A Lithium Battery Charging Boost Module	1	200	175
3D Printing		1200	950
Miscellaneous		500	500
	Total	11500	10365

Total Estimated Project Cost = BDT. 11500/-

Total Actual Cost of Project = BDT. 10365/-

% of Error in Cost Estimation = $(11500-10365)/10365 \times 100 \%$
= 10.95%

Total capstone project cost = 10365+ 5225 = 15590

The inclusion of both estimated and final costs in a project analysis serves to demonstrate the fundamental principles of planning, execution, and fiscal accountability. The estimated cost represents

the initial budgetary forecast, based on preliminary research and projected component prices, and it establishes a baseline for the project's financial scope. On the other hand, the final cost reflects the actual expenditure incurred during the procurement and implementation phases, capturing the real-world realities of sourcing, negotiation, and design adjustments. The comparison between these two figures is crucial as it provides a clear measure of the project's financial accuracy and management efficiency. A lower final cost indicates successful value engineering and cost-optimization efforts, while a higher final cost would necessitate an explanation for the overrun

2.5. P.E.S.T. Analysis

A PEST analysis was conducted to evaluate external factors. These factors consist of success, adoption and sustainability of the cuffless blood pressure monitoring system. This analysis considers Political, Economic, Social, and Technological aspects.

Political Factors

Healthcare devices are closely regulated to ensure patient safety and data privacy. Governments often set certification standards like FDA approval in the USA, CE marking in Europe, or local medical regulations. For our project, meeting biomedical safety standards is essential, particularly for non-invasive electrical measurements on human subjects. Furthermore, political efforts that promote telemedicine and digital healthcare policies in developing countries can help increase the use of low-cost monitoring solutions. On the other hand, delays in approval processes might slow down commercialization.

Economic Factors

One of the main reasons for adoption is affordability. In many developing countries, patients cannot afford costly continuous monitoring systems. Our prototype costs much cheaper than commercial cuffless monitors priced. This makes it a practical option for both low-income households and large public health programs. Additionally, the global wearable healthcare market is growing quickly, creating investment opportunities for low-cost alternatives [24]. However, factors like production scale-up, sensor availability, and import taxes can affect the final price.

Social Factors

Hypertension is a growing public health issue, affecting more than 1.3 billion adults worldwide (WHO, 2023) [25]. People are becoming more aware of the importance of preventive healthcare and early diagnosis, especially after the COVID-19 pandemic. Patients are looking for wearable, non-invasive, and comfortable devices for monitoring at home. A cuffless blood pressure device eliminates the discomfort of inflation cuffs, making it easier to use continuously in daily life. Additionally, cultural acceptance, user education, and trust in medical-grade accuracy are key for broader adoption.

Technological Factors

Rapid advances in miniaturized sensors, low-power microcontrollers, and wireless IoT connectivity have led to the creation of compact, affordable, and high-performance biomedical devices. In this project, technologies like the PPG MAX30102 module, Max86150, biometric hub MAX82664 and ESP8266 microcontroller allow for real-time data collection and wireless transmission [26], [27]. Additionally, improvements in cloud computing, mobile health apps, and AI-based health analytics can further improve the usefulness of these devices by enabling predictive diagnosis and remote doctor consultations. However, challenges such as motion artifacts, calibration needs, and signal noise pose risks that must be addressed.

2.6. Professional Responsibilities

Ensuring ethical, environmental, and social integrity was central to the Cuffless Blood Pressure Monitoring System project. These responsibilities showed our commitment to patient safety, affordability, and responsible technology use. We focused on a safe, non-invasive design to protect human health. We chose low-cost and accessible components to support healthcare fairness. Our signal acquisition and data processing methods respect privacy and ethical standards. Clear documentation and teamwork were essential throughout the project, reinforcing accountability and professionalism. Ultimately, this project highlights the engineer's role in improving healthcare innovation while benefiting society by making continuous blood pressure monitoring more accessible, sustainable, and inclusive.

2.6.1. Norms of Engineering Practice

A professional role outlines an engineer's duties to society, clients, and coworkers. In biomedical engineering projects like this one, these standards are crucial because they relate to patient health, data privacy, and public safety. The professional work on this project adhered to accepted engineering standards and ethical guidelines to ensure reliability and safety.

Ethical Responsibility:

The project made sure that the system was non-invasive and safe for human subjects. Data collected from sensors, including PPG, ECG, was managed with integrity and transparency, preventing misrepresentation of results. Ethical responsibility also involves protecting potential patient confidentiality in future implementations.

Accountability

Each team member took responsibility for their assigned tasks, including simulation, hardware prototyping, signal processing, and testing. They reported results honestly, highlighting challenges like motion artifacts and calibration errors, allowing stakeholders to make informed decisions.

Ongoing Learning

Team members actively learned how to use biomedical instrumentation, digital signal processing, and embedded system design to stay skilled. Ongoing literature review and simulation-based experimentation helped them adopt the latest practices in cuffless BP monitoring.

Respect for Diversity

The project aimed to serve patients from various cultural and economic backgrounds. It focused on being cost-effective and easy to use. This approach promotes inclusivity and accessibility, particularly in healthcare settings with limited resources.

Compliance with Legal Standards

The design follows biomedical device safety guidelines, specifically ISO/IEEE standards, to reduce risks to patients. Future commercialization will need to comply with regulatory requirements such as FDA (USA) or CE marking (Europe).

Fostering a Collaborative Environment

The project needed teamwork among different fields, including biomedical instrumentation, electronics, and c++ and python language. Team members worked together. If expanded, working with medical professionals and regulatory experts would help confirm clinical-grade validation.

Documentation and Intellectual Property

All simulation codes, circuit designs, and test results were documented to keep things clear and reproducible. We thought about protecting intellectual property with patents or copyrights for future commercialization. At the same time, we balanced this with open-source practices to support research growth.

2.6.2. Individual Responsibilities and Function as Effective Team Member

This is a capstone project, so all the group members work as a team. We divided the whole research work into different parts and assigned them to different individuals.

Table 2.3: Contribution Table

Name	Contribution
ABRAR ASIF	Book Writing, Data collecting, Flow chart, Data Analysis, Data Tables
MD ROBIUL ISLAM ROMAN	Team management, Book Writing, System Code, 3D Design,
M. FARHAD	Book Writing, Data collecting, Hardware setup, 3D Printing
IFTEKHAR ALAM	Book Writing, Data collecting. Poster presentation, Block Diagram

2.7. Management Principles and Economic Models

The successful execution of the Cuffless Continuous Blood Pressure Monitoring System project was achieved by using key engineering management principles and solid economic decision-making models. The team adopted a structured framework. This approach helped them manage complexity, make the best use of resources, reduce risks, and keep costs in check during the design and implementation process.

2.7.1. Engineering Management Principles Applied

Several fundamental engineering management principles were applied during the project to guide planning, execution, and monitoring:

1. Work Breakdown Structure (WBS)

The project was broken down into smaller, manageable work packages. Tasks like literature review, simulation, hardware prototyping, signal processing, and testing were clearly defined, with responsibilities assigned to specific team members. This increased clarity, accountability, and parallel task execution.

2. Gantt Chart for Scheduling

A Gantt chart tracked milestones and deadlines over the six-month timeline. Dependences between activities, such as needing to assemble hardware before testing, were clearly mapped. This allowed the team to adjust workloads proactively and avoid delays.

3. Risk Management Framework

The team identified risks early, including inaccurate sensor readings, motion artifacts, component unavailability, and calibration errors. The identified risk was systematically evaluated and that helped us to evaluate their probability and impact. Mitigation plans included using alternative sensors, applying signal filtering techniques, and exploring backup procurement options.

4. Iterative Development Approach

The project used a step-by-step development cycle, with regular weekly review sessions that acted as short sprints. This setup allowed for fast testing of both simulation models and hardware modules. The approach helped the team tackle technical challenges promptly and make ongoing improvements during the process.

5. Team Resource Allocation

Each member was given roles based on their expertise in areas like simulation, hardware, signal processing, or documentation. This optimized the use of team members and ensured that tasks were completed.

6. Communication and Documentation:

Weekly team meetings, progress tracking, and shared repositories ensured clear communication and transparency. Documentation included simulation codes, design diagrams, and testing reports, which promoted reproducibility and effective knowledge transfer.

2.7.2. Economic Perspective and Models

Alongside management principles, economic decision-making tools were applied to ensure feasibility and cost-effectiveness:

1. Sensors Sourcing Strategy:

A decision was made to utilize inexpensive, commercially available sensors instead of developing custom modules. This approach was selected to minimize initial costs and significantly reduce development time, which is essential for efficient prototype development. Components such as the MAX30102 for PPG and ECG signals were chosen for their affordability and reliability.

2. Cost-Benefit Performance (CBF):

Each design choice was assessed based on its cost-to-performance ratio. For instance, selecting the ESP32 microcontroller over pricier options provided wireless connectivity and enough processing power at a much lower cost. The advantages of being affordable, portable, and offering continuous monitoring greatly outweighed the slight extra calibration difficulties.

3. Budget Forecasting and Monitoring:

An initial budget was set up to cover sensors, the microcontroller, and various components. Weekly tracking made sure that spending stayed within budget, and contingency funds helped avoid overspending.

4. Resource Optimization:

The team made use of open-source software tools (VS studio, Arduino IDE), buying sufficient components when possible and popular microcontrollers for resource optimization. This cut costs without losing accuracy and efficiency.

5. Sustainability Considerations:

The project focused on long-term sustainability by using low-power parts, rechargeable power supply options, and durable sensors. By keeping the device affordable and energy-efficient, the solution helps healthcare access in low-resource areas while being ready for future growth.

2.8. Summary

This chapter outlines the project management structure for developing the Cuffless Blood Pressure Monitoring System. It covers project management topics like cost analysis, schedule management, and SWOT and PEST analyses. It includes the project life cycle, the roles of different stakeholders, and the combination of hardware, software, and signal processing components. SWOT and PEST analyses assess both internal strengths and external challenges, such as technology, economy, and societal issues. Economic factors show that effective design and lean engineering help reduce costs while maintaining reliable performance. The project follows strong engineering management principles, including quality assurance, risk management, team collaboration, key performance indicators, and strict adherence to design specifications. Sustainability is encouraged through cost-benefit analysis and life cycle costing models to ensure the healthcare solution is affordable, accessible, and viable in the long run. The project is relevant to modern healthcare innovation with sound economic feasibility, as it addresses professional responsibilities like ethics, stakeholder engagement, and the inclusion of new technologies, which are crucial for the system's success.

Chapter 3

METHODOLOGY AND MODELING

3.1. Introduction

This chapter explains the methodology and modeling that had been used for design, flowchart, and implementation for the cuffless blood pressure monitoring system. As there are 2 prototypes of this project, we used two measurement techniques for continuous blood pressure measurement. So, the systems are developed by integrating measurement techniques – Blood Pressure Trending (BPT) and Pulse Transit Time (PTT). By using these techniques our objective is to make a prototype that helped us make a system to give continuous, real-time, and comfortable BP monitoring without the need for external compression.

The first prototype used Blood Pressure Trending (BPT) to measure blood pressure. Blood pressure trending is a continuous calibration-based trending algorithm that is employed to adjust baseline shifts. By monitoring the changes in BP, the system ensures more stable and reliable reading. Using Pulse Express Pulse-Ox & Heart Rate Sensor with Biometric sensor hub MAX32664 we find heart rate and SPO2 (oxygen saturation) and estimated blood pressure [28]. The IC that is used is combined of Pulse Express Pulse-Ox & Heart Rate Sensor (MAX30102) and biometric sensor hub (MAX32664). The ESP8266 microcontroller was used to process the data and gives us a reading. A TFT LCD display was used to show the readings of the data.

The second prototype used Pulse transit time (PTT) to measure blood pressure. It has been reported that the pulse transit time (PTT), the interval between the peak of the R-wave in electrocardiogram (ECG) and the fingertip Photoplethysmogram (PPG), is related to arterial stiffness, and can be used to estimate the Systolic blood pressure (SBP) and Diastolic blood pressure (DBP). This phenomenon has been used as the basis to design portable systems for continuously cuff-less blood pressure measurement, benefiting numerous people with heart conditions [17]. The sensor “Integrated Photoplethysmogram and Electrocardiogram Bio-Sensor Module for Mobile Health” Max86150 is used to find electrocardiogram (ECG) and Photoplethysmogram (PPG) [29]. Raspberry Pi Zero was a microcontroller to process the data and gives us a reading. A 3.5-inch LCD TFT display was used to show the readings of the data.

3.2. ECG Signal

The ECG is a signal that shows the electrical activity of the heart. The heart doesn't just beat mechanically; it is managed by a unique electrical system. The heart creates small electrical impulses with each beat. These impulses move through the heart muscle, making it contract and pump blood. As these electrical impulses travel throughout the body, they also reach the surface of the skin. Special sensors called electrodes detect these signals. Electrodes are usually placed on the chest, arms, or legs. They measure the voltage difference between two points on the skin. Since the electrical activity of the heart changes with each part of the heartbeat, the recorded signal forms a repeating waveform known as the ECG signal.

The ECG waveform is made up of different sections:

1. P-wave: It is caused by the atria (upper chambers) contracting to push blood into the ventricles.
2. QRS complex: It is caused by the ventricles (lower chambers) contracting strongly to pump blood into the lungs and body. This part has the largest signal because the ventricles are powerful.
3. T-wave: It shows the recovery or relaxation of the ventricles after contraction.

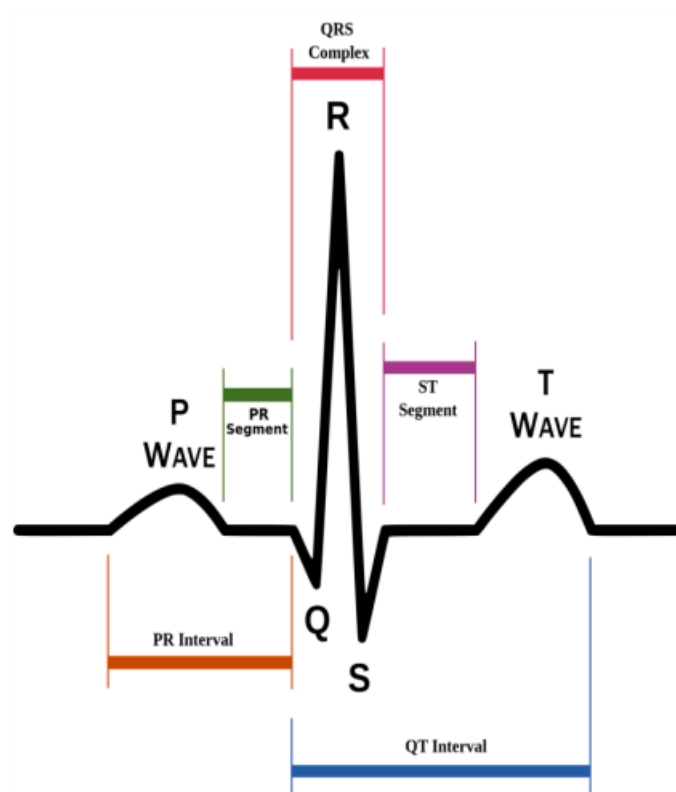


Fig. 3.1. Normal ECG signal with its different features [30]

The PR interval on an ECG measures the time from the start of the P-wave to the beginning of the QRS complex. It shows how long it takes for the electrical signal to move from the atria, through the AV node, and into the ventricles. This interval includes both atrial depolarization and a brief pause at the

AV node, which gives the ventricles time to fill with blood before they contract. A normal PR interval lasts about 0.12 to 0.20 seconds. If the PR interval is shorter than normal, it might mean the electrical impulses are bypassing the AV node. A longer PR interval could indicate a conduction delay or AV block [30].

The QT interval is measured from the start of the QRS complex to the end of the T-wave. It reflects the total time it takes for the ventricles to depolarize (contract) and then repolarize (relax). In other words, the QT interval shows the complete cycle of ventricular electrical activity, from activation to recovery. The QT interval length varies with heart rate, but it is typically considered normal between 0.35 and 0.44 seconds. A prolonged QT interval can be dangerous because it raises the risk of irregular heart rhythms like Torsades de Pointes. On the other hand, a shortened QT interval may be linked to metabolic conditions such as high calcium levels [30].

3.3. PPG Signal

Photoplethysmography (PPG) measures changes in blood volume in the small blood vessels under the skin. When the heart pumps, blood volume in the arteries and capillaries rises and falls with each beat. These changes affect how much light is absorbed or reflected as light passes through or onto the skin. The PPG waveform shows these variations and is directly tied to the cardiac cycle, reflecting the phases of systole (contraction) and diastole (relaxation).

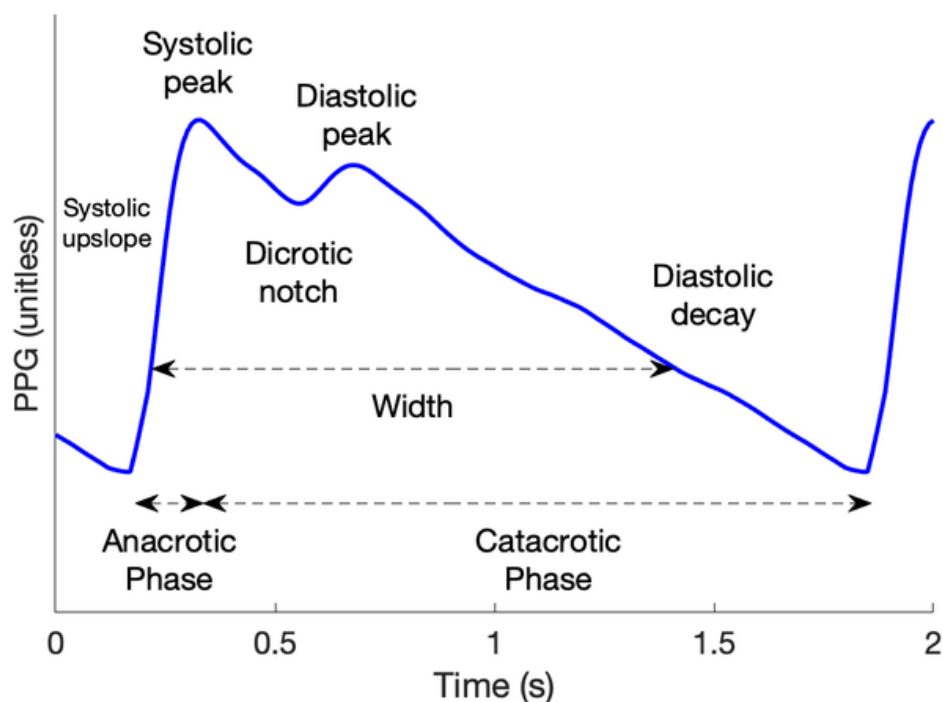


Fig 3.2: PPG SIGNAL [31]

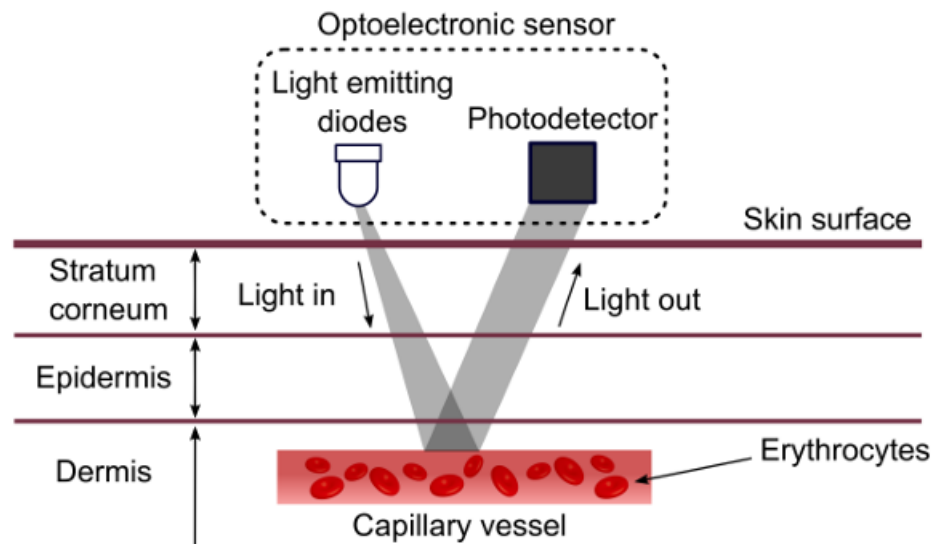


Fig 3.3: PPG Working principle [32]

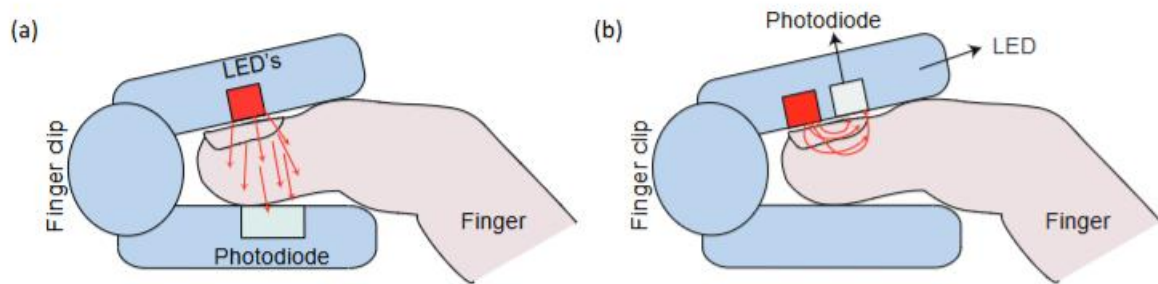


Fig 3.4: a) Transmission Mode b) Reflection Mode [31]

A PPG sensor works by shining light from a light-emitting diode (LED) into the skin and detecting the returning light with a photodetector. If the sensor is in transmission mode, the LED and detector are placed on opposite sides of the tissue (such as a fingertip), so the detector measures the light that passes through. In reflection mode, both the LED and detector are placed on the same side (such as on the wrist), and the detector measures the light reflected. As blood volume changes with every heartbeat, the amount of light absorbed or reflected also changes. These variations are converted into an electrical signal by the photodetector, which forms the PPG signal. This raw signal can then be processed to extract useful cardiovascular parameters.

3.4.PTT (Pulse transit time) Principle

In this project to find blood pressure the technique came from Vasilevskyi et al [33], which presented different mathematical models for finding blood pressure from ECG and PPG signals. Using MAX86150 sensor the ECG and PPG signals were extracted.

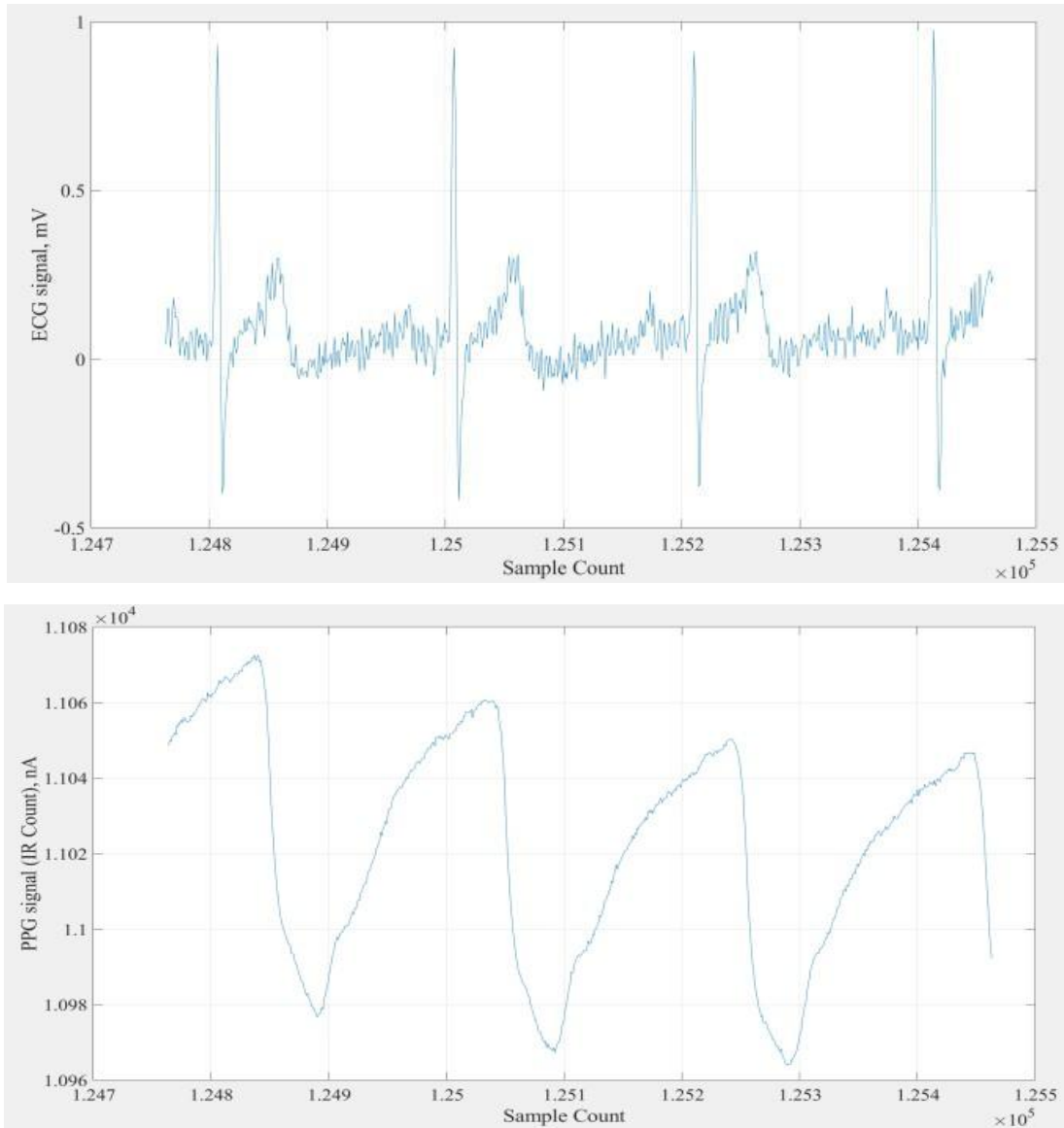


Fig 3.5: Extracted ECG signal and PPG signal [33].

The demonstrated PPG signals that are taken by MAX86150 are an inverse waveform compared to those reported in the literature [31]. This discrepancy may be attributed to variations in the equipment and software used for measuring ECG and PPG signals. From the extracted signal, two-time variations are taken which are t_1 and t_2 . Here t_1 is the time from the ECG peak to next PPG upper peak. t_2 is time from the ECG peak to PPG foot. In Figure 3.6 a picture demonstration given. This time difference between t_2 and t_1 is PTT (pulse transit time). [33].

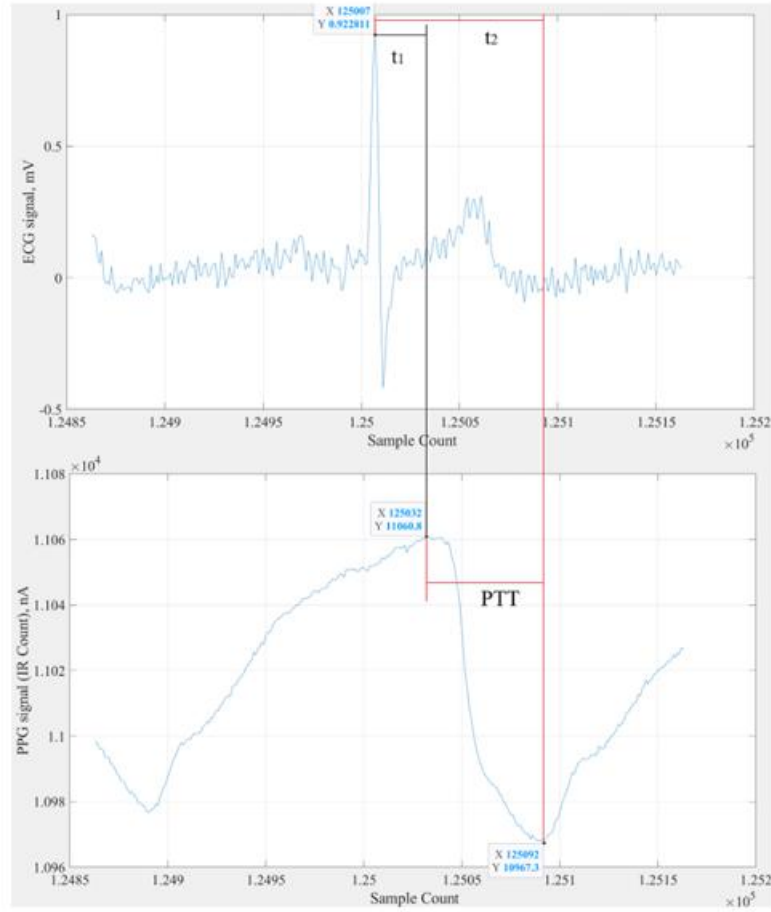


Figure 3.6 PTT measurement from ECG and PPG signal [33]

$$\mathbf{PTT = t_2 - t_1} \quad (3.1)$$

The measurements of SBP (Systolic blood pressure) and DBP (Diastolic blood pressure) formula and a_1 , b_1 has taken from the literature [33]. For SBP a linear model has used for measurement.

$$\mathbf{SBP = a_1 * PTT + b_1} \quad (3.2)$$

For measuring DBP (Diastolic blood pressure) using normal BP values to first measure MBP_{normal} (Mean blood pressure) using $SBP_{normal} = 120$ mmHg, and $DBP_{normal} = 80$ mmHg.

$$\mathbf{MBP_{normal} = \frac{SBP_{normal}}{3} + 2 \frac{DBP_{normal}}{3}} \quad (3.3)$$

The MBP_{normal} value come as 93.33 mmHg. Using the value of SBP that comes from linear model DBP can be expressed in terms of normal value of MBP_{normal} according to equation 5.

$$\mathbf{DBP = 1.5 \left(MBP_{normal} - \frac{SBP}{3} \right)} \quad (3.4)$$

MBP can be change for different blood pressure. We estimate mean arterial pressure (MBP) and diastolic blood pressure (DBP) from measured systolic blood pressure (SBP), a linear regression model was

developed in Python using scikit-learn. The dataset contained 1,015 paired SBP–DBP measurements [41]. The original data did not include MBP, it was calculated for each record using the standard clinical formula.

$$\text{MBP} = \text{DBP} + \frac{1}{3} (\text{SBP} - \text{DBP}) \quad (3.5)$$

A simple linear regression was then trained on the entire dataset, using SBP as the input feature and MBP as the target. The final model, which captures the direct relationship between SBP and MBP and saved. Once the predicted MBP is available, DBP can be approximated from the relationship.

$$\widehat{\text{DBP}} = 1.5 \times \left(\widehat{\text{MBP}} - \frac{\text{SBP}}{3} \right) \quad (3.6)$$

This approach gives us Systolic blood pressure and Diastolic blood pressure.

3.5. Block Diagram and Working Principle

This section shows the block diagram, flowchart and working principle of both prototypes.

3.5.1 Prototype using BPT (Blood Pressure trending)

This is the block diagram of the device that uses BPT (Blood pressure trending).

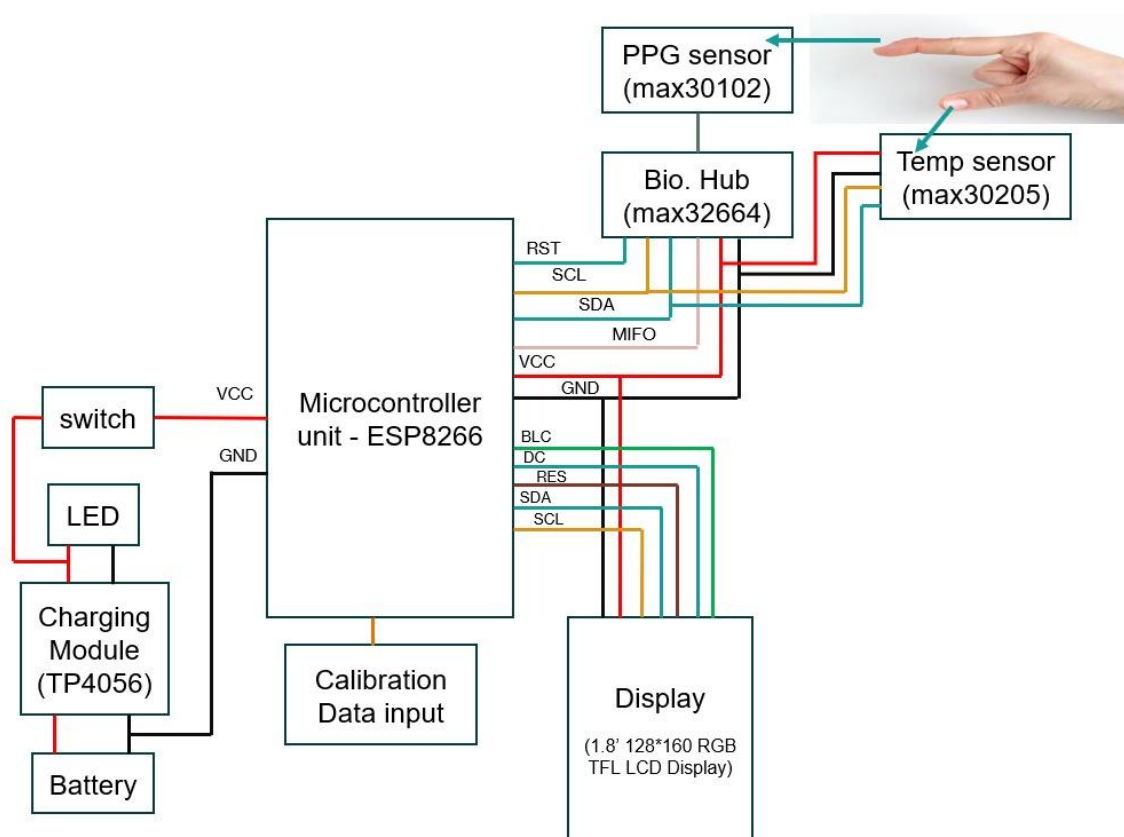


Fig 3.7: Block Diagram of Prototype using blood pressure trending.

The device uses MAX32664 that communicates with the MAX30102 through I2C to perform finger-based heart rate, blood oxygen saturation (SpO2), and blood pressure monitoring (Systolic blood

pressure and Diastolic blood pressure). The embedded algorithm uses digital filtering, pressure/position compensation, advanced R-wave detection, and automatic gain control to determine the heart rate in beats per minute while minimizing power. Also, the Analog Devices sensor hardware has built-in ambient light rejection to minimize background noise. SpO2 results are reported as percentage of hemoglobin that is saturated with oxygen. The calibration values for SpO2, Heart rate, Systolic blood pressure and Diastolic Blood pressure configuration should be determined before deploying the product. Estimated blood pressure is reported for systolic and diastolic blood pressure. Blood pressure cuff measurements are used to set the blood pressure calibration data in the firmware [28].

The MAX30205 temperature sensor measures temperature and converts the data into digital form. An I2C-compatible two-wire serial interface allows access to conversion results. The device accepts standard I2C commands to read the data, set the overtemperature alarm (OS) trip thresholds, and configure other characteristics. While reading the temperature register, any changes in temperature are ignored until the read is completed. The temperature register is updated for the new temperature measurement upon completion of the read operation [34].

Flow Chart

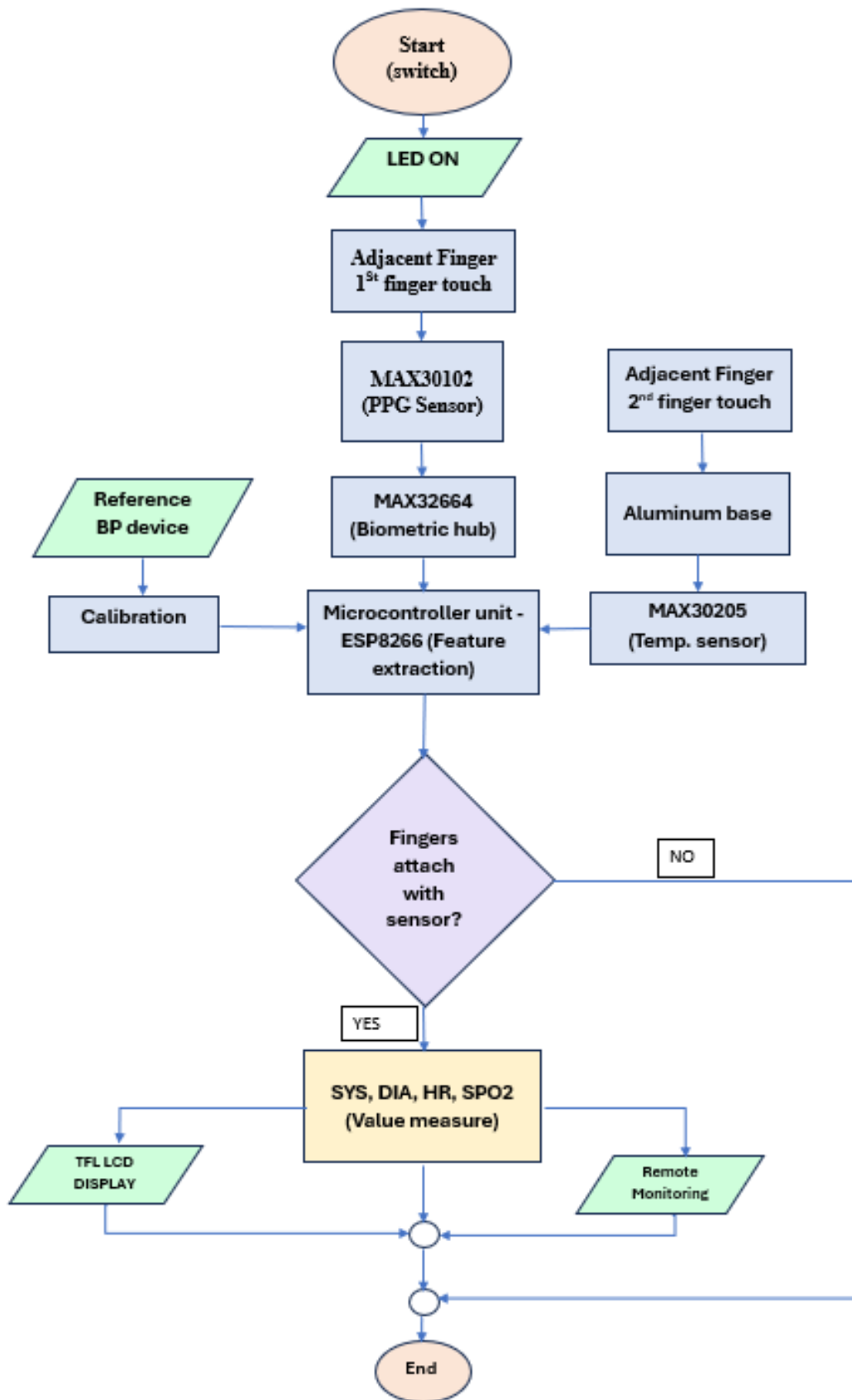


Fig 3.8: Flow Chart of Prototype using blood pressure trending.

Working Procedure

Figure 3.8 gives a detailed explanation of how the project works. The operation starts when the device is switched on using the ON/OFF switch. This activates the system and is confirmed by the LED indicator light. After that all the sensor and operating system were also on. Using Omron Digital Blood Pressure Monitor we take 3 calibrated data (for Systolic and Diastolic). After editing the code with reference calibrated data, the code is then input into the ESP8266 microcontroller. Using two fingers one on Pulse Express Pulse-Ox & Heart Rate Sensor with (MAX32664) another then another finger Skin Temperature Sensor (MAX30205). All the sensor reading is then collected and processed by the ESP8266 microcontroller. The microcontroller executes the embedded algorithms to compute the final health parameters which gives us Heart rate, SpO₂ and estimated blood pressure. The results are shown in real time on the 1.8-inch TFT LCD display making the device very user friendly. Using ESP8266 Wi-fi capability the result can also be showed in Mobile. For portability, the entire system is powered by a 3.7V, 3000 mWh rechargeable Li-ion battery, which provided several hours of uninterrupted operation during testing. To ensure continuous usability, we integrated the TP4056 charging module, enabling safe and efficient recharging of the battery through a USB port. Through this our project successfully delivered heart rate, SpO₂, temperature, and estimated blood pressure in real time.

3.5.2 Prototype using PTT (Pulse transit time)

This is the block diagram of the device that uses PTT (Pulse Transit Time).

Block diagram Overview

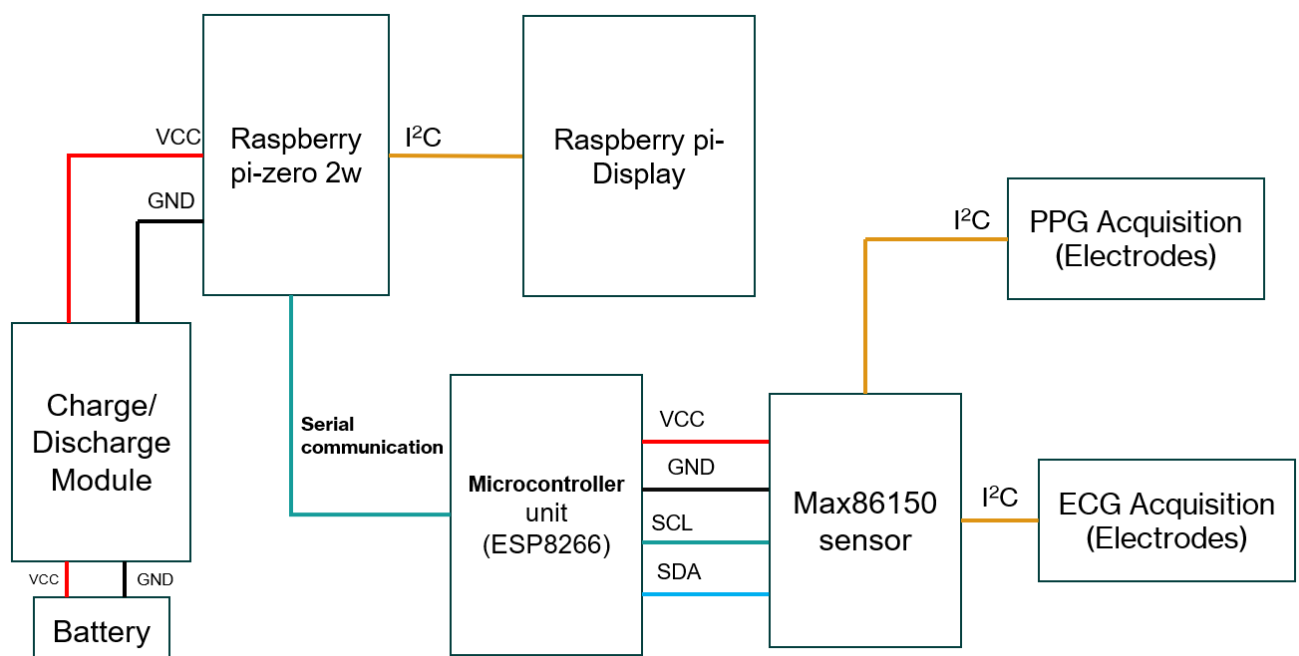


Fig 3.9: Block Diagram of Prototype using PTT

This block diagram shows the design of a cuffless blood pressure monitoring device that works based on Pulse Transit Time (PTT). It combines various hardware parts for continuous and non-invasive

blood pressure measurement. The system runs on a rechargeable battery that powers the whole setup through a charge/discharge module. This ensures a stable power source and safe operation. The central processing unit is the Raspberry Pi Zero 2W. It handles data processing and overall system management. It connects to a Raspberry Pi display using the I²C protocol to show real-time ECG, PPG signals, and estimated blood pressure readings. The Raspberry Pi also communicates with a microcontroller unit (ESP8266) through serial communication. This serves as a link between the sensors and the main processor. In summary, this system combines power management, sensing components, data collection, wireless communication, and a display unit into a compact assembly. It removes the need for inflatable cuffs, enabling continuous, real-time monitoring in a wearable and convenient format. This makes it especially suitable for current healthcare applications like remote patient monitoring, fitness tracking, and early detection of cardiovascular issues.

3.2.2.1. Flow Chart

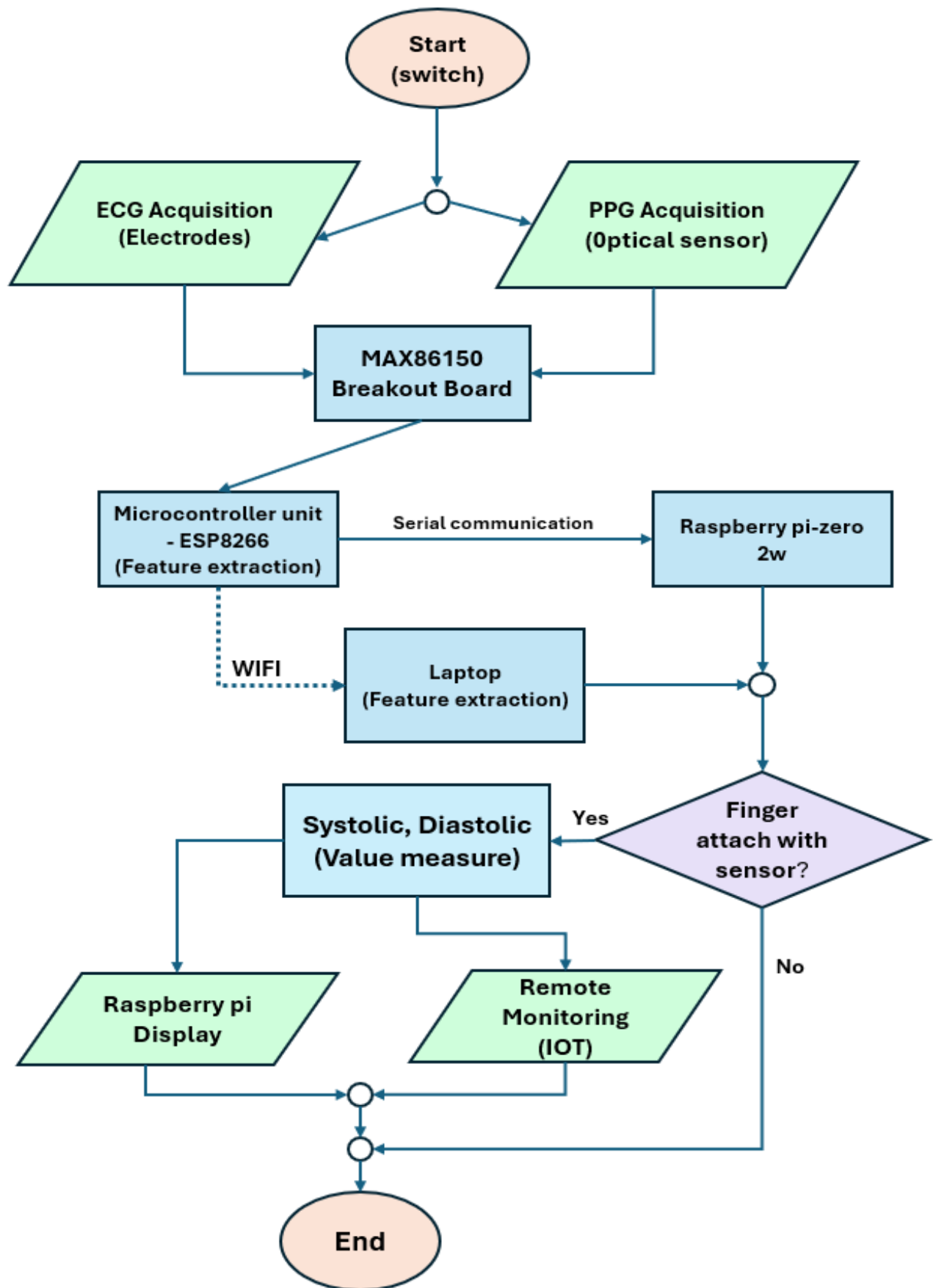


Fig 3.10: Flow Chart of Prototype using PTT.

Working Procedure

The procedure starts when the user turns on the device. At this point, both the ECG electrodes and the optical PPG sensor activate and connect to the MAX86150 breakout board. This board collects synchronized ECG and PPG signals. The raw signals then travel to the microcontroller unit (ESP8266), where initial feature extraction takes place. The microcontroller identifies key points in the waveforms, including the R-peak in the ECG signal and the baseline of the PPG waveform. It records their timestamps to calculate the time interval between them, called Pulse Transit Time (PTT). The processed data is sent through serial communication to the Raspberry Pi Zero 2W for further calculations and display. Before proceeding, the system checks that the user's finger is correctly placed on the sensor. If it is not, measurements are paused until there is proper contact.

Once valid signals are confirmed, the Raspberry Pi does calculations to find systolic and diastolic blood pressure from the PTT value using specific calibration equations. These equations create a direct link between PTT and blood pressure since PTT is inversely related to arterial stiffness and pressure. The calculated SBP and DBP values appear on the Raspberry Pi display for real-time monitoring. At the same time, the results can be sent through IoT-enabled remote monitoring for long-term observation and clinical use. This cycle continues as long as the finger stays in place, continuously and non-invasively updating the blood pressure readings. When the measurement is finished, the process ends.

Signal Processing

The raw physiological signals acquired from the MAX86150 sensor are often contaminated with noise, baseline shift and motion artifacts which often hinder us getting accurate data extraction. To address this problem the ECG signal is processed using a digital bandpass Butterworth filter with a passband of 5–15 Hz. The frequency range is carefully selected because of the QRS complex (The QRS complex is a term used in electrocardiograms (ECGs) that refers to the combined Q wave, R wave, and S wave) which contains the most diagnostically relevant information for heartbeat detection, lies within this band. The lower cutoff removes slow baseline wander. The upper cutoff eliminates high-frequency noise and muscle interference. The filtering uses a zero-phase forward and backward algorithm (filtfilt) to keep the true location of R-peaks without causing phase distortion. The PPG signal reflects changes in blood volume, and it is less affected by high-frequency noise. However, it is vulnerable to motion artifacts. Instead of pre-filtering, a prominence-based peak detection algorithm with distance constraints is used to identify systolic peaks and diastolic troughs reliably. This preprocessing makes sure that both ECG and PPG signals are clean and dependable for the next analysis.

Feature Detection (Peak Detection in ECG and PPG)

After filtering, feature detection is done to extract meaningful physiological markers. For the ECG signal, a Pan-Tompkins-inspired approach is used. The filtered ECG is differentiated, squared, and passed through a moving window integrator to emphasize rapid changes linked to ventricular depolarization. Candidate R-peaks are identified using an adaptive thresholding technique based on robust z-scoring. They are further refined by looking for the maximum within a short window around each candidate. A refractory period of 250 ms is enforced to avoid false double detections. For the PPG signal, systolic peaks (maximum blood volume points) and diastolic troughs (minimum blood volume points) are found using the (find_peaks) algorithm with limits on minimum peak prominence and inter-peak distance. These detected features are aligned with the ECG R-peaks to calculate the pulse transit time (PTT) and the timing interval t_1 . These values are then used in blood pressure estimation models.

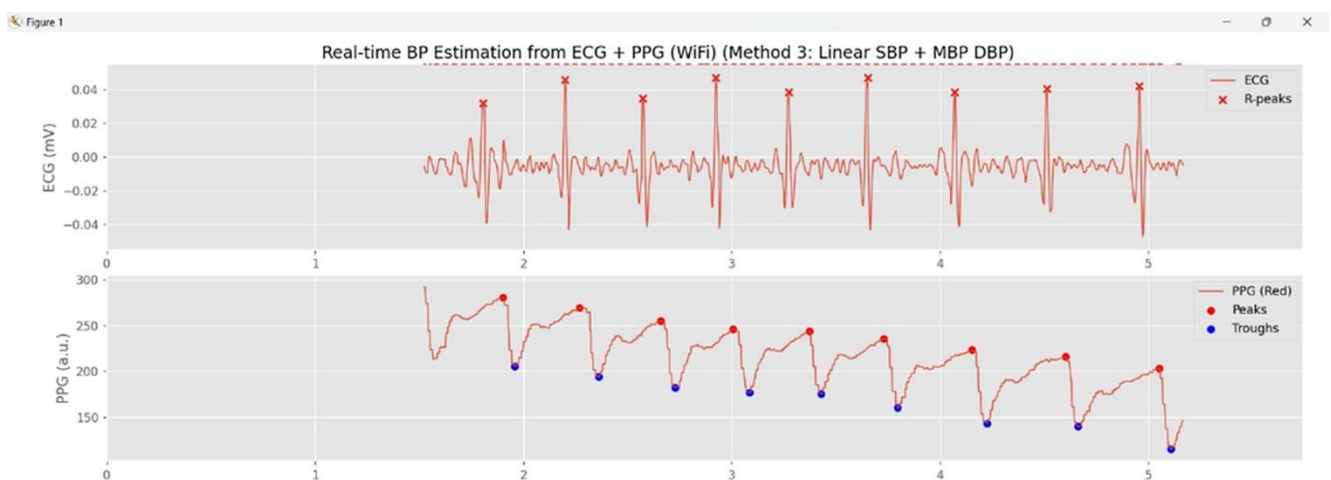


Fig 3.11: Signal Processing and Feature Detection

3.6. Modeling

Prototype 1 (BPT) Modeling

The 3D model shows the casing of the project. The shape has been built in a way that the inner and outer view can be seen and planned accordingly. There is no cut on the below side of the box. At the upper surface of the box there is a big cut for the display. There are also little cuts for temperature sensor led light and on/off switch. The rest of the components are inside of the box. As visibility is not needed no cut is not provided. The body dimensions are 20×12×5 cm the big cut for screen which is 46 mm × 46 mm. The sensor and battery module will be inside the box. The switch controls as a power switch for the whole device.



Fig 3.12: Inside and Upper side of the case of prototype 1

The choice of this type of architecture is to carry without any discomfort.

Prototype 2 (PTT) Modeling

A 3D model was designed to make the prototype. The prototype was carefully designed to store all the components inside so that it can be portable.

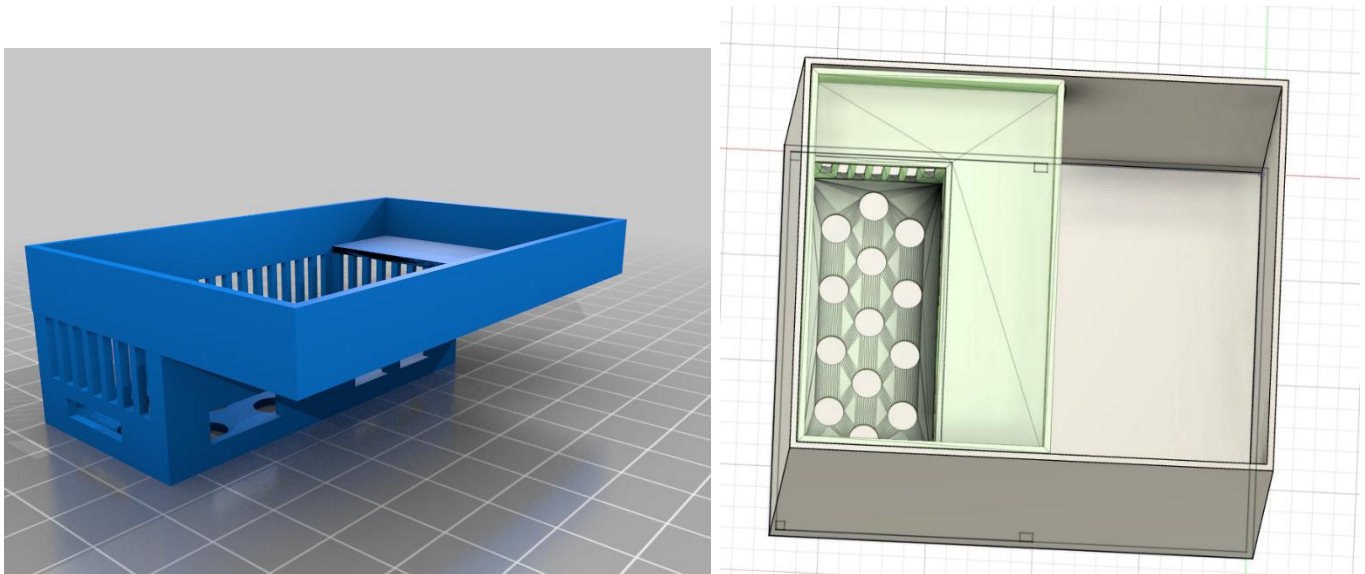


Fig 3.13: Raspberry pi holder and full box design



Fig 3.14: Designed box for prototype 2

3.7. Summary

This chapter explained the methods and models for two different prototypes of a cuffless blood pressure monitoring system. The first prototype used the Blood Pressure Trending (BPT) method. The hardware integrated a MAX30102 optical sensor and a MAX32664 biometric hub to capture physiological signals. An ESP8266 microcontroller processed these signals. The device was calibrated with a reference blood pressure monitor. The final calculated values for heart rate, SpO₂, and estimated blood pressure appeared on a TFT screen. The second prototype was built on the Pulse Transit Time (PTT) principle. It used a MAX86150 sensor to collect simultaneous ECG and PPG signals. A Raspberry Pi Zero processed these signals, using digital filtering and peak detection algorithms to calculate the PTT for blood pressure estimation. Both systems focused on portability and user comfort. They featured custom 3D-printed enclosures to neatly house all electronic components. The modeling and design process successfully created two functional, self-contained prototypes for non-invasive, continuous blood pressure monitoring.

Chapter 4

PROJECT IMPLEMENTATION

4.1. Introduction

This chapter presents the implementation aspect of the “Development of a Feasible Cost-effective Cuffless Blood Pressure Monitoring System Using Different Non-invasive Sensing Technologies Blood Pressure Trending, Pulse Transit Time” project, marking the transition from the theoretical framework and modeling in Chapter 3 to the actual hardware realization. It documents the structured execution of the project’s primary components, which encompassed circuit development, sensor integration, firmware programming, and enclosure fabrication.

For prototype 1 The hardware assembly involved embedding key components such as the ESP8266 microcontroller, TFT display module, rechargeable lithium-ion battery with charging circuitry, and biomedical sensors (SpO₂, heart rate, Blood pressure and temperature) into a custom-designed enclosure. All components were carefully soldered and organized the electrical interconnections on a prototyping board to ensure stability and signal accuracy, following the design specifications established earlier. On the software side, a code was developed for the ESP8266 firmware to acquire sensor data, process readings in real time, and display outputs on the TFT screen and configured Wi-Fi connectivity to enable data transmission over a local network, allowing for remote access and monitoring.

For prototype 2, the hardware integration involved assembling the MAX86150 breakout board for ECG and PPG signal acquisition, the ESP8266 microcontroller for initial feature extraction, and the Raspberry Pi Zero 2W for further processing and display. The system ran on a rechargeable lithium-ion battery connected through a charge/discharge module to ensure reliable and portable operation. All components were linked according to the block diagram. The MAX86150 sensor communicated via the I²C interface with the microcontroller and Raspberry Pi. The extracted features were sent through serial communication to Raspberry Pi, which calculated systolic and diastolic values. The processed results appeared on a Raspberry Pi-based display and were also sent through IoT functionality for remote monitoring.

On the software side, the ESP8266 was programmed to gather raw ECG and PPG signals from the MAX86150, perform initial filtering, and send the data. The Raspberry Pi ran algorithms for peak detection and feature extraction to calculate blood pressure trends. Additionally, Wi-Fi connectivity was set up to allow real-time data transfer to a laptop or cloud server, enabling local visualization and remote

access. This combined hardware and software setup ensured that the system followed the planned methodology shown in the flowchart and block diagram, allowing for effective physiological data collection, processing, and monitoring.

4.2. Required Tools and Components

Here both software tools and hardware components are discussed for both Prototype 1 and Prototype 2

Software Tools

ARDUINO IDE

The Arduino IDE served as the main programming environment for the ESP8266 microcontroller. It offered a straightforward and effective interface to write, compile, and upload firmware directly to the hardware. In this project, the Arduino IDE managed real-time signal acquisition from the sensors, filtered the raw data, and transmitted it for further processing. Its lightweight design, built-in libraries, and ease of integration with embedded systems made it a great choice for overseeing the microcontroller's operations and ensuring smooth interaction between hardware components.

Visual Studio (VS Studio)

Visual Studio was used to run Python code on the Raspberry Pi. It provided a flexible and strong software environment for managing and processing system-level data. With its Python integration, Visual Studio enabled the creation and execution of complex algorithms that improved the prototype's functionality beyond what the ESP8266 could handle on its own. In this project, it mainly managed communication tasks, data parsing, and other processing that supported the microcontroller's role. Its debugging features and modular design helped ensure that the software ran efficiently and reliably.

Microsoft Excel

Microsoft Excel served as the data analysis and visualization platform for this project. The measurements collected from the sensors were imported into Excel, where they were systematically organized for further analysis. Using its calculation tools, Excel was used to process raw data, generate mapping tables, and validate the accuracy of blood pressure estimations. Additionally, its charting features allowed the creation of plots and graphical representations that provided clear insights into system performance. In this project, Excel played a vital role in bridging raw hardware data with meaningful, interpretable results for evaluation and presentation.

Hardware components

Rechargeable Li-ion Battery (3.7V, 3000 mWh)

A rechargeable Li-ion battery at 3.7V and 3000 mAh is used as the main power source for the device. Lithium-ion batteries are known for their high energy density, lightweight, and long cycle life, which makes them ideal for portable health devices.



Fig 4.1: Lithium-Ion battery [35]

Table 4.1 Technical Specification 3.7 Volt 3000mWh Battery [35]

Specification	Value
Brand	GEEPAS
Model	ICR-18650-3000mAh
Nominal Voltage	3.7V
Capacity	3000mAh
Dimension	Diameter: 18mm, Length: 65mm
Cut-Off Voltage	3.0V
Cycle Life	≥1500 times
Net Weight	42g
Operating Temperature	Charging: 0-45°C, Discharging: -20-60°C
Shipment Weight	0.045 kg
Shipment Dimensions	12 x 5 x 5 cm

Charging Port and Charging Module (TP4056)

The charging system consists of a standard charging port and the TP4056 charging module. The TP4056 is a lithium-ion battery charger IC that manages safe charging and discharging cycles, preventing overcharging and ensuring battery longevity.

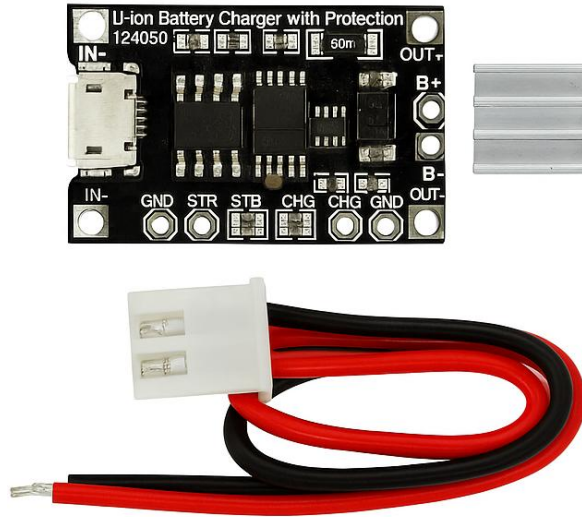


Fig 4.2: Charging Module [36]

Table 4.2 Technical Specification of TP4056 [36]

Specification	Value
Material	Fiber
Color	Black
Dimensions	28x17x4mm
Charging current	1A (maximum)
Maximum charging voltage	4.2V (1.5% accuracy)
Input voltage	4.5V-5.5V
Shipment Weight	0.008 kg

LED Indicator

The LED indicator provides a visual system status, such as power ON, charging state, or sensor operation. Though small, it plays an important role in user interaction by giving immediate feedback about whether the device is active or idle.

ON/OFF Switch

The ON/OFF switch is a very important component that allows the user to power the device on or off as needed. By cutting off the battery supply when not in use, it prevents unnecessary power drain and extends the operational life of the system.

Microcontroller Unit (ESP8266)

The ESP8266 microcontroller serves as the central processing unit of the system. It collects data from the sensors then processes the information and then displays the results on the LCD screen. The ESP8266 is also equipped with Wi-Fi capability, which means it can potentially transmit data to external devices or cloud servers for remote monitoring.

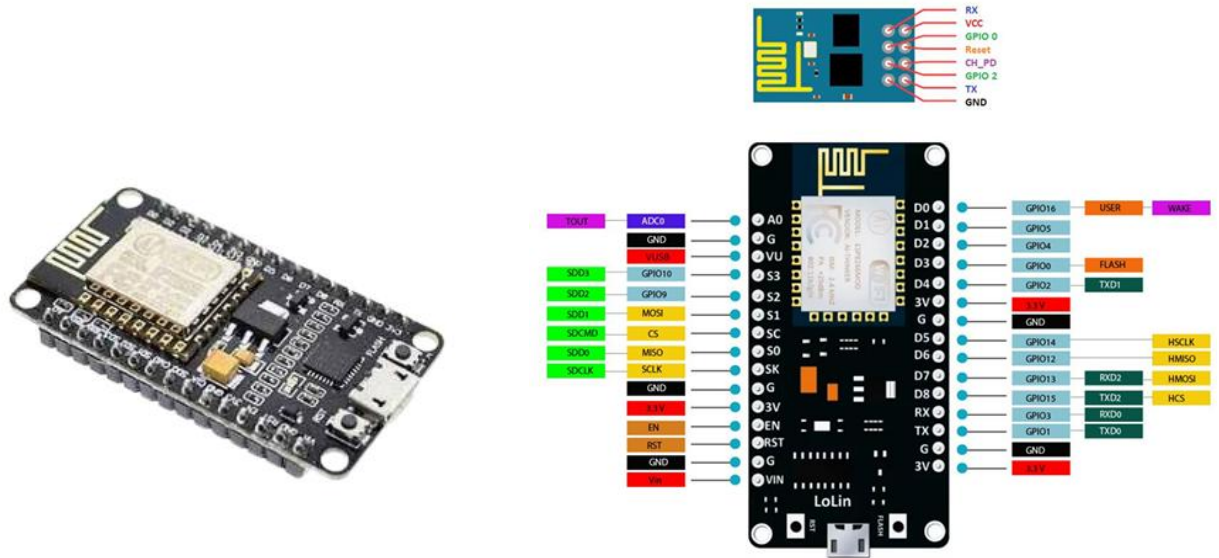


Fig :4.3 ESP 8266 Node MCU with Pin diagram [37]

Table 4.3 Technical Specification ESP 8266 Node MCU [37]

Content	Description
Drive Type	Dual high-power H-bridge
Power Input (V)	4.5 to 9
CI Voltage (V)	3.3
Flash size (MB)	4
Operating Temperature (C)	-40 to 125
Transfer Rate (Kbps)	110 to 46800
AD0	1 Channel ADC
Length (mm)	49
Width (mm)	24.5
Height (mm)	13
Shipment Weight	0.06 kg
Shipment Dimensions	6 X 4 X 2 cm

Calibration Port

The calibration port is included in the system for the purpose of calibration and debugging. It allows developers to fine-tune sensor performance and ensures accurate measurements by adjusting or updating system parameters and code.

Pulse Express Pulse-Ox & Heart Rate Sensor with MAX32664:

The Pulse Express module combines the MAX30102 optical sensor with the MAX32664 biometric sensor hub to give us the data on heart rate, blood oxygen saturation (SpO₂) and estimated blood pressure. The MAX30102 is a high-sensitivity pulse oximeter that uses integrated red and infrared LEDs along with a photodetector to measure heart rate and blood oxygen saturation (SpO₂). The MAX32664 acts as the embedded processor, running exclusive algorithms calculates heart rate, SpO₂ and estimate blood pressure. With an ultra-low power design and a simple I2C interface, the sensor can be easily connected to microcontrollers such as the ESP8266.



Fig 4.4 MAX32664 Sensor [28]

Skin Temperature Sensor (MAX30205)

The MAX30205 is a high-accuracy digital temperature sensor specifically designed for body temperature measurement, offering precision up to ± 0.1 °C. In this project, the sensor is mounted on an aluminum base to enhance heat conduction from the skin surface, ensuring stable and accurate readings when the user places their finger on it.



Fig 4.5: MAX30205 Sensor with aluminum hub [34]

1.8" 128×160 RGB TFT LCD Display

The 1.8-inch 128×160 RGB TFT LCD display is a compact screen capable of showing text and graphical data in full color. With its 128×160-pixel resolution, it provides a clear and user-friendly interface to present health parameters such as systolic and diastolic blood pressure, heart rate, oxygen saturation, and body temperature.



Fig 4.6: TFT LCD Display [38]

Table 4.4 Technical Specification TFT LCD Display [38]

Specification	Value
Driver IC	ST7735
Input Voltage (V)	3.3 to 5
Pixel Resolution	128 x 160
Interface Type	SPI
Length (mm)	45
Width (mm)	53
Height (mm)	10
Weight (gm)	16
Shipment Weight	0.02 kg
Shipment Dimensions	10 x 10 x 3 cm

Raspberry Pi Zero

Raspberry Pi Zero 2 W is the latest product in our most affordable range of single-board computers. The board incorporates a quad-core 64-bit Arm Cortex-A53 CPU, clocked at 1GHz. At its heart is a Raspberry Pi RP3A0 system-in-package (SiP), integrating a Broadcom BCM2710A1 die with 512MB of LPDDR2 SDRAM. The upgraded processor provides Raspberry Pi Zero 2 W with 40% more single-

threaded performance, and five times more multi-threaded performance, than the original single-core Raspberry Pi Zero.

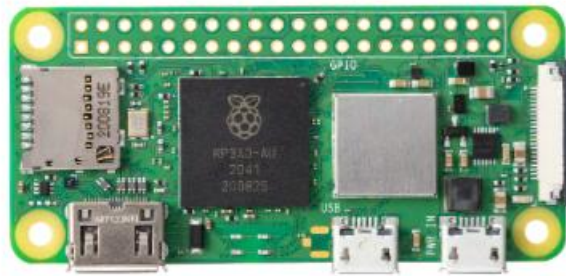


Fig 4.7: Raspberry Pi Zero [39]

Table 4.5 Technical Specification Raspberry Pi Zero [39]

Specification	Details
Form factor	65mm × 30mm
Processor	Broadcom BCM2710A1, quad-core 64-bit SoC (Arm Cortex-A53 @ 1GHz)
Connectivity	• 2.4GHz IEEE 802.11b/g/n wireless LAN, Bluetooth 4.2, BLE, onboard antenna • 1 x USB 2.0 interface with OTG • HAT-compatible 40-pin I/O header footprint microSD card slot • Mini HDMI port • CSI-2 camera connector
Connectivity Memory	512 MB
Video	HDMI interface and Composite video
Multimedia	H.264, MPEG-4 decode (1080p30) H.264 encode (1080p30) OpenGL ES 1.1, 2.0 graphics
Input power	5V DC 2.5A
Operating temperature	-20C to +70C
Production lifetime	Raspberry Pi Zero 2 W will remain in production until at least January 2030

Integrated Photoplethysmogram and Electrocardiogram Bio-Sensor Module for Mobile Health sensor (MAX86150 sensor)

The MAX86150 is an integrated electrocardiogram, pulse oximeter, heart rate monitor sensor module. It includes internal LEDs, photodetector, and low-noise electronics with ambient light rejection. The MAX86150 helps ease design-in to mobile and wearable devices. The MAX86150 operates on a 1.8V supply voltage with a separate power supply for the internal LEDs. Communication to and from the

module is entirely through standard I2C-compatible interface. The module can be shut down through software with near zero standby current, allowing the power rails to always remain powered [29].



Fig 4.8: MAX86150 sensor [29]

4.3. Implemented Models

This section describes the model and indicates what equipment is being used here.

4.3.1. Simulation Model

There is no simulation model for this Project

4.3.2. Hardware Model

4.3.2.1 Prototype 1 (Blood pressure trending)

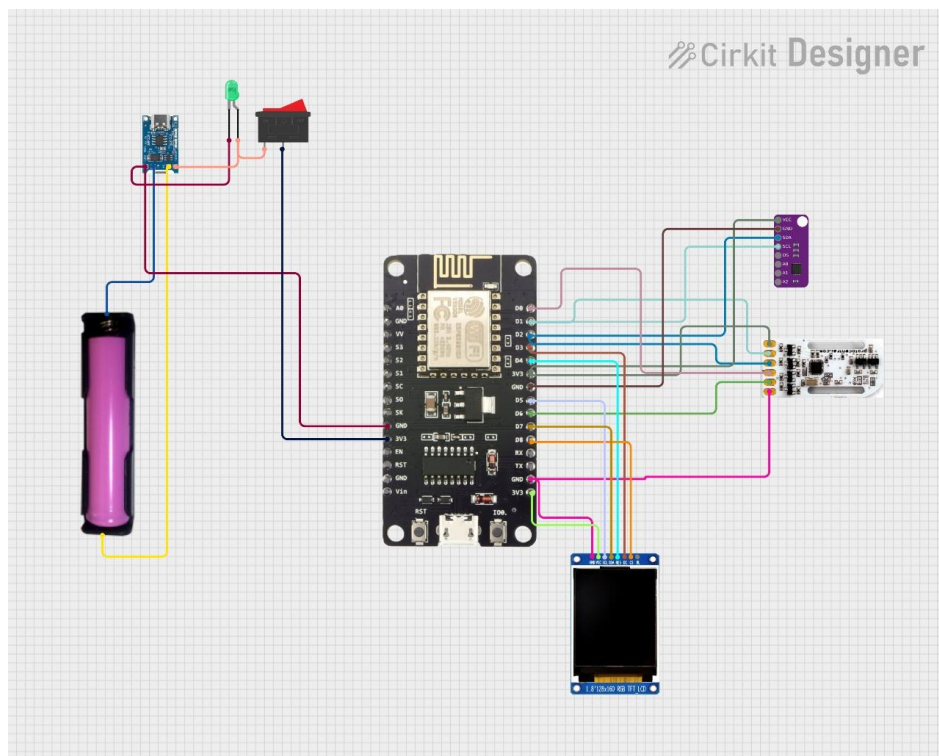


Fig 4.9: Circuit Diagram of the Prototype 1(Blood pressure trending)



Fig 4.10: Outer view of the device (Prototype 1)

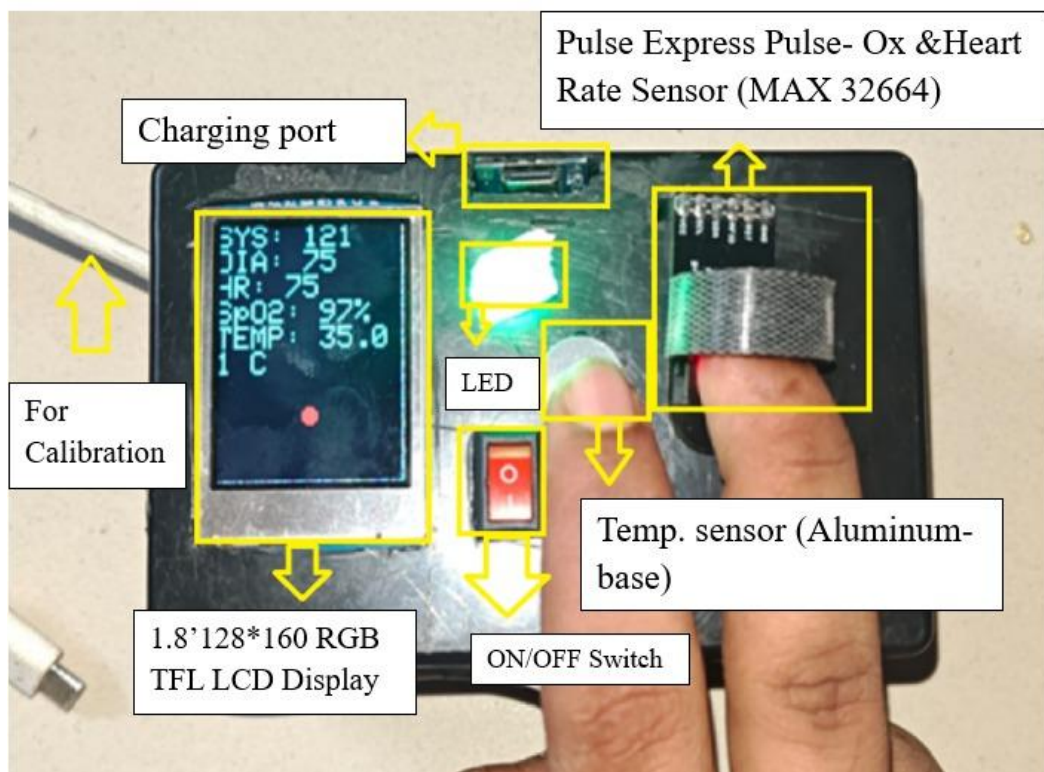


Fig 4.11: Outer view of the device with BP reading (Prototype 1)

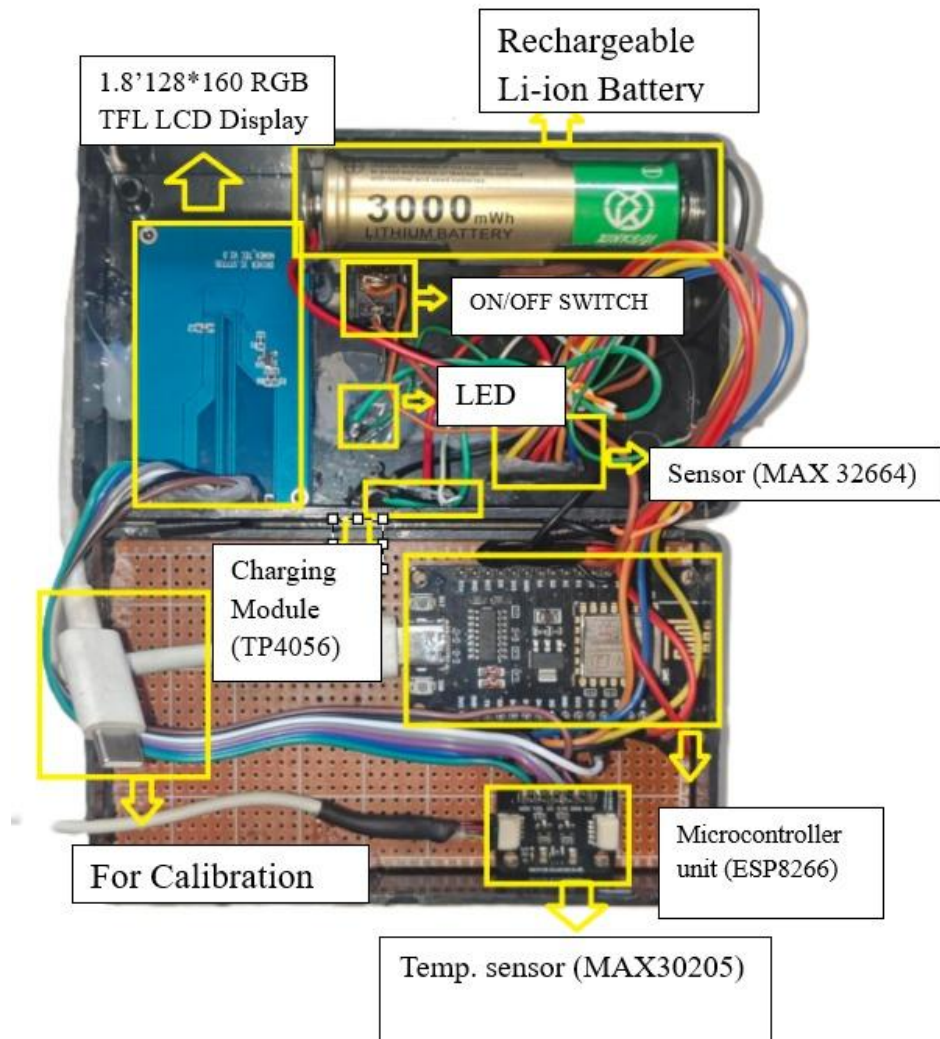


Fig 4.12: Inner view of the device (Prototype 1)

Firstly, a digitalized circuit diagram was drawn as an architecture for implementation into hardware model. The system was mainly powered by a 3.7V 3000mAh Li-ion battery, which formed the core of the hardware unit. It provided portable and stable power to the entire setup. To ensure safe and reliable recharging, the battery was integrated with a TP4056 charging module that included protection against overcharging and over-discharging. This design prevented battery damage and lowered the risk of overheating during extended use. An LED indicator was connected to the TP4056 module to show the charging and operational status of the device. This feature allowed users to easily check whether the system was powered on or charging. Additionally, an ON/OFF switch was installed along the battery connection to the microcontroller. This gave users the ability to manually turn the device on or off as needed. The combination of the power source, charging circuit, LED indicator, and switch was carefully connected to ensure both usability and safety, reducing the chances of short circuits or overcurrent events.

The main part of the project was the ESP8266 microcontroller. It received power from the Li-ion battery through the regulated output of the TP4056 circuit. ESP8266 acted as the brain of the system. It received,

processed, and transmitted health data. Two biomedical sensors connected through the I²C interface included the Pulse Express Pulse-Ox & Heart Rate Sensor with MAX32664 and the Skin Temperature Sensor (MAX30205). These sensors were placed to gather real-time physiological data when a user put their finger on the designated contact area. To avoid inaccuracies, the wiring was kept short and secure, which helped reduce noise in the sensor readings. ESP8266 processed the collected data and sent it to a TFT LCD display connected via SPI communication. This allowed the results to be shown in real time. Users could conveniently monitor their heart rate, SpO₂, estimated blood pressure, and temperature without relying on any external devices. The whole setup was designed carefully, paying attention to power distribution, grounding, and insulation to ensure both user safety and reliable device performance.

4.3.2.1 Prototype 2 (Pulse transit Time)



Fig 4.13: Outer view of the device with BP reading (Prototype 2)

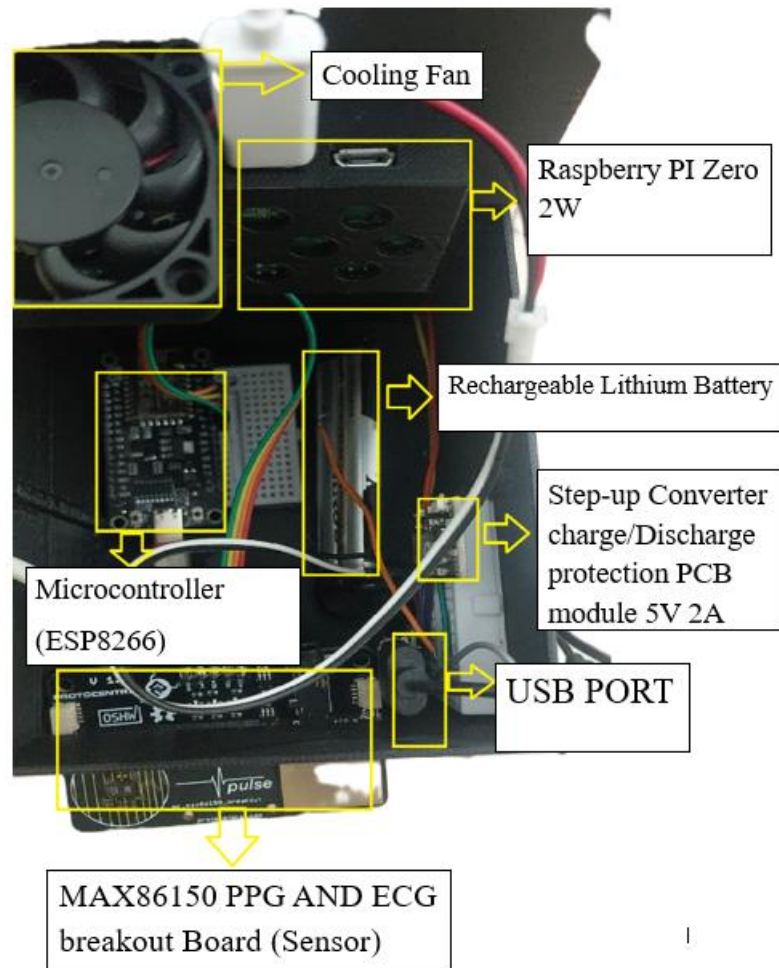


Fig 4.14: Inner view of the device (Prototype 2)

The device was designed and built using the system layout shown in the block diagram. Its main goal is to acquire ECG and PPG signals in a portable and effective way. The heart of the system is the MAX86150 sensor, which captures both ECG and PPG signals through specific electrodes. The ECG electrodes connect to the sensor to record the heart's electrical activity, while the PPG electrodes measure blood volume changes using optical signals. Combining both types of signal acquisition in one sensor module reduces complexity and improves integration. Data processing and management happens on the Raspberry Pi Zero 2W, the main processing unit. This module connects to a display using the I²C protocol, enabling real-time visualization of the acquired signals. A microcontroller unit, the ESP8266, is also included to help with communication and preprocessing tasks. It ensures smooth data transmission from the sensor to the Raspberry Pi, while also filtering out noise and artifacts from the raw biomedical signals. The entire setup runs on a rechargeable battery linked through a charge/discharge module. This module supplies stable power to all components and manages the charging process, ensuring nonstop functionality during continuous monitoring. Using I²C communication greatly simplifies the wiring between modules and reduces the overall hardware size, making the design compact and reliable. With this configuration, the device can continuously acquire

accurate ECG and PPG signals, process the data, and display it in real time. The system shows how modern microcontrollers and sensors can work together to create biomedical devices that are efficient, portable, and suitable for research or clinical use. This design highlights the importance of low power use, stable communication, and effective signal acquisition, offering a practical example of translating biomedical engineering concepts into working prototypes for real-world health monitoring solutions.

4.4. Engineering Solution in accordance with professional practices

The proposed solution was developed by following established engineering practices. This approach ensures that the system is reliable and safe for real-world use. The integration of a Li-ion battery with the TP4056 charging module complied with electrical safety guidelines. This included protection against overcharging, over-discharging, and short circuits—essential measures for battery-powered biomedical systems. The use of an ESP8266 microcontroller provided modularity, scalability, and adherence to common communication protocols, which makes the design open to future improvements. Sensor connections were made according to I²C/SPI industry standards, ensuring accuracy, low latency, and minimal signal interference.

From a biomedical engineering perspective, precautions were taken to avoid direct high-voltage exposure to the user by using only medically safe, low-power sensors (MAX32664), aligning with the principle of patient safety and risk minimization. Furthermore, the hardware layout considered proper grounding, insulation, and component placement, which are critical professional practices in embedded system design. The data display on a TFT LCD was organized in a user-friendly format, reflecting human-centered engineering design principles.

Overall, the solution demonstrates alignment with professional codes of practice such as safety, reliability, ethical responsibility, and usability, ensuring that the system is not only technically functional but also ethically and socially responsible in a healthcare-related application.

4.5. Summary

This chapter described how we implemented two cuffless blood pressure monitoring prototypes. We successfully assembled the hardware for the first prototype, which is based on Blood Pressure Trending (BPT). This involved integrating an ESP8266 microcontroller, a MAX32664 biometric sensor, a MAX30205 temperature sensor, and a TFT display into a custom enclosure. We used a rechargeable Li-ion battery along with a TP4056 charging module to provide portable power. The system was programmed with the Arduino IDE to gather sensor data, process it, and show the results. The second prototype, which uses Pulse Transit Time (PTT), was built around a Raspberry Pi Zero and a MAX86150 sensor for collecting ECG and PPG signals. We focused on strong signal processing algorithms that

filter out noise and accurately detect peaks for PTT calculation. Both systems were built using professional engineering practices. We prioritized electrical safety, signal quality, and user-centered design to create functional, reliable, and cost-effective devices that are now ready for testing and evaluation.

Chapter 5

RESULTS ANALYSIS & CRITICAL DESIGN REVIEW

5.1. Introduction

The chapter represents the outcome of the hardware-based tests that were conducted on prototype 1 and prototype 2. The results for prototype 1 are Systolic Blood pressure, Diastolic blood pressure, mean arterial pressure, skin temperature, heart rate and oxygen saturation (SpO₂) and for prototype 2 the result is Systolic blood pressure, Diastolic blood pressure, Mean arterial pressure and Heart rate. The objective of this chapter is to verify whether these results are validated and acceptable. The parameters were observed carefully. For validating data on Systolic and Diastolic blood pressure both cuffless Data (from the prototype) and cuff data (from the omron BP Device) were taken at the same time. Both data were taken in EXCEL to calculate MAPE (Mean Absolute Percentage Error). The MAPE gives us the accuracy of the blood pressure contrast of prototype and Analog blood pressure device). Some extra parameters like heart rate, skin temperature and Oxygen saturation (SpO₂) were also integrated in device so that they can use more multipurpose way and just not only for blood pressure.

5.2. Results Analysis

This section critically analyzed the obtained results and provide a detailed explanation of irregularities.

5.2.1. Simulated Results

There is no simulation result for this Project.

5.2.2. Hardware Results

5.2.2.1 Prototype 1 (Blood pressure trending)

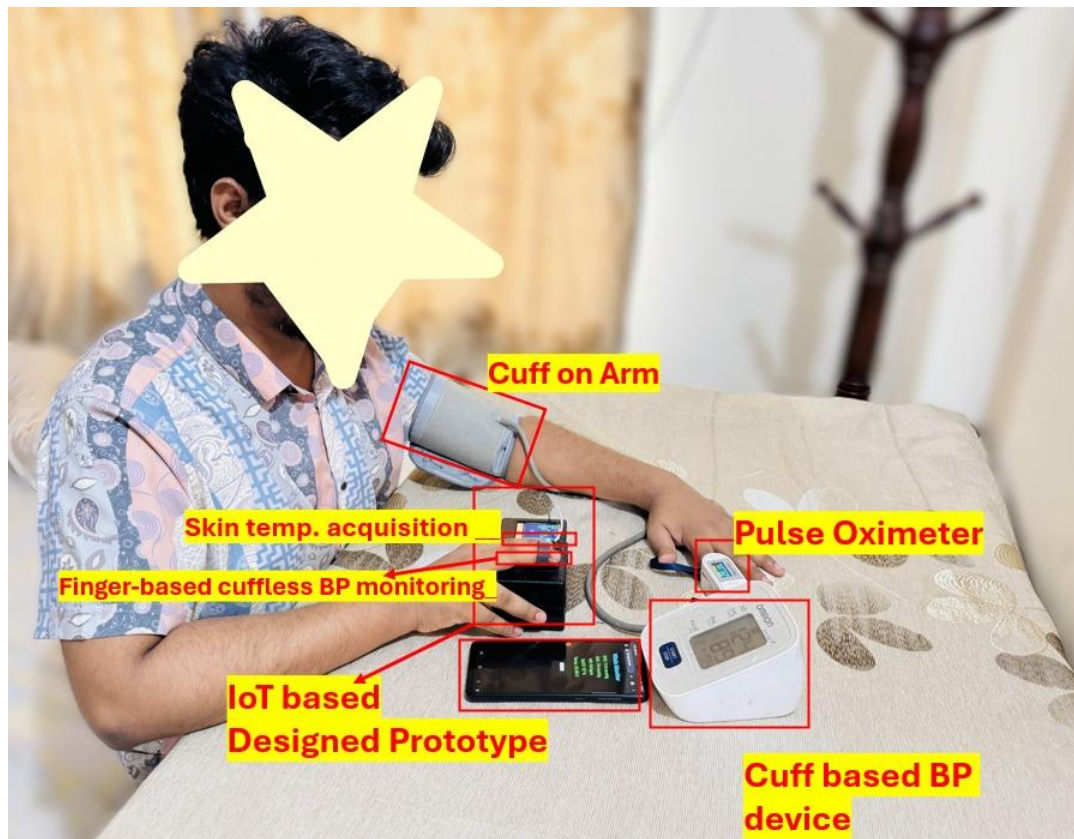


Fig 5.1: Data taken for prototype 1

The test involved 10 volunteers in the 20-24 age range equally divided between 5 male and 5 female students as well as another 10 volunteers aged 48 to 55 age range including 5 males and 5 females. None of the participants reported any notable respiratory illnesses.

Table 5.1: Data table for 20-24 Age using BPT prototype.

Subject	Data Taken	Gender	Age	Weight (KG)	Height (CM)	BMI	SYS Cuffless	DIA Cuffless	DIA Cuff	MAP Cuffless	MAP Cuff	MAP Cuffless	MAPE SYS	MAPE DIA	MAPE SYS OVERALL	MAPE DIA OVERALL	MAPE MAP OVERALL	HR Cuffless	HR Cuff	HR Cuffless	HR MAPE OVERALL	SPO2 Cuffless	SPO2 Cuff	SPO2 MAPE OVERALL
Person 1	1	F	22	68	167.64	24.1966	104	107	71	66	82	79.67	5.89	4.4	3.47	5.01	2.07	86	89	3.6	2.48	98	99	0.83
	2						118	110	65	67	82.67	81.33						84	87			99	99	
	3						100	108	65	63	76.67	78						80	83			97	99	
Person 2	1	F	21	50	154.94	20.8278	103	98	71	63	81.67	74.67	2.92	6	3.47	5.01	4.43	80	82	2.12	2.48	99	100	1.01
	2						101	102	68	66	79	78						78	80			99	100	
	3						103	100	79	76	87	84						77	78			99	100	
Person 3	1	F	21	80	170.18	27.6231	107	106	74	72	85	83.33	2.36	3.92	3.47	5.01	2.49	90	87	2.64	2.48	98.9	97.6	0.95
	2						97	102	73	72	81	82						88	85			98.5	97.9	
	3						101	102	78	72	85.67	82						85	84			97.5	96.6	
Person 4	1	F	21	50	162.56	18.9209	110	108	77	78	88	88	3.25	4.07	3.47	5.01	3.21	87	90	2.68	2.48	98	97.6	0.99
	2						99	104	74	76	82.33	85.33						86	88			98	97.1	
	3						104	107	73	79	83.33	88.33						88	86			97	98.6	
Person 5	1	F	22	47	160.02	18.3548	123	124	85	82	97.67	96	2.64	4.11	3.47	5.01	3.27	78	75	2.99	2.48	99	100	0.54
	2						115	120	80	84	91.67	96						76	77			99.6	100	
	3						109	112	79	82	89	92						79	76			99.8	100	
Person 6	1	M	21	78	177.8	24.6735	107	113	72	73	83.67	86.33	5.22	4.26	3.47	5.01	1.94	80	81	1.66	2.48	97.6	96.6	0.89
	2						109	114	79	74	89	87.33						83	82			98	98.6	
	3						110	116	79	75	89.33	88.67						79	77			98	99	
Person 7	1	M	20	70	170.18	24.1702	139	135	85	80	103	98.33	2.71	5.17	3.47	5.01	4.07	80	78	2.87	2.48	98.5	97.3	0.75
	2						134	130	82	78	99.33	95.33						79	77			98	98	
	3						132	129	84	80	100	96.33						84	87			98	97	
Person 8	1	M	24	50	167.64	17.7916	106	109	70	75	82	86.33	4.97	8.22	3.47	5.01	6.81	80	82	2.09	2.48	98	96.7	0.68
	2						107	113	69	75	81.67	87.67						79	81			98	98	
	3						108	115	68	74	81.33	87.67						81	82			98	97.3	
Person 9	1	M	24	88	172.72	29.4983	123	125	70	69	87.67	87.67	1.64	2.69	3.47	5.01	0.97	90	88	1.84	2.48	96	96.7	1.03
	2						126	125	75	72	92	89.67						91	89			99	98	
	3						120	123	76	74	90.67	90.33						92	93			96	97.3	
Person 10	1	M	24	84	172.72	28.1575	130	126	80	72	96.67	90	3.12	7.28	3.47	5.01	5.4	85	81	2.35	2.48	99	98.2	0.4
	2						126	122	75	71	92	88						85	84			99	98.7	
	3						129	125	77	72	94.33	89.67						86	85			99	99.1	
													Accuracy		96.53	94.99				Accuracy	97.52		Accuracy	99.17

For the Table 5.1 and Table 5.2 results, the all-over errors were calculated using Mean Absolute Percentage Error (1).

$$MAPE = \frac{1}{n} \sum_{t=1}^n \left| \frac{Cuffless - Cuff}{Cuff} \right| * 100 \quad (5.1)$$

The cuffless value is the value that we get from the MAX32664 sensor and the cuff value we get from hardware machine and n is the total number of observations.

Table 1 consists of 10 volunteers (young age range 21-25) with data of Blood pressure (Systolic blood pressure and Diastolic blood pressure), Heart Rate and Oxygen saturation and MAP (Mean arterial pressure). The formula for MAP is

$$MAP = DBP + \frac{SBP - DBP}{3} \quad (5.2)$$

MAP = Mean Arterial Pressure, DBP = Diastolic Blood Pressure, SBP = Systolic Blood Pressure

The MAPE percentage of each individual participant and for all the participants are also presented in the table. The MAPE% for systolic Blood Pressure is 3.47%, for diastolic blood pressure is 5.01%, for mean arterial pressure is 3.47%, for heart rate is 2.19% and for oxygen saturation is 0.83%. So, the accuracy for Systolic blood pressure is 96.53%, accuracy for diastolic blood pressure is 94.99%, accuracy for MAP is 96.53%, accuracy for heart rate is 97.52% and accuracy for oxygen saturation is 99.17%. Table 1 data is also plotted in fig 5.2 to fig 5.5.

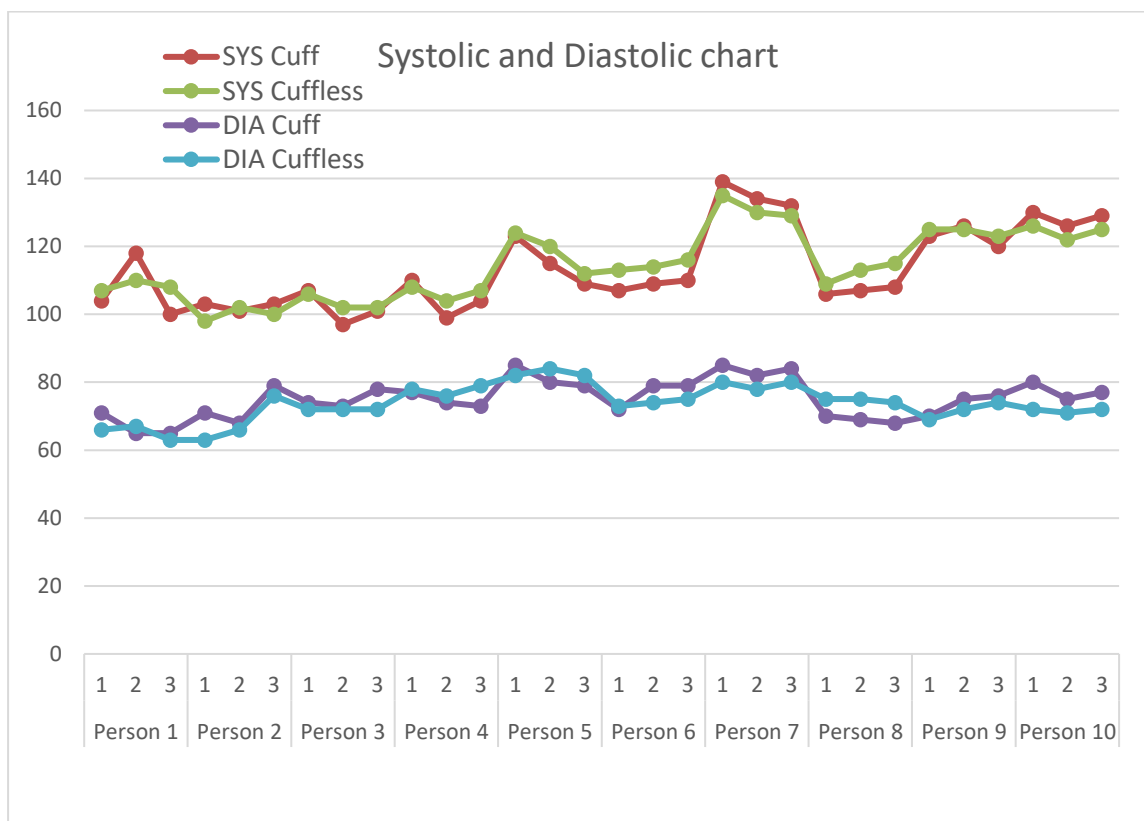


Fig 5.2: Systolic and Diastolic data for cuff and cuffless (Age 20-24)

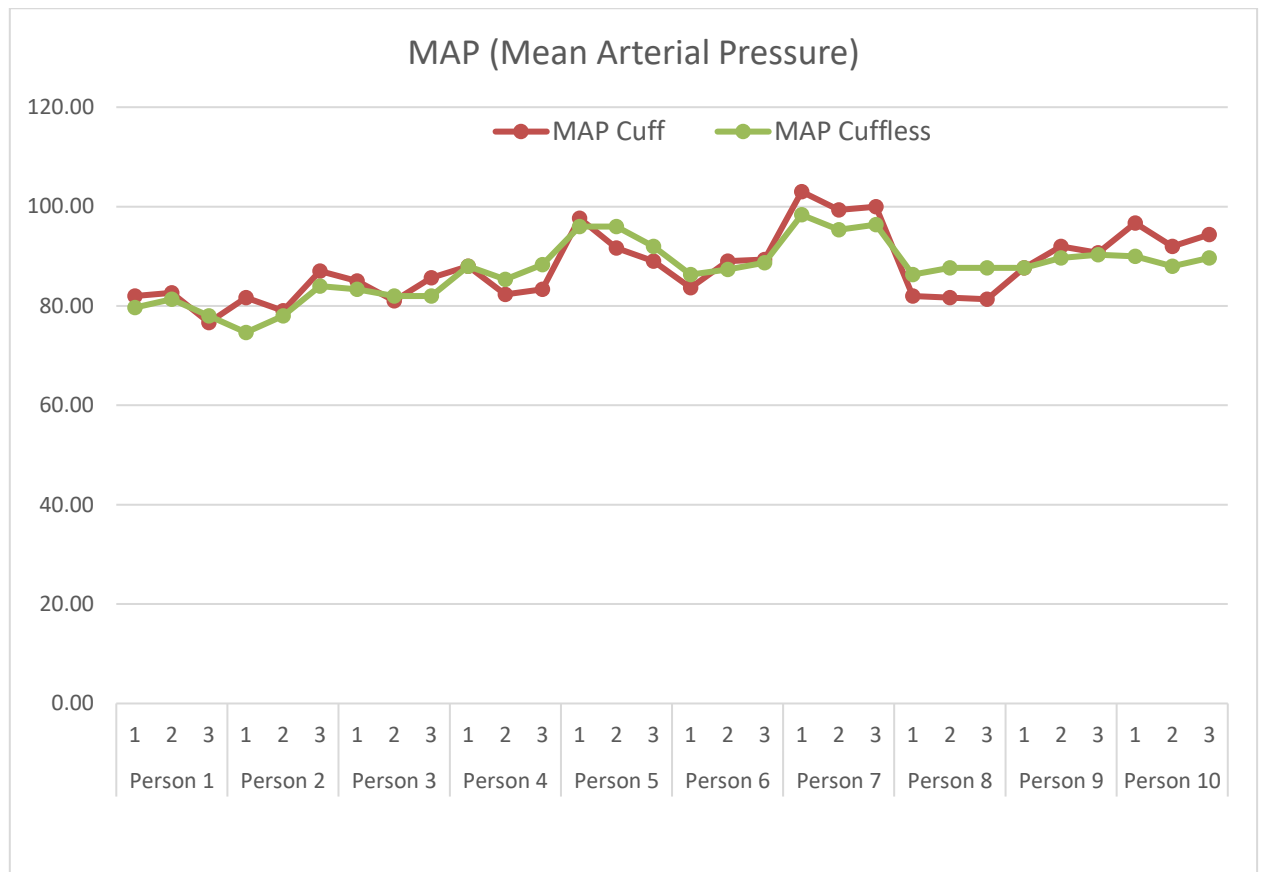


Fig 5.3: Mean Arterial Pressure (MAP) data for cuff and cuffless (Age 20-24)

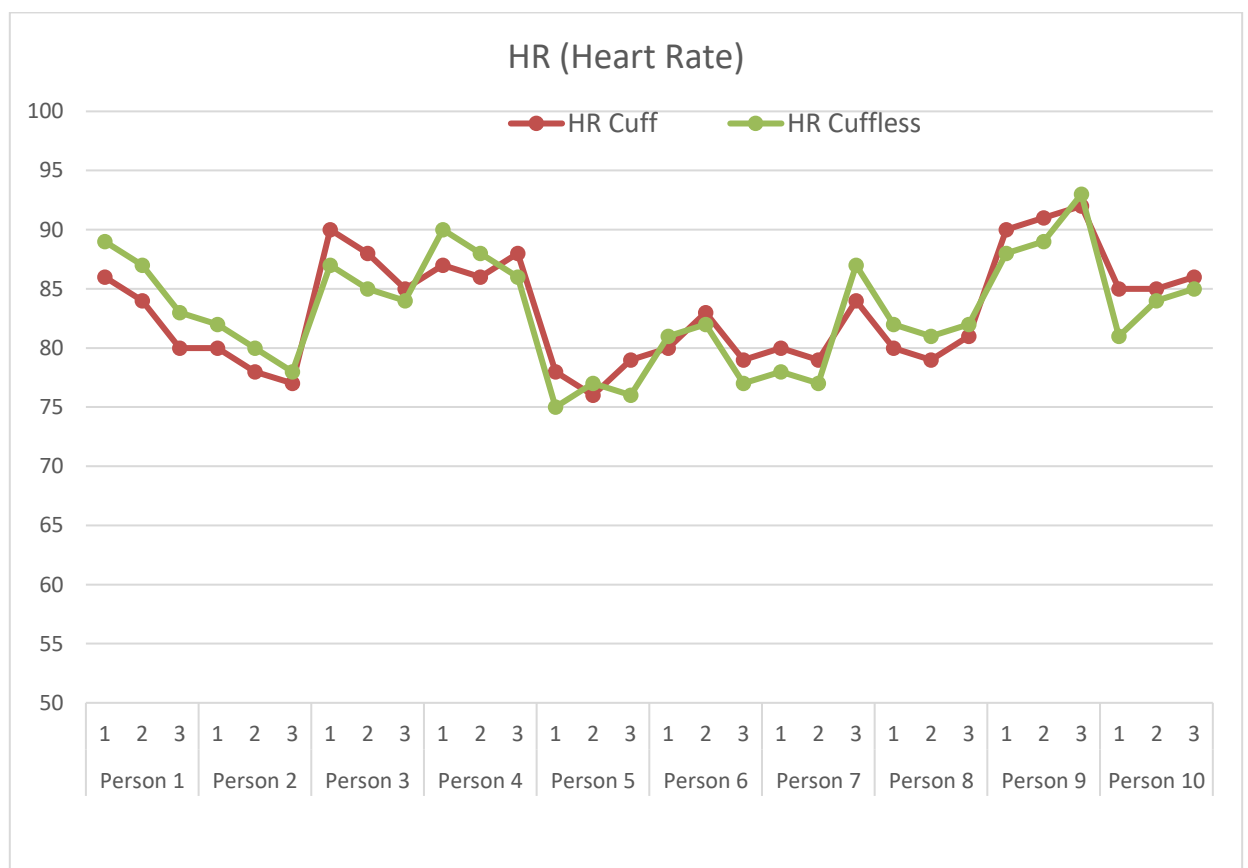


Fig 5.4: Heart rate data for cuff and cuffless (Age 20-24)

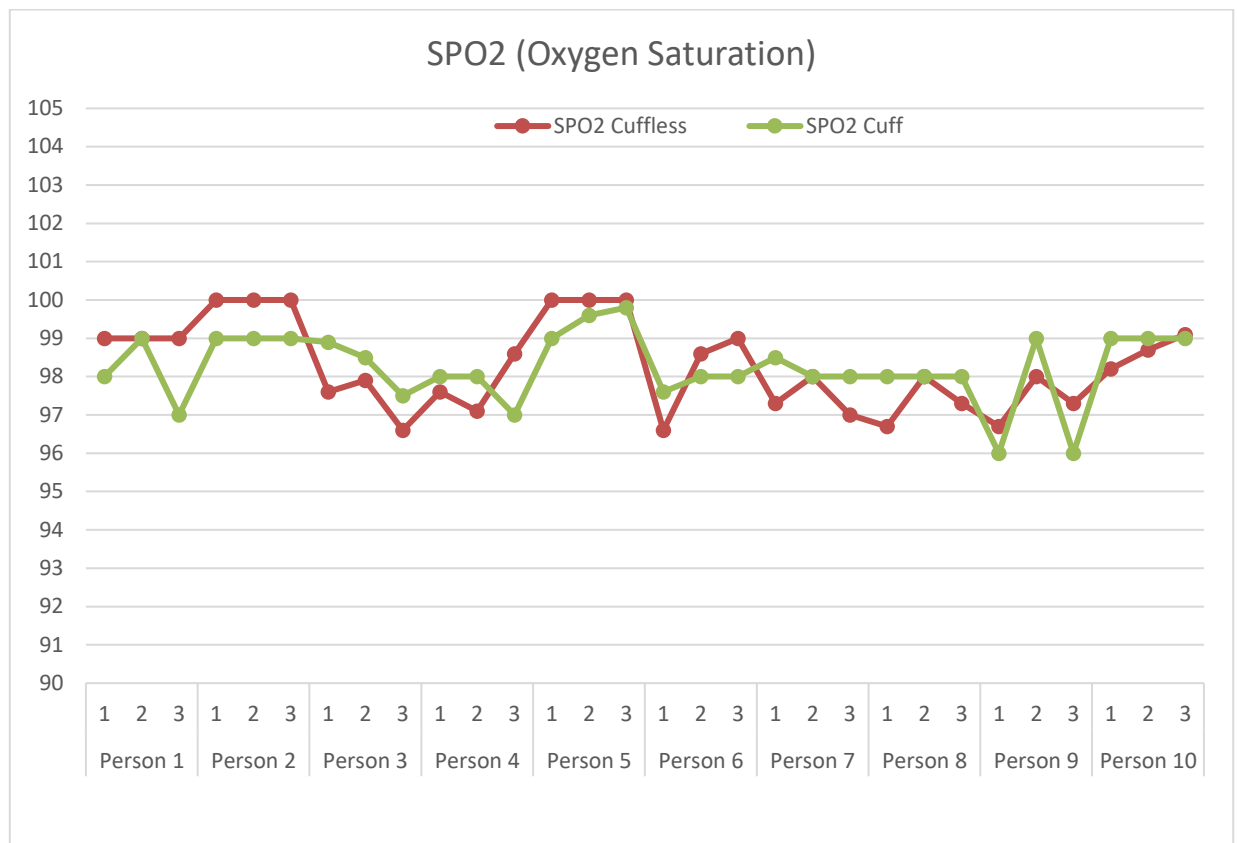


Fig 5.5: Oxygen Saturation (SpO2) data for cuff and cuffless (Age 20-24)

Table 5.2 consists of 10 volunteers (elderly age range 45-65) with data of Blood pressure (Systolic blood pressure and Diastolic blood pressure), Heart Rate and Oxygen saturation and MAP (Mean arterial pressure). The MAPE percentage of each individual participant and for all the participants are also presented on the table. The MAPE% for systolic Blood Pressure is 4.71%, for diastolic blood pressure is 5.09%, for mean arterial pressure is 3.99%, for heart rate is 2.19% and for oxygen saturation is 0.7%. So, the accuracy for Systolic blood pressure is 95.29%, accuracy for diastolic blood pressure is 94.91%, accuracy for MAP is 96.01%, accuracy for heart rate is 97.81% and accuracy for oxygen saturation is 99.30%. Table 5.2 data is also plotted in fig 5.6 to fig 5.9.

Table 5.2: Data table for 48-55 Age using BPT prototype.

Subject	Data-Taken	Gender	Age	WEIGHT (KG)	Height (CM)	BMI	SYS Cuff	SYS Cuffless	DIA Cuff	DIA Cuffless	MAP Cuff	MAP Cuffless	MAPE SYS	MAPE DIA	MAPE SYS OVERALL	MAPE DIA OVERALL	MAPE MAP	MAPE MAP OVERALL	HR Cuff	HR Cuffless	HR MAPE	HR MAPE OVERALL	SPO2 Cuff	SPO2 Cuffless	SPO2 MAPE	SPO2 MAPE OVERALL
Person 1	1	M	53	75	168	26.57	115	108	80	78	91.67	88	6.28	3.26			4.52		84	85			98	99		
	2						118	110	82	79	94	89.33							86	88	2.42		99	99	0.68	
	3						117	110	83	80	94.33	90							80	83			98	99		
Person 2	1	F	48	73	155	30.39	120	112	75	74	90	86.67	5.73	3.01			4.22		79	77	2.59		99	100	1.01	
	2						122	115	77	74	92	87.67							80	82			99	100		
	3						125	119	79	76	94.33	90.33							73	75			99	100		
Person 3	1	M	52	75	172	25.35	124	118	81	84	95.33	95.33	5.04	4.47			3.32		90	90	0		99	98	1.14	
	2						126	119	82	78	96.67	91.67							91	91			99	98		
	3						127	121	83	79	97.67	93							91	91			99	97.6		
Person 4	1	F	47	95	162	36.2	130	124	77	78	94.67	93.33					2.31		94	94			98	98		
	2						131	125	79	78	96.33	93.67	5.09	0.85					94	95	0.71		98	97	0.68	
	3						132	124	79	79	96.67	94							93	94			98	99		
Person 5	1	F	45	47	152.4	20.24	115	110	75	72	88.33	84.67	4.16	4.32			2.01		80	84			98.6	98.2	0.48	
	2						111	116	77	74	88.33	88							78	81	4.91		97.4	96.6		
	3						110	106	79	83	89.33	90.67							85	80			98.1	97.9		0.7
Person 6	1	M	56	84	176.78	26.88	131	125	82	76	98.33	92.33					6.45		84	82			97.5	96.2	0.8	
	2						130	122	81	77	97.33	92	5.89	6.9					85	87	2.84		95	96		
	3						130	121	83	76	98.67	91							79	82			99	99		
Person 7	1	M	48	70	170.18	24.17	133	129	85	79	101	95.67					5.25		73	74			98.6	97.8	0.61	
	2						132	128	82	78	98.67	94.67	3.29	6.79					79	77	2.97		98	98		
	3						130	125	83	76	98.67	92.33							80	84			98	97		
Person 8	1	F	49	68	167.64	24.2	106	113	70	74	82	87					5.36		80	82	2.09		98	97.3	0.48	
	2						112	107	69	75	83.33	85.67	4.85	8.23					79	81			98	98		
	3						115	119	68	75	83.67	89.67							81	82			98	97.3		
Person 9	1	M	54	72	172.72	24.13	137	132	90	85	105.67	100.67	3.01	7.4			4.04		90	88			96	96.7	0.75	
	2						131	132	91	83	104.33	99.33							91	89	1.84		99	98		
	3						130	136	89	82	102.67	100							92	93			96	96.5		
Person 10	1	F	55	61	157.48	24.6	105	101	68	72	80.33	81.67	3.8	5.66			2.44		85	83	1.56		99	98.2	0.4	
	2						107	102	72	76	83.67	84.67							85	84			99	98.7		
	3						103	106	72	76	82.33	86							86	85			99	99.1		
															95.29	94.91		96.01			Accuracy	97.81			Accuracy	99.3

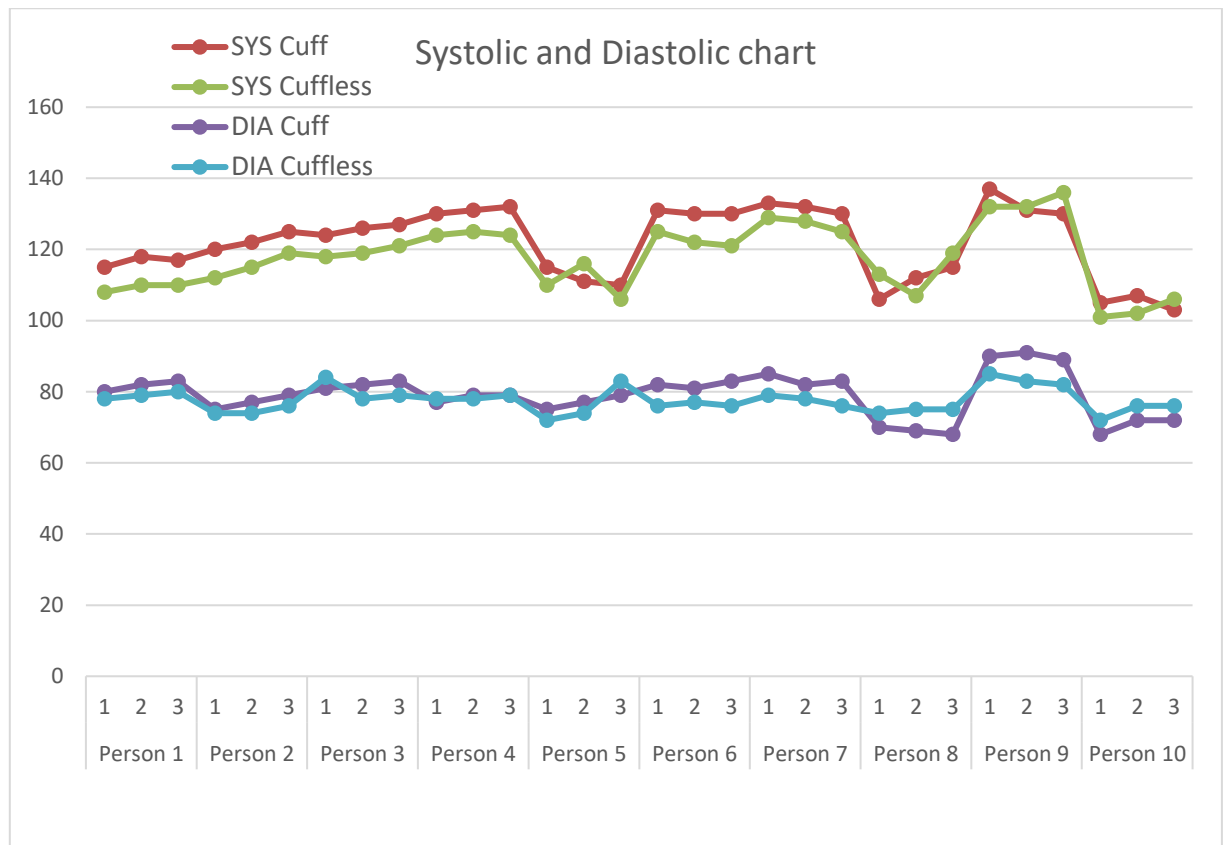


Fig 5.6: Systolic and Diastolic data for cuff and cuffless (Age 44-55)

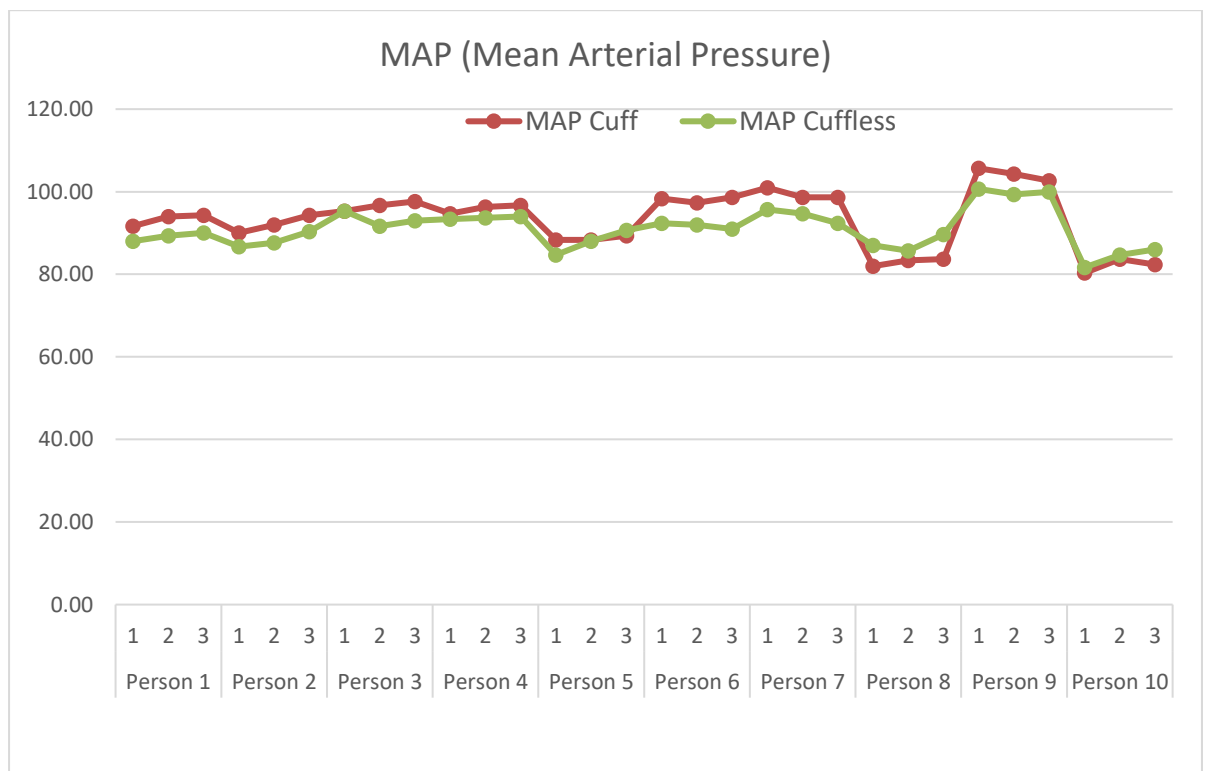


Fig 5.7: Mean Arterial Pressure (MAP) data for cuff and cuffless (Age 44-55)

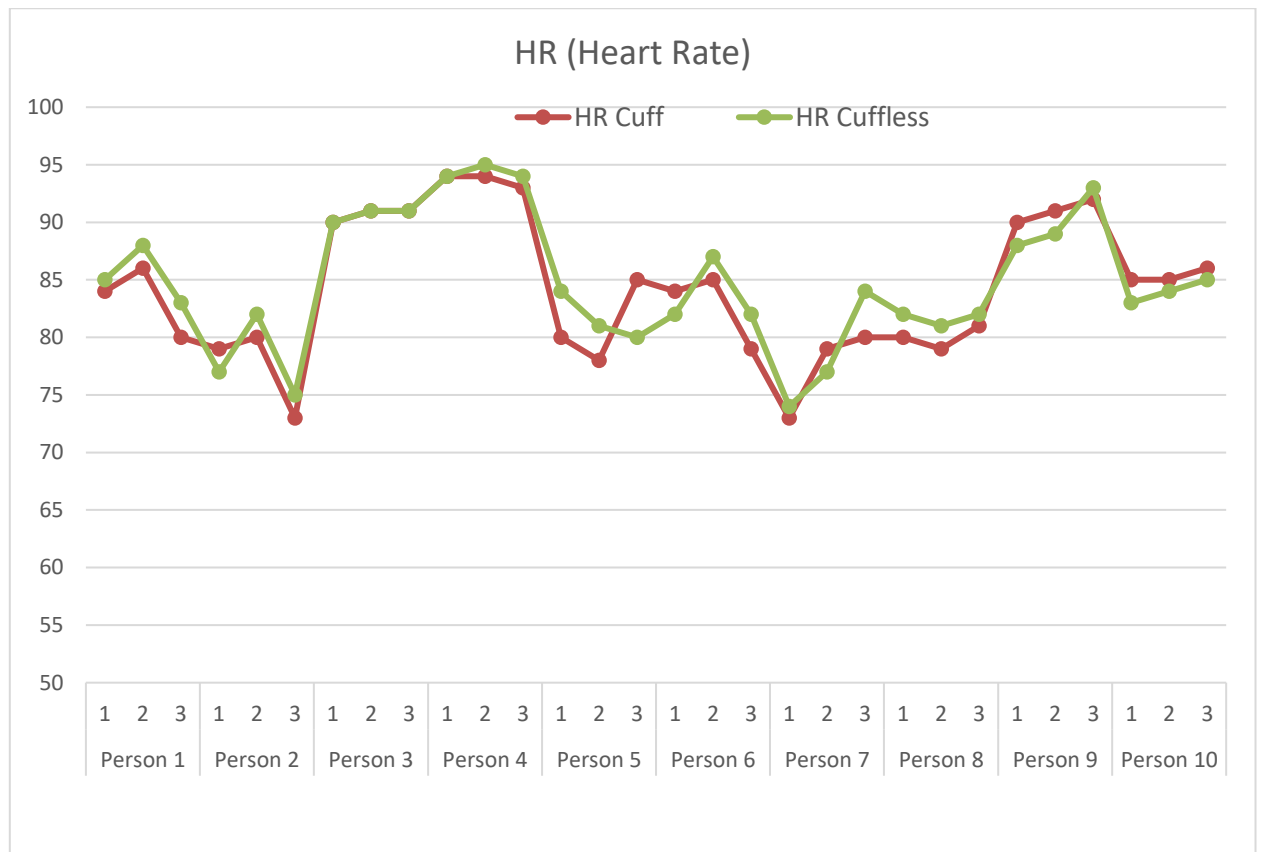


Fig 5.8: Heart rate data for cuff and cuffless (Age 44-55)

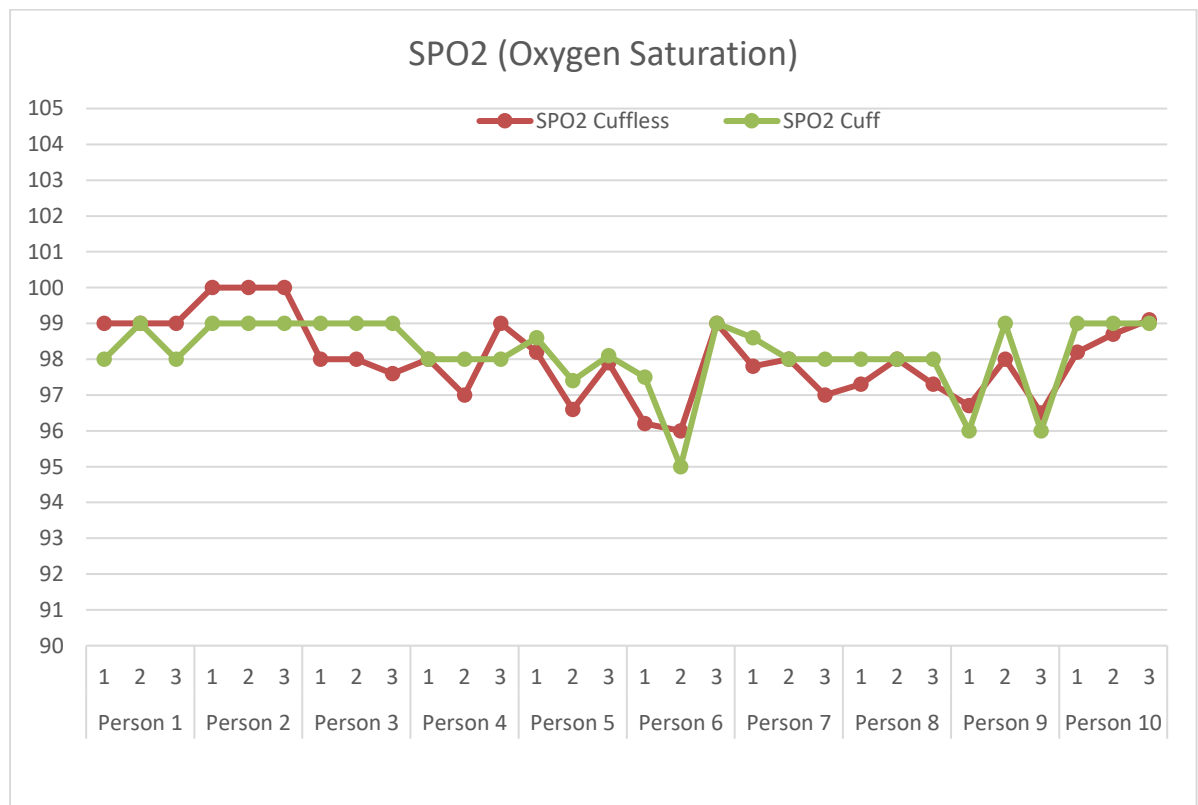


Fig 5.9: Oxygen Saturation (SpO2) data for cuff and cuffless (Age 44-55)

Each person's Blood pressure, Heart rate, skin temperature and Oxygen saturation results vary slightly depending on the measurement environment and body conditions. This could be related to a number of variables, including exercise level, stress, or time of day. From the two data table (Table 5.1 and Table 5.2) analyses it can be said that the device is more effective for young people. As the Blood pressure, Heart rate, skin temperature and Oxygen saturation is variedly different from one another in young age group. On the other hand, in young people the variation is not that much different from one another thus giving less error comparatively from young people. The MAP rate MAPE% is also low compared with old age people. So, the device is more compatible for young care services.

Uncertainty analysis

Table 5.3: Data taken for uncertainty analysis.

Data- Taken	Method BPT (SBP/DBP, mmHg) P1 (24 YEAR MALE)	Method BPT (SBP/DBP, mmHg) P2 (24 YEAR MALE)
1	124/77	128/85
2	126/79	124/80
3	127/77	123/76
4	128/75	126/84
5	123/75	126/81
6	129/80	124/79
7	124/77	123/73
8	127/76	126/78
9	125/76	125/81
10	121/74	126/76
11	126/78	123/72
12	125/74	127/79
13	127/74	124/74
14	123/81	124/78
15	124/76	127/78
16	128/74	124/75
17	130/78	124/82
18	130/75	124/76
19	134/75	123/79
20	132/83	123/78
Mean, mmHg	126.65/76.7	124.7/78.2
uA , mmHg	0.43/0.48	0.35/0.77

For P1

Summary: SBP = 126.65 ± 10.77 mmHg (with 95% confidence)

DBP = 76.7 ± 6.558 mmHg (with 95% confidence)

For P2

Summary: SBP = 124.7 ± 10.53 mmHg (with 95% confidence)

DBP = 78.2 ± 6.760 mmHg (with 95% confidence)

FOR P1:

Given Data:

SBP (systolic): 124, 126, 127, 128, 123, 129, 124, 127, 125, 121, 126, 125, 127, 123, 124, 128, 130, 130, 134, 132

DBP (diastolic): 77, 79, 77, 75, 75, 80, 77, 76, 76, 74, 78, 74, 74, 81, 76, 74, 78, 75, 75, 83

Step 1: Calculate the average ($BP(\text{avg})$)

For SBP:

$$\begin{aligned} \text{SBP}_{\text{avg}} &= \\ &= \frac{129 + 124 + 127 + 125 + 121 + 126 + 125 + 127 + 123 + 124 + 128 + 130 + 130 + 134 + 132}{20} \end{aligned}$$

$$\text{SBP}_{\text{avg}} = \frac{2533}{20} = 126.65$$

For DBP:

$$\text{DBP}_{\text{avg}} = \frac{77 + 79 + 77 + 75 + 75 + 80 + 77 + 76 + 76 + 74 + 78 + 74 + 74 + 81 + 76 + 74 + 78 + 75 + 75 + 83}{20}$$

$$\text{DBP}_{\text{avg}} = \frac{1534}{20} = 76.7$$

Step 2: Calculate uA (standard uncertainty of type A)

$$uA = \left[\frac{\sum (BP_i - BP(\text{avg}))^2}{n(n-1)} \right]^{0.5}$$

So, $\sum (SBP_i - \text{SBP}_{\text{avg}})^2 = 200.55$

Now,

$$uA_{\text{SBP}} = \left[\frac{200.55}{20 \times 19} \right]^{0.5} = \left[\frac{200.55}{380} \right]^{0.5} = (0.527763)^{0.5} = 0.7275$$

So, $\sum (DBP_i - \text{DBP}_{\text{avg}})^2 = 120.2$

Now,

$$uA_{DBP} = \left[\frac{120.2}{20 \times 19} \right]^{0.5} = \left[\frac{120.2}{380} \right]^{0.5} = (0.316316)^{0.5} = 0.5624$$

Step 3: Calculate relative uncertainty $\widetilde{uA}$

$$\widetilde{uA} = \left(\frac{uA}{BP(\text{avg})} \right) \times 100\%$$

For SBP:

$$\widetilde{uA}_{SBP} = \left(\frac{0.7275}{126.65} \right) \times 100\% = 0.005744 \times 100\% = 0.5744\%$$

For DBP:

$$\widetilde{uA}_{DBP} = \left(\frac{0.5624}{76.7} \right) \times 100\% = 0.007333 \times 100\% = 0.7333\%$$

Step 4: Given $\gamma_1 = 4.3\%$, $\gamma_2 = 0.08\%$

Step 5: Calculate combined relative uncertainty $\widetilde{uC}$

$$\widetilde{uC} = \sqrt{\widetilde{uA}^2 + \gamma_1^2 + \gamma_2^2}$$

For SBP:

$$\widetilde{uC}_{SBP} = \sqrt{(0.5744)^2 + (4.3)^2 + (0.08)^2} = \sqrt{0.3299 + 18.49 + 0.0064} = \sqrt{18.8263} = 4.3395\%$$

For DBP:

$$\widetilde{uC}_{DBP} = \sqrt{(0.7333)^2 + (4.3)^2 + (0.08)^2} = \sqrt{0.5377 + 18.49 + 0.0064} = \sqrt{19.0341} = 4.3631\%$$

Step 6: Expansion factor $kp = 1.96$ (for 95% confidence)

Step 7: Calculate expanded uncertainty U

$$U = kp \times u_c \times BP(\text{avg})/100\%$$

uncertainty: $u_c = \widetilde{uC} \times BP(\text{avg})/100$

Then,

$$U = kp \times u_c = kp \times \frac{\widetilde{uC} \times BP(\text{avg})}{100}$$

For SBP:

$$u_{c,SBP} = \frac{4.3395 \times 126.65}{100} = \frac{549.6}{100} = 5.496$$

$$U_{SBP} = 1.96 \times 5.496 = 10.77$$

For DBP:

$$u_{c,DBP} = \frac{4.3631 \times 76.7}{100} = \frac{334.6}{100} = 3.346$$

$$U_{DBP} = 1.96 \times 3.346 = 6.558$$

Final Results:

For SBP (systolic):

- Average: 126.65
- uA : 0.7275
- $\widetilde{uA}$: 0.5744%
- $\widetilde{uC}$: 4.3395%
- Expanded uncertainty U : 10.77

For DBP (diastolic):

- Average: 76.7
- uA : 0.5624
- $\widetilde{uA}$: 0.7333%
- $\widetilde{uC}$: 4.3631%
- Expanded uncertainty U : 6.558

Summary: SBP = 126.65 ± 10.77 mmHg (with 95% confidence)

DBP = 76.7 ± 6.558 mmHg (with 95% confidence)

FOR P2:

Given Data:

SBP (systolic): 128, 124, 123, 126, 126, 124, 123, 126, 125, 126, 123, 127, 124, 124, 127, 124, 124, 124, 123, 123

DBP (diastolic): 85, 80, 76, 84, 81, 79, 73, 78, 81, 76, 72, 79, 74, 78, 78, 75, 82, 76, 79, 78

Number of readings, $n=20$

Step 1: Calculate the average (BP_{avg})

For SBP: List: 128, 124, 123, 126, 126, 124, 123, 126, 125, 126, 123, 127, 124, 124, 127, 124, 124, 124, 123, 123

So, sum = 2494

$$SBP_{avg} = \frac{2494}{20} = 124.7$$

For DBP: List: 85, 80, 76, 84, 81, 79, 73, 78, 81, 76, 72, 79, 74, 78, 78, 75, 82, 76, 79, 78

Sum = $85 + 80 + 76 + 84 + 81 + 79 + 73 + 78 + 81 + 76 + 72 + 79 + 74 + 78 + 78 + 75 + 82 + 76 + 79 + 78$

So, sum = 1564

$$DBP_{avg} = \frac{1564}{20} = 78.2$$

Step 2: Calculate uA (standard uncertainty of type A)

$$uA = \left[\frac{\sum (BP_i - BP_{\text{avg}})^2}{n(n-1)} \right]^{0.5}$$

So, $\sum (SBP_i - SBP_{\text{avg}})^2 = 46.20$

Now,

$$uA_{SBP} = \left[\frac{46.20}{20 \times 19} \right]^{0.5} = \left[\frac{46.20}{380} \right]^{0.5} = (0.1215789)^{0.5} = 0.3487$$

So, $\sum (DBP_i - DBP_{\text{avg}})^2 = 223.20$

Now,

$$uA_{DBP} = \left[\frac{223.20}{20 \times 19} \right]^{0.5} = \left[\frac{223.20}{380} \right]^{0.5} = (0.587368)^{0.5} = 0.7664$$

Step 3: Calculate relative uncertainty $\widetilde{uA}$

$$\widetilde{uA} = \left(\frac{uA}{BP_{\text{avg}}} \right) \times 100\%$$

For SBP:

$$\widetilde{uA}_{SBP} = \left(\frac{0.3487}{124.7} \right) \times 100\% = 0.002797 \times 100\% = 0.2797\%$$

For DBP:

$$\widetilde{uA}_{DBP} = \left(\frac{0.7664}{78.2} \right) \times 100\% = 0.009800 \times 100\% = 0.9800\%$$

Step 4: Given $\gamma_1 = 4.3\%$, $\gamma_2 = 0.08\%$

These are relative uncertainties from other sources.

Step 5: Calculate combined relative uncertainty $\widetilde{uC}$

$$\widetilde{uC} = \sqrt{\widetilde{uA}^2 + \gamma_1^2 + \gamma_2^2}$$

For SBP:

$$\begin{aligned} \widetilde{uC}_{SBP} &= \sqrt{(0.2797)^2 + (4.3)^2 + (0.08)^2} = \sqrt{0.07824 + 18.49 + 0.0064} = \sqrt{18.57464} \\ &= 4.3104\% \end{aligned}$$

For DBP:

$$\widetilde{uC}_{DBP} = \sqrt{(0.9800)^2 + (4.3)^2 + (0.08)^2} = \sqrt{0.9604 + 18.49 + 0.0064} = \sqrt{19.4568} = 4.4110\%$$

Step 6: Expansion factor $kp = 1.96$ (for 95% confidence)

Step 7: Calculate expanded uncertainty U

First, find absolute combined uncertainty: $u_c = \frac{\widetilde{uC} \times BP_{\text{avg}}}{100}$

Then,

$$U = kp \times u_c = kp \times \frac{\widetilde{uC} \times BP(\text{avg})}{100}$$

For SBP:

$$u_{c,SBP} = \frac{4.3104 \times 124.7}{100} = \frac{537.4}{100} = 5.374$$

$$U_{SBP} = 1.96 \times 5.374 = 10.53$$

For DBP:

$$u_{c,DBP} = \frac{4.4110 \times 78.2}{100} = \frac{344.9}{100} = 3.449$$

$$U_{DBP} = 1.96 \times 3.449 = 6.760$$

Final Results:

For SBP (systolic):

- Average: 124.7 mmHg
- uA : 0.3487
- $\widetilde{uA}$: 0.2797%
- $\widetilde{uC}$: 4.3104%
- Expanded uncertainty U : 10.53 mmHg

For DBP (diastolic):

- Average: 78.2 mmHg
- uA : 0.7664
- $\widetilde{uA}$: 0.9800%
- $\widetilde{uC}$: 4.4110%
- Expanded uncertainty U : 6.760 mmHg

Summary: SBP = 124.7 ± 10.53 mmHg (with 95% confidence)

DBP = 78.2 ± 6.760 mmHg (with 95% confidence)

5.2.2.2 Prototype 2 (Pulse Transient Time)

For Table 5.3, the all-over errors were calculated using Mean Absolute Percentage Error by using equation 6. The cuffless value is the value that we get from the MAX86150 sensor and the cuff value we get from hardware machine (Omron).

Table 5.3 consists of 10 volunteers (young age range 22-25) with data of Blood pressure (Systolic blood pressure and Diastolic blood pressure), Heart Rate and MAP (Mean arterial pressure). The formula for MAP was measured by using equation 7.

MAP = Mean Arterial Pressure, DBP = Diastolic Blood Pressure, SBP = Systolic Blood Pressure

Table 5.4: Data Table for 22-24 age using Prototype 2 (PTT)

Subject	Data-Taken	Gender	Age	WEIGH T (KG)	Height (cm)	BMI	SYS Cuff	SYS Cuffless	DIA Cuff	DIA Cuffless	MAP Cuff	MAP Cuffless	MAPE SYS	MAPE DIA	MAPE OVERAL	MAPE DIA OVERAL	MAPE MAP OVERAL	HR Cuff	HR Cuffless	HR MAPE	HR MAPE OVERALL	
Person 1	1	M	24	85	170	29.41	114	120	80	70	91.33	86.667	✓	✓	3.60	9.17	3.60	I _r	79	80	✓	4.03
	2						112	118	83	75	92.67	89.333							80	81		
	3						116	123	83	89	94.00	100.333	6.63	9.17					80	82		
	4						113	125	78	70	89.67	88.333							77	85		
	5						119	126	80	75	93.00	92.000							84	88		
Person 2	1	M	25	76	167.64	27.04	114	120	80	83.8	91.33	95.87	✓	✓	4.41	6.12	4.41	I _r	74	76.8	✓	2.38
	2						112	125	78	84	89.33	97.67							74	77		
	3						108	122	83	71	91.33	88.00	9.02	6.12					77	78.4		
	4						110	120	78	77	88.67	91.33							77	77.8		
	5						113	120	82	80	92.33	93.33							82	81		
Person 3	1	M	22	85	172	28.73	120	126.5	73	76.7	88.67	93.30	✓	✓	3.59	5.85	3.59	I _r	63	62	✓	1.93
	2						121	114	74	83	89.67	93.33							63	61		
	3						118	126	76	79.8	90.00	95.20	4.41	5.85					63	62		
	4						120	120.9	77	80	91.33	93.63							61	63		
	5						121	125	80	77.5	93.67	93.33							69	69		
Person 4	1	M	23	95	170	32.87	131	125	86	82	101.00	96.33	✓	✓	4.71	5.84	4.71	I _r	79	77	✓	2.37
	2						125	120	86	83.4	99.00	95.60							77	74.5		
	3						132	126	81	83.2	98.00	97.47	4.59	5.84					73	73.6		
	4						133	125	78	74.2	96.33	91.13							74	76.9		
	5						132	127	86	74	101.33	91.67							75	76		
Person 5	1	M	24	78	170	26.99	124	129	83	72	96.67	91.00	✓	✓	4.56	5.56	4.56	I _r	63	63	✓	3.987
	2						123	110	79	77.1	93.67	88.07							65	58.9		
	3						121	118	78	82	92.33	94.00	5.80	5.56					66	62.9		
	4						115	120	77	82.4	89.67	94.93							64	64.7		
	5						119	110	78	78	91.67	88.67							70	68.6		
Person 6	1	F	24	59	165	21.67	107	103	74	88	85.00	93.00	✓	✓	7.39	14.58	7.39	I _r	76	82	✓	6.67
	2						108	102	80	73	89.33	82.67							85	76		
	3						110	101	71	82	84.00	88.33	6.09	14.58					80	78		
	4						115	105	72	82	86.33	89.67							77	75		
	5						117	112	82	69	93.67	83.33							82	90		
Person 7	1	F	25	52	162.5	19.69	111	116	85	74	93.67	88.00	✓	✓	8.48	12.73	8.48	I _r	87	83	✓	2.91
	2						101	110	77	66	85.00	80.67							85	87.5		
	3						103	110	78	87	86.33	94.67	7.41	12.73					88	85.7		
	4						100	110	70	77	80.00	88.00							91	95		
	5						102	109	74	85	83.33	93.00							89	89		
Person 8	1	F	23	50	152.4	21.53	117	101	76	85	89.67	90.33	✓	✓	6.42	11.78	6.42	I _r	85	80	✓	6.15
	2						111	104	84	69	93.00	80.67							84	80.5		
	3						115	100	80	89	91.67	92.67	9.72	11.78					80	78		
	4						109	115	78	85	88.33	95.00							83	77		
	5						109	120	78	85	88.33	96.67							82	73		
Person 9	1	F	24	45	160.02	17.57	119	111	85	79	96.33	89.67	✓	✓	7.21	7.15	7.21	I _r	91	87.9	✓	5.56
	2						114	119	78	82	90.00	94.33							84	88		
	3						112	122	73	79	86.00	93.33	7.29	7.15					88	83		
	4						117	125	75	77	89.00	93.00							90	88		
	5						115	126	71	80	85.67	95.33							92	81.2		
Person 10	1	F	23	60	162.56	22.71	126	115	85	69	98.67	84.33	✓	✓	9.88	12.18	9.88	I _r	91	97	✓	4.43
	2						125	115	89	77	101.00	89.67							98	90		
	3						114	121	73	77	86.67	91.67	6.81	12.18					92	96		
	4						114	120	73	85	86.67	96.67							96	97		
	5						118	125	75	80	89.33	95.00							99	97		
Accuracy															93.222	90.903	93.976	96.013				

The test involved 10 volunteers in the 20-24 age range equally divided between 5 male and 5 female students. For each volunteer 5 set data taken None of the participants reported any notable respiratory illnesses.

The MAPE percentage of each individual participant and for all the participants are also presented in the table. The MAPE% for systolic Blood Pressure is 6.778%, for diastolic blood pressure is 9.097%, for mean arterial pressure is 6.024% and for heart rate is 3.987% So the accuracy for Systolic blood pressure is 93.222%, accuracy for diastolic blood pressure is 90.903%, accuracy for MAP is 93.976% and accuracy for heart rate is 96.013%. Table 5.3 data is also plotted in fig 5.10 to fig 5.12.

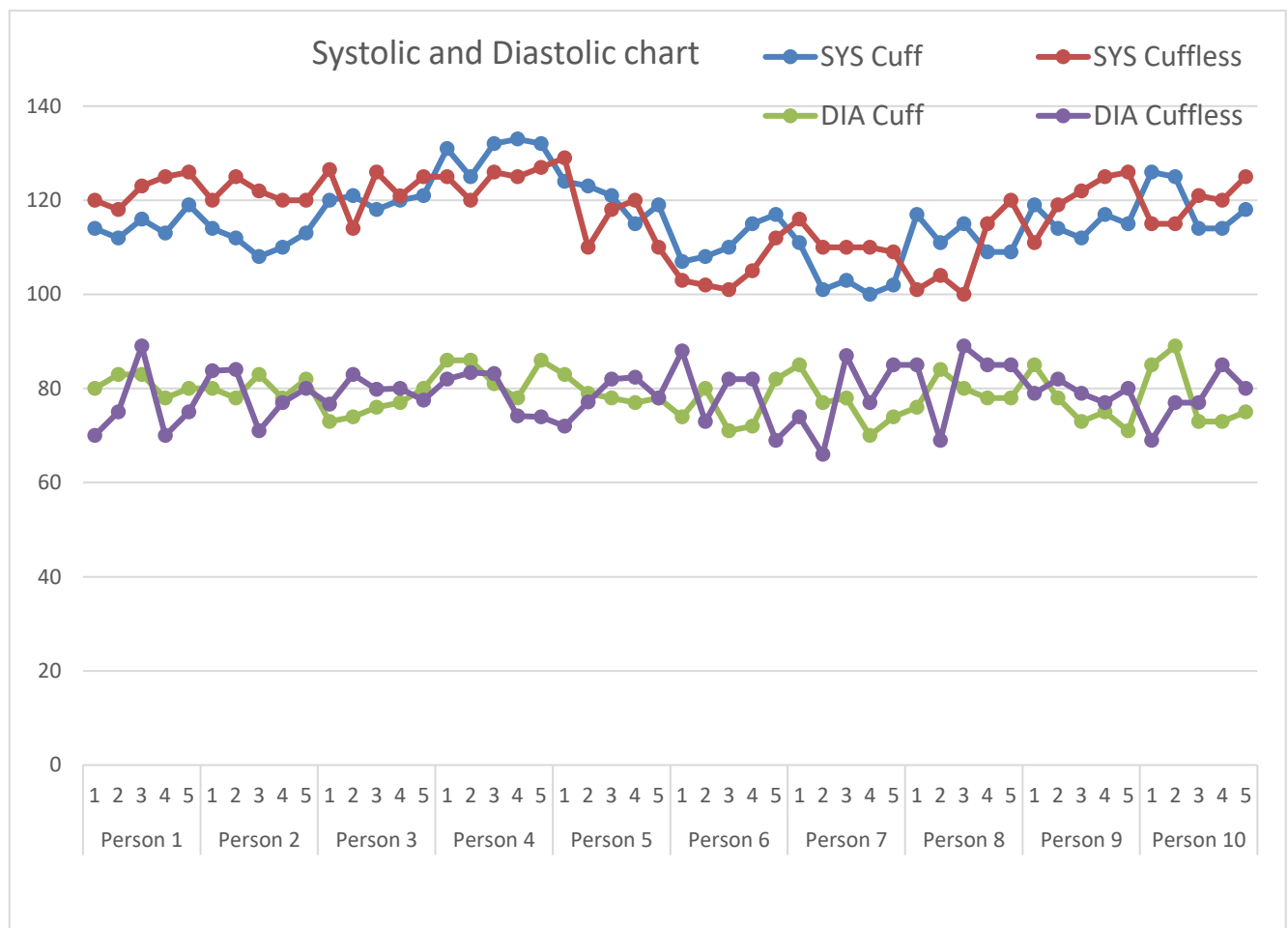


Fig 5.10: Systolic and Diastolic data for cuff and cuffless (Age 22-25)

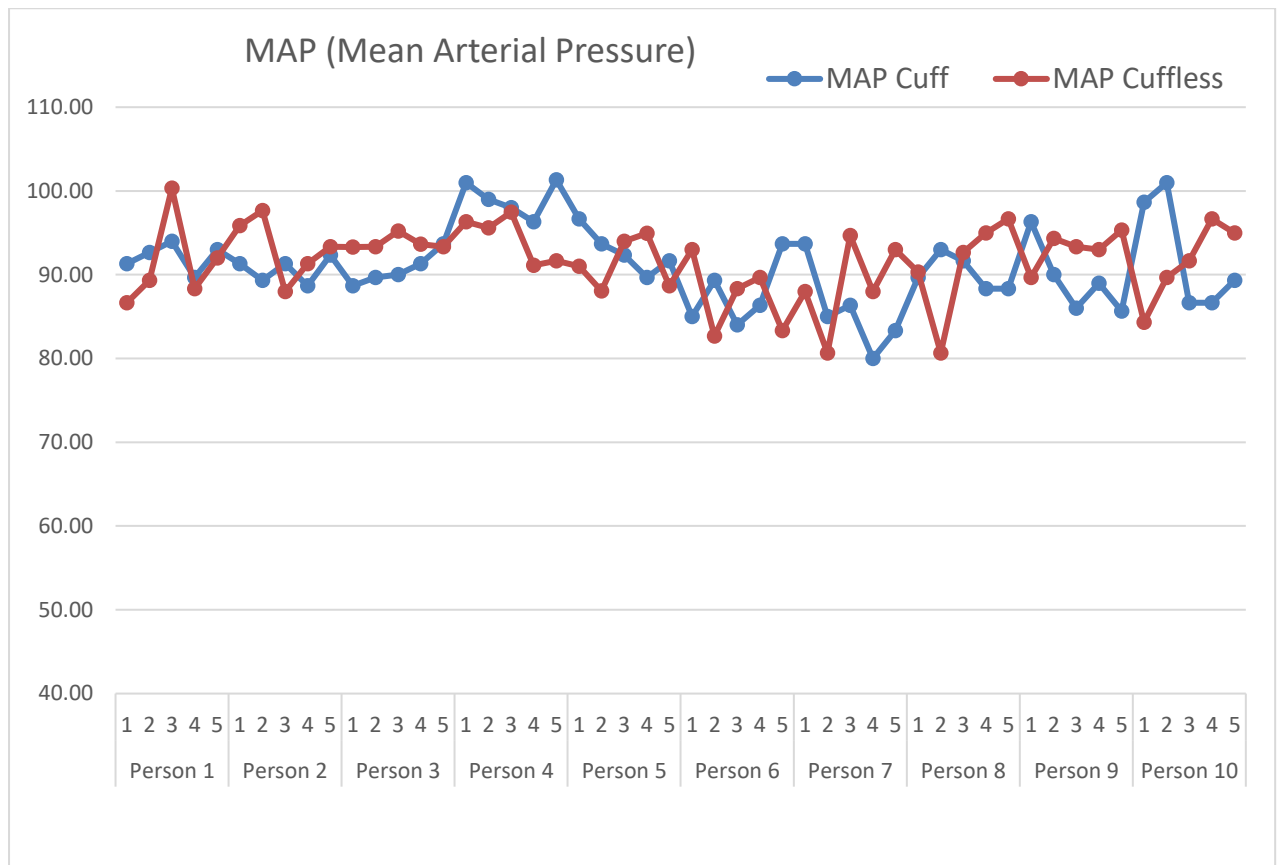


Fig 5.11: Mean Arterial Pressure (MAP) data for cuff and cuffless (Age 22-25)

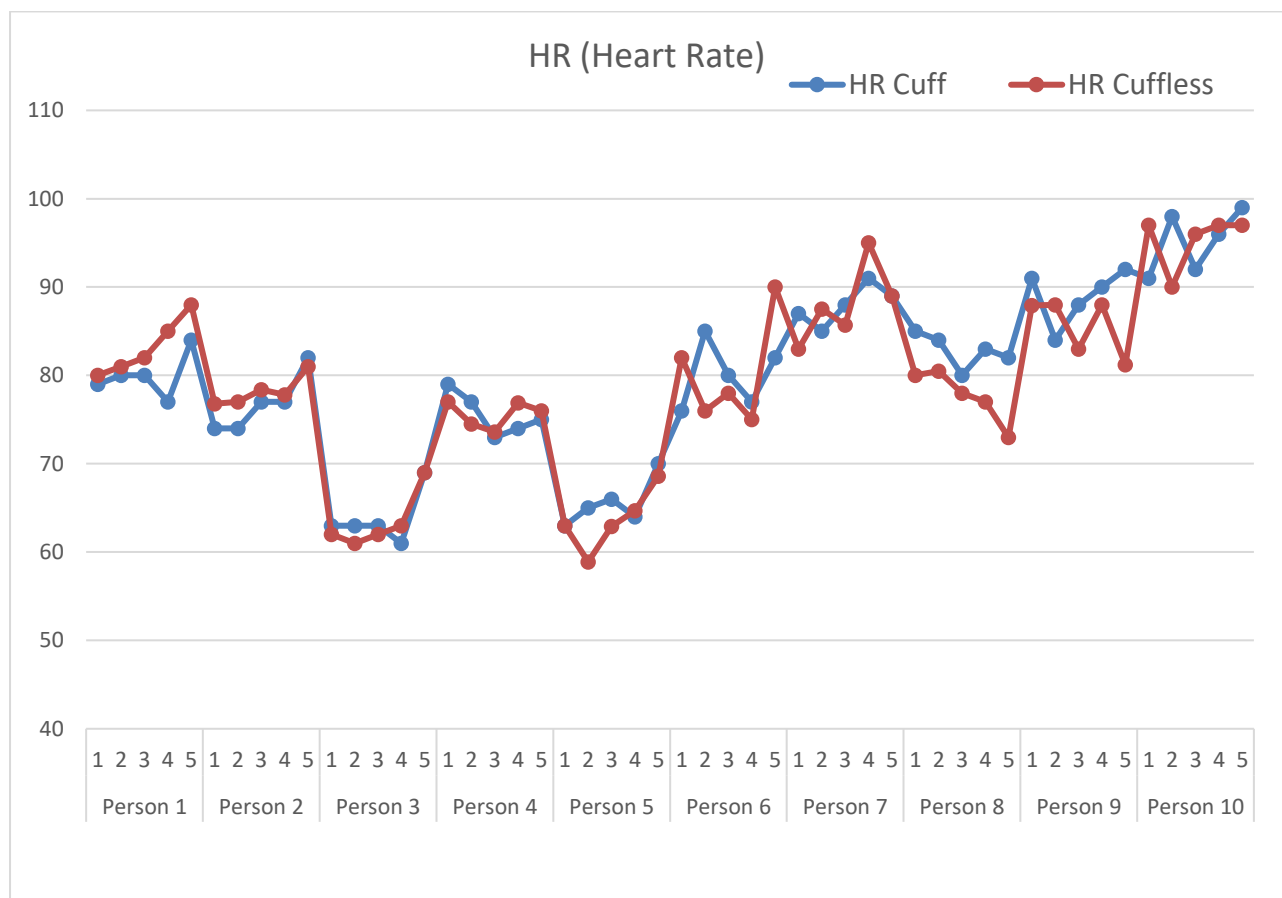


Fig 5.12: Heart rate data for cuff and cuffless (Age 22-25)

The figures 5.10 to 5.12 give us an overview of how the data of cuff and cuffless value are varied from each other. The error can come from various reasons such as environment and body condition. This could be related to a number of variables, including exercise level, stress, or time of day.

5.3. Comparison of Results

Table 5.5: Comparison of Results between Cuff device, Prototype 1 (BPT) and Prototye 2 (PPT)

Data - taken	Cuff device Sys/Dia, HR	Prototype 1 (BPT) Sys/Dia, HR	Prototye 2 (PPT) Sys/Día, HR
1	117/81, 67	117/82,65	117/81.4,70
2	118/71,68	118/72,70	116.2/81.9,65.9
3	120/70,68	120/72,70	124/78,67.9
4	112/72,68	112/73,69	118/80,64
5	116/74,69	115/72,67	119.9/81.1,71.4
6	115/74,67	116/75,69	112/83,71
7	116/72,70	118/73,72	113/83,65

5.4. Summary

This chapter presents the results and analysis from testing the two prototypes. The first prototype, based on Blood Pressure Trending (BPT), was validated against a standard cuff-based device. The testing involved 20 volunteers from two age groups, 20-24 and 48-55. The system showed high accuracy, with a Mean Absolute Percentage Error (MAPE) of 3.47% for systolic pressure, 5.01% for diastolic pressure, and 0.83% for SpO₂ in the younger group. The older age group had slightly higher error rates, but all measured parameters remained within an acceptable clinical range. This suggests the device works well, especially for younger users. The results show that the prototypes effectively offer a viable, cost-friendly option for cuffless blood pressure monitoring, meeting the main goals of the project.

Chapter 6

CONCLUSION

6.1. Summary of Findings

The result of cuffless blood pressure measurement with non-invasive method device is working correctly. This device makes it easier for Blood pressure measurement. This satisfies the need for its intended user, saves their time and money and gives people more flexibility. The sensors MAX3266 and Max86150 have given us clear signal which helped us with very good blood pressure measurement. The devices not only measure blood pressure but also heart rate and oxygen saturation. The cuffless blood pressure measurement system is very friendly, and a user can carry it with them everywhere. The device can detect blood pressure within a limited margin of error as it can't show exactly the measure that a cuff device shows. To continually enhance the designs, it should be encouraged that the healthcare industry, technologists, designers, and elderly aged groups collaborate.

6.2. Novelty of the work

The novelty of this work lies in developing a prototype for continuous blood pressure measurement by simultaneously recording ECG and PPG signals using MAX32664 and MAX86150 sensor. Unlike traditional methods that depend on cuff-based approaches, this prototype introduces a cuffless measurement system. This change helps avoid the discomfort and limitations common in standard blood pressure monitoring. Additionally, the design focuses on portability and compactness, making the device small, lightweight, and easy to carry. These unique features position the proposed prototype as a more user-friendly, non-invasive, and efficient option for real-time blood pressure monitoring.

6.3. Cultural, Societal, and Environmental Factors and Impacts

6.3.1. Cultural Factor

- **Privacy:** In many cultures it might be offensive and untrustworthy of blood pressure that is monitored without direct contact. There might be concerns about who has access to this data and how it's being used.
- **Touch vs. Non-Touch:** In some cultures, there might be a preference for human contact during medical checkups. A contactless system might seem impersonal or less caring.
- **Equity and Accessibility:** The system is designed to be cost effective and inclusive, allowing all people from different socioeconomic backgrounds to benefit.

6.3.2. Societal and Economical Impacts

- **Accessibility:** This system could be a big benefit for people who have difficulty going to clinics or hospitals regularly. It could also be useful in places with limited medical staff.
- **Hygiene:** This system could reduce the spread of germs in hospitals and clinics, especially beneficial during outbreaks.
- **Cost-Effectiveness:** If widely used, this system could potentially lower healthcare costs, allowing for remote monitoring and early detection of health issues.

6.3.3. Environmental Factors and Impacts

- **Energy Efficiency:** The system is carefully designed with low power sensors, and rechargeable batteries thus reducing overall energy consumption and supporting sustainability.
- **Reduction of Waste:** Conventional cuff-based devices need regular replacement, but this system minimizes disposable parts and thereby lowers medical costs. The device is also made of ecofriendly materials.
- **Digital Record-Keeping:** As the device automatically stores patient data electronically, the system reduces the need for human hand and paper based medical records contributing to a greener healthcare ecosystem.

6.4. Limitations of the Work

Single-person measurement: The device is limited to measuring the blood pressure of only one person at a time. So, it cannot be used in groups or clinical settings where multiple people may need to be tested simultaneously.

Low data availability: The dataset that has been taken from this device to measure accuracy is very small. Because of this the result is not highly reliable to validate the device performance with the standards for Bangladesh perspective Directorate General of Drug Administration (DGDA).

Signal interruption: The accuracy of the device depends on clean ECG and PPG signals. Whenever there is noise and interruption in the signal, the device may produce unreliable blood pressure measurement that reduces its dependability and accuracy.

PPG skin effect: One of the main limitations of this project is the PPG skin effect. Different characteristics of human skin can impact the accuracy of the measured signals. Factors like skin tone, surface temperature, sensor attachment, motion, and differences in skin tissue all distort the readings.

People with darker skin tones may see lower signal quality because of higher melanin absorption. Additionally, changes in skin temperature can affect how light interacts with tissue.

6.5. Future Scopes

- **Multi-person measurement:** The system can be upgraded to handle multiple users thus allowing several people to measure their blood pressure efficiently in a clinical or household environment.
- **Larger dataset collection:** The dataset can be expanded with diverse age groups, health conditions and various demographics that will help and validate the system better resulting in higher accuracy and reliability.
- **Improved signal processing:** For better signal processing advanced filtering and noise-reduction techniques can be implemented to minimize the effect of signal interruptions in ECG and PPG, ensuring more stable and precise measurements.
- **Mobile applications:** A mobile app can be made for real-time monitoring, data storage, and trend analysis, enabling users to track long-term blood pressure variations conveniently.
- **AI and machine learning enhancement:** One of the big upgrades can be AI and machine learning integration. Merging with AI and machine learning, the model can predict blood pressure with more accuracy by learning from larger datasets, making the device more intelligent and adaptive over time.

6.6. Sustainability

The project contributes to multiple Sustainable Development Goals (SDGs), particularly:



Fig 6.1: SDG goals [40]

SDG 3 Good Health and Well-being: The project directly supports good health and well-being. By enabling continuous, non-invasive blood pressure monitoring it gives us a big leap of BP measurement. Hypertension is a major cause of cardiovascular diseases. Early detection with this device can prevent serious complications. It makes health monitoring easier and enables individuals to take proactive steps toward better well-being.

SDG 8: Decent Work and Economic Growth: This innovation can open new doors in the healthcare device industry. As the need for affordable health monitoring increases, the device can help create jobs in manufacturing, research, and healthcare services. It encourages economic growth by promoting affordable medical technology that can be made and shared widely.

SDG 10 Reduced Inequalities: By designing the system to be cost-effective and easy to use, it tackles healthcare accessibility issues. Many people in rural or low-income areas cannot afford regular clinical check-ups or advanced devices. This project helps reduce these inequalities by making blood pressure monitoring available to more people, ensuring that even underserved communities can benefit from modern healthcare technology.

6.7. Conclusion

This project met its main goal of designing and implementing a practical, cost-effective, cuffless blood pressure monitoring system. The work involved developing two different prototypes, each using a unique non-invasive sensing method to provide a comfortable and continuous monitoring option without an inflatable cuff. The first prototype employed the Blood Pressure Trending (BPT) technique. A hardware system was created by combining a MAX30102 optical sensor, a MAX32664 biometric hub, and a MAX30205 temperature sensor. An ESP8266 microcontroller controlled the setup. The device was calibrated against a standard Omron blood pressure monitor and was placed in a custom-designed, portable enclosure powered by a rechargeable battery. Extensive testing on 20 volunteers showed the system's high accuracy confirming its reliability for real-time health parameter estimation. The second prototype was based on the Pulse Transit Time (PTT) principle. This system used a MAX86150 sensor to obtain simultaneous ECG and PPG signals, which a Raspberry Pi Zero 2W processed. Considerable effort went into applying digital signal processing algorithms to filter out noise and accurately detect physiological peaks for precise PTT calculation, allowing for blood pressure estimation.

In conclusion, the project resulted in two functional prototypes that effectively turned theoretical models into practical, user-friendly devices. The results support the initial idea that accurate, cuffless blood pressure monitoring is possible using low-cost, commercially available components. The project achieved its technical goals and followed professional engineering practices, providing a strong foundation for future development in portable healthcare technology.

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Appendix A

Datasheet of the ICs used

Click [here](#) to ask an associate for production status of specific part numbers.

MAX32664

Ultra-Low Power Biometric Sensor Hub

General Description

The MAX32664 is a sensor hub family with embedded firmware and world-class algorithms for wearables. It seamlessly enables customer-desired sensor functionality, including communication with Analog Devices' optical sensor solutions and delivering raw or calculated data to the outside world. This is achieved while keeping overall system power consumption in check. The device family interfaces to a microcontroller host through a fast-mode slave I²C interface for access to raw and processed sensor data as well as field updates. A firmware bootloader is also provided.

The MAX32664 Version A supports the MAX30101/MAX30102 high-sensitivity pulse oximeter and heart-rate sensor for wearable health for finger-based applications. A master mode I²C interface for communication with sensors is provided.

The MAX32664 Version B supports the MAX86140/MAX86141 for wrist-based applications. A master mode SPI interface for communication with sensors is provided.

The MAX32664 Version C supports the MAX86140/MAX86141 for wrist-based applications and MAXM86161 for ear-based applications. The device provides either a master mode SPI or an I²C interface for communication with sensors.

The MAX32664 Version D supports the MAX30101/MAX30102 high-sensitivity pulse oximeter and heart-rate sensor for wearable health for finger-based applications. A master mode I²C interface for communication with sensors is provided. Version D also supports estimated blood pressure monitoring.

The wearable algorithms in the MAX32664 sensor hub support a directly connected accelerometer. They also allow feeding of X, Y, and Z samples from a host-connected accelerometer. This architecture provides robust detection and compensation of motion artifacts in captured samples.

The tiny form factor 1.6mm x 1.6mm WLP or 3mm x 3mm TQFN allows for integration into extremely small application devices.

Applications

- Wearable Fitness
- Hearables
- Wearable Medical
- Portable Medical
- Mobile Devices

Benefits and Features

- Biometric Sensor Hub Enables Faster Time to Market
- Finger-Based (Version A) Algorithms Measure:
 - Pulse Heart Rate
 - Pulse Blood Oxygen Saturation (SpO₂)
- Wrist-Based (Version B) Algorithm Measures:
 - Pulse Heart Rate
- Wrist-Based or Ear-Based (Version C) Algorithms Measure:
 - Pulse Heart Rate
 - Pulse Blood Oxygen Saturation (SpO₂)
- Finger-Based (Version D) Algorithms Measure:
 - Pulse Heart Rate
 - Pulse Blood Oxygen Saturation (SpO₂)
 - Estimated Blood Pressure
- Both Raw and Processed Data Are Available
- Basic Peripheral Mix Optimizes Size and Performance
 - One Slave I²C for Communication with a Host Microcontroller
 - One Master I²C for Communication with Sensors (Versions A, C, and D)
 - One Master SPI for Communication with Sensors (Versions B and C)
 - 32.768kHz RTC
 - FIFO Provides Minimal Host Interaction
 - Bootloader Facilitates Secure, Authenticated Firmware Upgrades

MAX86150

Integrated Photoplethysmogram and Electrocardiogram Bio-Sensor Module For Mobile Health

General Description

The MAX86150 is an integrated electrocardiogram, pulse oximeter, heart rate monitor sensor module. It includes internal LEDs, photodetector, and low-noise electronics with ambient light rejection. The MAX86150 helps ease design-in to mobile and wearable devices.

The MAX86150 operates on a 1.8V supply voltage with a separate power supply for the internal LEDs. Communication to and from the module is entirely through a standard I²C-compatible interface. The module can be shut down through software with near zero standby current, allowing the power rails to remain powered at all times.

Applications

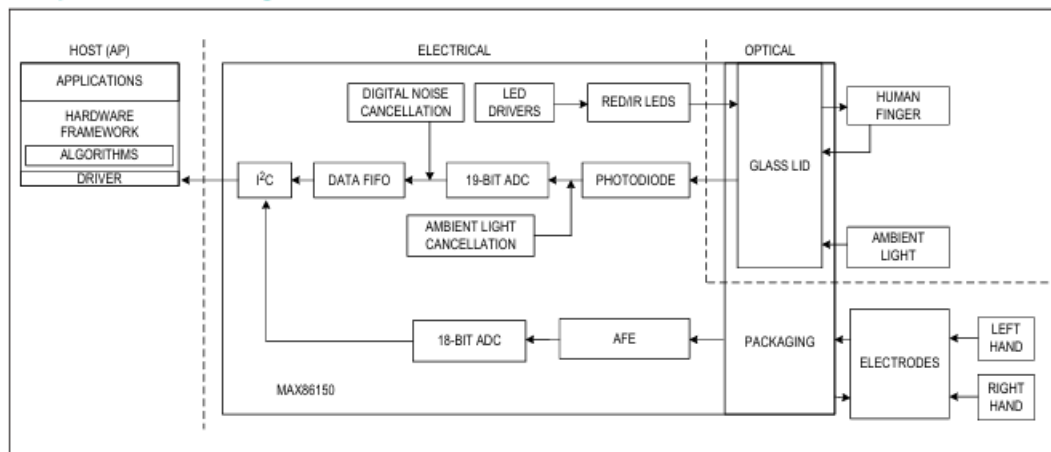
- Smartphones
- Tablets
- Wearable Devices
- Fitness Assistant Devices

Benefits and Features

- Electrocardiogram (ECG) Optimized for Dry Electrode Operation
- Reflective Heart Rate Monitor and Medical-Grade Pulse Oximeter
- Miniature 3.3mm x 5.6mm x 1.3mm 22-pin Optical Module
 - Optical-Grade Glass for Long-Term Optimal and Robust Performance
- Ultra-Low Power Operation for Mobile Devices
 - Ultra-Low Shutdown Current (0.7μA Typical)
- High SNR and Robust Ambient Light Cancellation
- -40°C to +85°C Operating Temperature Range

Ordering Information appears at end of data sheet.

Simplified Block Diagram



Appendix B

iThenticate Plagiarism Report



Page 2 of 112 - Integrity Overview

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